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Unit 1 Development of Nursing

1. State the contributions of Florence Nightingale to the development of nursing;
2. Identify the four concepts essential to nursing;
3. State the aims of nursing.

Nursing has been called the oldest art and the youngest profession. The word **nurse** evolved from the Latin word *nutricius* which means "nourishing". When looking at the historical development of nursing, most *nurse historians*① agree that nursing or the care of the ill and injured has been done since the beginning of human life and has generally been a woman荫 role. A mother caring for a child in a cave is an example of early nursing.

Nursing before 19th century was developed slowly and fastidiously. *No special training was provided or required for the person who served as nurses, and the image of nurses was bound to the servant or simple career.*②

*The emergence of professional nursing is usually attributed to the influence of Florence Nightingale.*③ Florence Nightingale, who was born in 1820 in a wealthy and intellectual family in England, was well educated and her education included several ancient and modern languages, literature, philosophy, history, science, mathematics, religion, art and music. It was expected that she would follow the usual path of a wealthy woman of those days: to be married, bearing children and maintaining an elegant home. However, Nightingale chose to become a nurse, and such decision was shaped by three major influences in her life. Firstly, she was dissatisfied with what she viewed as the dull, routine life-style of the up-class women of her day. Secondly, she had received a "classical education which is equal to that of most men of her day". This education provided her with an understanding of the circumstances of the world in which she lived. Thirdly, she became aware of the inadequate care being provided in hospitals as she accompanied her mother on visits to the ill, and she believed that she was "called by God to help others and to improve the well-being of mankind".

At that time, hospitals were terrible places and nurses were, in most cases, the dregs of society. Thus, hospitals were certainly not places for up-class women to go and her family was

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① 护理史学家。
② 从事护理工作的人没有也无需经过任何特殊的训练，护士的形象与仆人或从事简单职业的人相提并论。
③ 专业护理的出现多归因于南丁格尔的影响。

in an uproar over her decision. Nightingale studied nursing in secret and she visited *Kaiserswerth*① and received training there for three months at the age of 31.

The outbreak of the *Crimean War*② provided Nightingale an opportunity to organize a small band of nurses to work for the British hospital at *Scutari in Turkey*③. When they arrived, what she found was a hospital" so crowded that patients lay on the floor, still in bloody uniforms, bath equipment, cutlery, and laundry facilities were either non-existent or nearly so". With great compassion, Nightingale and her nurses began to work, they changed the military hospitals by setting up diet kitchens, a laundry, recreation centers, and reading rooms, and organizing classes for orderlies. Through her efforts, Florence Nightingale made a great difference in the care of the British soldiers, and her efforts were largely responsible for *dramatic reductions in the death rate*④ of soldiers from 42 percent to 2 percent. *This great success changed the prejudices against women and elevated the status of all nurses*.⑤

After the war, Nightingale returned to England. *She continued to devote herself to nursing by establishing school of nursing based on her knowledge of what was effective nursing*.⑥ The Nightingale Training School for Nurses at St. Thomas' Hospital in London was the first nursing school which could provide systematic training programs for nurses in the world. And eventually, many schools in Europe and America used the Nightingale model for nursing education.

In addition, Florence Nightingale was a productive writer. She wrote and published countless articles and papers. Her most famous written documents were *Notes on Nursing*⑦ and *Notes on Hospital*⑧. Through those publications, she shared her ideas about nursing and nursing education, thus *her contributions to nursing are numerous and far-reaching*⑨.

Modern nursing has been developed rapidly since Florence Nightingale established professional nursing and now its unique body of knowledge has been formulated and the function or focus of nursing has been defined.

In defining the knowledge base of nursing, four main concepts need to be briefly introduced here since they dominate and shape *professional nursing practice*⑩. The four concepts are: the person, environment, health, and nursing.

Person: As the recipient of care, the person is the focus of professional nursing. *The person is viewed as an integrated whole which consists of bio-psycho-social aspects*.⑪ Because

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① 德国凯撒斯畏斯城。
② 克里米亚战争。
③ 土耳其斯库塔里。
④ 死亡率显著下降。
⑤ 这一巨大成就改变了社会对妇女的偏见，也提升了所有护士的地位。
⑥ 她继续投身于护理事业，利用她对护理的认识建立了一所护理学校。
⑦ 《护理札记》。
⑧ 《医院札记》。
⑨ 她对护理事业的贡献是巨大而意义深远的。
⑩ 护理专业实践。
⑪ 人被视为是由生理 - 心理 - 社会诸方面构成的一个整体。

persons are greater than the sum of their parts,① we cannot simply learn about persons by viewing their components in isolation. The interactions of all the parts are what make persons unique.

Environment: Environment is an important concept in nursing because a healthy environment leads to healthy people. Physical, social, political, economic and cultural environments all influence health. A pleasant and comfortable environment has a significant relationship to health.

Health: Health has been defined in many ways. However, *nurses agree the definition from WHO, that is "health is a state of complete physical, mental, and social well-being, not merely the absence of disease or infirmity"*.②

Nursing: Nursing is both an art and a science. Art may be viewed as the systematic application of knowledge or skills. As a science and a profession, nursing is experiencing evolutional growth of its own. There is no single definition of nursing. However, *the theme of helping and caring is inherent in nursing*.③

The aims or the focuses of nursing have been clearly identified today. Nursing practice involves four areas related to health — *health promotion, health maintenance, health restoration, and care of the dying*④. Health promotion means helping people develop resources to maintain or enhance their well-being. Health maintenance nursing activities are those actions that help clients to maintain their health status. Health restoration means helping people to improve health following health problems or illness. Care of the dying involves comforting and caring for people of all ages while they are dying.

Daily Language

1. When is the Nurses Day?
   护士节是哪一天？
2. It is on May 12 each year.
   每年 5 月 12 日为护士节。
3. Why do people choose the day?
   为什么选择那一天？
4. We choose that day in honor of Florence Nightingale, who established modern nursing. She was born on May 12th.
   我们选择 5 月 12 日是为了纪念南丁格尔，她创立了现代护理学。南丁格尔出生在 5 月 12 日。

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① 整体"人"的功能大于其各方面机能的简单相加。
② 护理多采用 WHO 对健康的定义："健康不仅是没有疾病或虚弱，它是一种生理、心理、社会适应的完全良好状态。"
③ 帮助与照顾始终是护理固有的主题。
④ 增进健康、保持健康、恢复健康以及临终关怀。

Unit 2 Health and Illness

1. Define health and illness;
2. Describe the characteristics of health definition of WHO;
3. Differentiate illness and disease;
4. Identify the factors that influence health.

Health is a multidimensional concept. Traditionally, health has been defined as a state of presence or absence of disease. *Nightingale defined health as a state of "being well and using individual's every power the person has to the fullest extent"*.① In 1980, health was defined by the *American Nurses Association*② (ANA) in its social policy as "*a dynamic state of being in which the developmental and behavioral potential of an individual is realized to the fullest extent possible*."③

The World Health Organization (WHO) takes a more holistic view of health, and defined health in 1947 as "*a state of complete physical, mental, and social well-being, and not merely the absence of disease or infirmity*."④ This is the most widely accepted definition of health. The natures of this definition are: to reflect concern for the individual as a total person functioning physically, psychologically, and socially; to place health in the context of environment; and to equate health with productive and creative living. Generally, health, which is a positive state, is *a wide variation in personal meaning and perceptions of health*.⑤ It considers the dimensions of the person, which include the physical, intellectual, emotional, socio-cultural, spiritual, and environmental aspects composing the whole person.

The definitions of illness are also individualized to each person who experiences an alteration in health. The terms "illness" and "disease" are often used to mean the same process, but they are different. Illness is the personal experience of feeling unhealthy. It is a state in which *the person's physical, emotional, intellectual, social, developmental, or spiritual function is reduced or impaired compared with previous experience*.⑥ It involves

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① 南丁格尔将健康定义为"身体处于良好状态并最大限度发挥个体所有的能力"。
② 美国护士协会。
③ (健康是)一个能尽可能发挥个体发展和行为潜能的动态过程。
④ (健康是)一种生理、心理和社会适应的完好状态,而不仅仅是没有疾病或虚弱。
⑤ 在个体的健康含意和感受方面的一个大范围的变异。
⑥ 人的生理、情绪、智力、社会、发展或精神活动方面与以前相比有所减退或削弱。

changes in the person菸 state of being and in social function, and is highly subjective; only the individual can say he or she is ill .

Disease is an interruption in the continuous process of health, manifested by abnormalities or disturbance in the structure and function of body parts, organs or systems .① When these abnormalities *appear clustered together*②, the signs and symptoms of a particular disease come into view . Disease can result in a reduction of capacities or a shortening of normal life span . Opposite to illness, disease is objective . It may be diagnosed by physicians through specific inspections .

A person may feel ill or just not feel well because of a disease . Whereas a person may also not feel ill even he has specific disease . Therefore, illness may or may not be related to disease .

Peopl菸 health is influenced by many factors, which are generally divided into 5 groups: physical, psychological or emotional factors, environmental factors, lifestyle, and social factors .

Physical factors include *genetic makeup*③ , age, race, developmental level, and sex . Genetic makeup affects biologic characteristics, innate temperament, activity level, and intellectual potential . The susceptibility of some specific diseases is related to genetic makeup .

Psychological or emotional factors contain *mind-body interactions*④ and self-concept . Interaction of individua菸 mind and body and emotional responses of individual may impact health status positively or negatively . The manners and attitudes with which people view and deal with situations can be affected by *self-concept*⑤ .

The environmental factors are housing, sanitation, climate, food, air, water, soil, sunshine and so on . Environmental pollutants, which can change normal components of life substances, include fire, smoking, waste water, exhaust gas, residue of industry and human life, radiation such as X-ray, electromagnetic wave and microwave, nitrogen oxides from vehicles, “ greenhouse effect” caused by carbon dioxide, pesticides and chemicals used by farmers and so on .

Lifestyle involves aspects of person菸behavior and surroundings that they control .⑥ It may affect health positively or negatively . Lifestyle with positive effects on health is a healthy lifestyle, such as regular exercise, alcohol and smoking avoidance, immunization update, regular dental checkups and so on . Lifestyles with potentially negative effects on health are often defined as risk factors, such as smoking, frequently staying up all night or lying in, etc .

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① 疾病是健康连续过程的中断，通过机体各部分、器官或系统的异常或紊乱表现出来。
② 集中出现。
③ 遗传构成。
④ 身心的交互作用。
⑤ 自我概念。
⑥ 生活方式涉及人的行为和可控制的环境等方面。

There are many social factors influencing individual's health, such as stable marriage or intimate family relationship, economic level, occupational situations and culture, health care system, social violence and traffic accident, people's cultural background, etc. These factors may be risk factors for illness, or healthy factors that can promote health.

Daily Language

1. You look pale. What's wrong with you?
   你脸色苍白，怎么了？
2. I didn't sleep a wink last night.
   我昨晚一夜没睡。
3. Do you have anything in your mind? You look so concerned!
   你在想什么？你看起来好像有心事。
4. I'm worried about my husband. His snoring is getting worse and worse.
   我为我的丈夫担心。他打鼾越来越严重了。
5. Maybe you need to consult a doctor about it.
   也许这个问题你需要看医生。
6. I am ill with a common cold.
   我生病感冒了。
7. Don't worry. I will make an arrangement for you to see the doctor.
   别担心，我会帮你预约医生的。
8. I don't feel well today. I think I am ill.
   我今天觉得不舒服，我想我是生病了。
9. He has been diagnosed to have pneumonia.
   他被诊断患了肺炎。

Vocabulary

1. multidimensional [ˌmʌltidaɪˈmenʃənəl, -daɪˈm-] *adj*. 多面的，多维的
2. holistic [həʊˈlɪstɪk] *adj*. 整体的，全盘的
3. infirmity [ɪnˈfɜːmɪti] *n*. 虚弱，衰弱，缺点
4. variation [ˌveərɪˈeɪʃn] *n*. 变更，变化，变异，变种
5. intellectual [ˌɪntəˈlektʃuəl] *adj*. 智力的，有智力的 *n*. 知识分子
6. spiritual [ˈspɪrɪtʃuəl] *adj*. 精神上的
7. abnormality [ˌæbnɔːˈmæləti] *n*. 变态；畸形，异常性
8. disturbance [dɪˈstɜːbəns] *n*. 骚动，动乱，打扰，干扰，骚乱，搅
9. inspection [ɪnˈspekʃn] *n*. 检查，视察；望诊

Unit 3 Health Promotion

1. Describe the purposes of health promotion;
2. Understand the roles of nurses in health promotion;
3. Apply the nursing process to health promotion.

Health promotion is defined as "activities directed toward increasing the level of well being." It is considered to be an approach to maintain healthier behavior for the population of society. *Positive health promotion is the process of enabling people to increase control over and improve their own health, aimed primarily at improving health potential and maintaining health balance.*①

The nurses play a very important role in health promotion. Changes in the health system, the demands of the society, environmental and social issues, and the increasing use of modern technology have all influenced the role of nurses. Today, more and more clients are spending less time in *acute care facilities*②. The focus of care is shifting from hospital setting to community and preventive nursing. *As the largest group of health care workers, nurses must prepare for shift in emphasis and anticipate the nursing services that consumers will require in society.*③ They must act as advocates, teachers, or coordinators of health care services.

In health promotion, a nurse must apply nursing process. The steps of nursing process are health assessment, formulation of a nursing diagnosis, development of a health promotion/protection plan, implementation of the plan, and evaluation.

A thorough assessment of the client's health status is basic to health promotion. Components of this assessment are the health history, physical examination, physical-fitness assessment, nutrition assessment, health risk appraisal, life-style assessment, health beliefs review, and life-stress review. Assessment skills are essential to provide the meaningful data needed for health planning.

Following assessment and summarizing of data, nursing diagnoses are identified. The nurse and client may agree on potential nursing diagnoses that will assist the client in decreasing

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① 健康促进是一个促使人们更好地控制和改善其健康状况的过程，其主要目的在于挖掘健康潜力，维持身体的平衡。
② 医疗急救机构。
③ 作为医务人员中最大的群体，护士必须对工作重点的转变做好准备，同时能预见服务对象对护理的要求。

the risk of developing specific disease. *Nursing diagnosis statements that describe motivation and wellness behavior*① will identify patient strengths, recognize self-care potential, reinforce healthy life-style, assist the nurse in providing comprehensive meaningful nursing care, and achieve the goals of health promotion and the prevention of illness. *During the diagnosis phase, the client has determined areas of potential problems and risks as well as areas of strength or positive health.*② The client then determines a health promotion plan.

A health promotion plan needs to be developed according to the needs, desires, and priorities of the client. The emphasis of health promotion is to strengthen *self-care*③ responsibility of patients or population. It is the patients who decide the goals of health, determine the health promotion plans, and take the responsibility for the success of the plans.

*During the planning process a nurse acts as a resource person rather than as an adviser or counselor.*④ The nurse provides information when asked, emphasizes the importance of small steps to behavior change, and reviews the client's goals and plans to make sure they are realistic, measurable, and acceptable to the client. The main activities in the planning phase include: identifying health care goals, identifying possible behavior changes, assigning priorities to behavior changes, making a commitment to behavior change, identifying effective reinforcements and rewards, *determining barriers to change*,⑤ and developing a schedule for implementing the behavior change.

Implementing is the "doing" part of behavior change. Self-responsibility is emphasized for implementing the plan. In this stage, the nurse will assist the patient to make decisions about their health problems. The nurse ascertains that the assessment has been accurate and complete and that the goals are individualized and attainable. The nursing strategies may include supporting, teaching, consulting, coordinating, facilitating, and enhancing the behavior change.

Evaluation takes place on an ongoing basis,⑥ both during the attainment of short-term goals and after the completion of long-term goals. In this stage, the patient may decide to continue with the plan, reorder priorities, change strategies, or revise the health promotion contract. Evaluation of the plan is a collaborative effort between the nurse and the patient or client.

*Health promotion programs can be categorized as information dissemination, health appraisal and assessment, life-suyle and behavior change, work-site health, and environmental control programs.*⑦ The nurse can help patients change their health behaviors

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① 描述动机和健康行为的护理诊断陈述。
② 在健康促进的诊断阶段，消费者已经确定其潜在的健康问题和危险因素以及健康的优势和积极的方面。
③ 自我护理。
④ 在计划阶段，护士是资源提供者而不是一名指导者或顾问。
⑤ 确认影响其行为改变的因素。
⑥ 评估应动态进行。
⑦ 健康促进项目可分为信息传授、健康测评和评估、生活方式及行为改变、工作环境健康以及环境控制几种类型。

by providing ongoing support, supplying additional information and health education .

In conclusion, the roles and functions of nursing activities in health promotion are as follows: 1) act as the model of healthy life-style behavior and attitudes; 2) facilitate client's involvement in the assessment, implementation, and evaluation of health goals; 3) teach client self-care strategies to enhance fitness; 4) improve nutrition of population; 5) manage stress, enhance interpersonal relationships, and assist individual, families, and communities to increase their levels of health; 6) teach clients to be effective health care consumers; 7) assist clients, families and communities to develop and choose health-promoting options; 8) guide client's development in effective problem solving and decision making; 9) reinforce client's personal and family health-promoting behaviors; and finally, 10) advocate in the community for changes that promote a healthy environment .

Daily Dialogue

Nurse: What can I do for you ?
我能帮您做点什么吗？

Patient: Would you please tell me something about diabetes ?
您能不能告诉我一些关于糖尿病的知识？

Nurse: Certainly . Diabetes is the metabolic disorder of carbohydrates owing to the low function of the pancreas .
当然可以。糖尿病是一种由于机体胰腺功能降低而引起的碳水化合物代谢紊乱。

Patient: Is it possible that I can recover from it ?
我能恢复健康（被治愈）吗？

Nurse: Yes . You can get it under control . Dietetic treatment is especially important .
是的，您的糖尿病能够得到控制。饮食治疗特别重要。

Patient: Can you give me some details about dietetic treatment ?
您能告诉我详细的饮食治疗方法吗？

Nurse: Dietetic treatment means several meals a day, but only a little each time . No sugar, less rice and flour .
饮食治疗指少食多餐。要求进食无糖、含米和面粉少的饮食。

Patient: If I feel very hungry, what should I do ?
如果我觉得很饿，该怎么办呢？

Nurse: You can eat some vegetables .
您可以吃些蔬菜。

Patient: Any drugs ?
需要吃药吗？

Unit 4 Nursing Process

1. List and briefly describe the 5 steps of the nursing process;
2. State the purposes of the nursing process;
3. Explain how Maslow荪 hierarchy of needs is used in nursing practice.

*The nursing process is a method for organizing nursing actions involving both intellectual and physical activities.*① As an integral part of nurse荪knowledge, the nursing process guides nurses in making decisions about nursing care and implementing these decisions. *It is a systematic process and requires problem solving, critical thinking, decision-making, and a substantial knowledge basis to identify the status of the client.*② The purposes of the nursing process include: 1) to identify actual or potential health problems; 2) to guide nurses in making decisions about nursing care; 3) to carry out the plan for care; 4) to evaluate the results of nursing intervention; 5) to develop alternative nursing strategies. The nursing process involves 5 steps: assessment, diagnosis, planning, intervention, and evaluation.

The first step of the nursing process is "assessment". It involves the collection of data about the client to formulate nursing diagnoses. Methods used to collect data include observation, interviewing, physical examination, and review of records. Observation involves the use of the senses of sight, smell, hearing, and touch. The nursing history is the tool used when interviewing. *The skills used in physical examination are inspection, palpation, percussion, and auscultation.*③ A review of both current and previous medical and nursing records is helpful when gathering information.

The second step is "nursing diagnosis". *It is a statement of an actual or potential health problem that a nurse is licensed and competent to treat.*④ It involves analysis of the data collected during the assessment step. The diagnostic statement should include the problem and its cause. The problem is the current or the potential client need that can be resolved by nursing interventions. The cause may be a direct or a contributing factor in the developing of the health problem.

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① 护理程序是组织护理活动的一种方法，包括脑力的和体力的活动。
② 护理程序具有系统性，要求护士具备解决问题、批判性思维、决策制定的能力，以及识别病人健康状况的基本的护理知识。
③ 体检的方法包括视诊、触诊、扣诊和听诊。
④ 护理诊断是陈述护士确立的并有能力解决的有关病人现存的或潜在的健康问题。

The third step is“ planning”. *Planning involves identifying goals and selecting nursing actions to resolve a client's health problems*.① Planning starts on the day of admission and continues as the client's needs and problems change. To focus the nurse's time and energy where it is needed most, the client's multiple problems are prioritized. *To establish priorities of care*,② the nurse lists the client's problems and then determines which ones require the most urgent attention.

Nurses have adopted Maslow's hierarchy of needs as a tool for setting priorities of care. Abraham Maslow, a psychologist, proposed that there are five levels of needs that motivate human behavior. He grouped them in tiers according to their significance: *physiologic needs* (*first level*), *safety and security needs* (*second level*), *needs for love and belonging* (*third level*), *needs for esteem and self-esteem* (*fourth level*), *and self-actualization needs* (*fifth level*).③ *Maslow believed that until the physiologic needs are satisfied, humans could not or would not seek to fulfill other less life-threatening needs*.④ Any problem that poses a threat to physiologic functioning is, therefore, ranked first in order of priority. Nursing diagnoses such as *Ineffective Breathing Pattern*⑤ and *Fluid Volume Deficit*⑥ demand the nurse's more attention than other diagnoses such as Anxiety or *High Risk for Injury*⑦ which compromise the second level needs of safety and security. *Parental Role Conflict and Social Isolation*⑧ are examples of nursing diagnoses that apply to the third level of love and belonging needs. Examples of nursing diagnoses that affect the fourth level of esteem and self-esteem needs are *Powerlessness and Ineffective Individual Coping*⑨. *Altered Growth and Development*⑩ and *Spiritual Distress*⑪ are example of nursing diagnoses that interfere with achieving an individual's fifth level of self-actualization needs. Sometimes two or more nursing diagnoses have equal priority since they fall within the same tier of needs within the hierarchy. Also, priorities can change from day to day or even hour to hour as new health problems develop or current ones change.

The forth step of the nursing process is“ intervention”(or implementation). It is the execution of the plans established for the client. Each nurse caring for a client promotes continuity of care by implementing the established care plan. The nurse may implement the plan herself, or she may delegate some tasks to other health team members. Professional nurses are responsible for coordinating, supervising, and implementing the client's plan of care.

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① 护理计划涉及确立护理目标以及选择能解决病人健康问题的护理措施。
② 建立护理的优先项目。
③ 生理的需要（第一层次）、安全的需要（第二层次）、爱与归属的需要（第三层次）、尊重他人和自尊的需要（第四层次）、自我实现的需要（第五层次）。
④ 马斯洛认为除非生理需要被满足，否则一个人不会也不可能去寻求满足对生命威胁小些的需要。
⑤ 清理呼吸道无效。
⑥ 体液不足。
⑦ 有受伤的危险。
⑧ 父母角色冲突与社交隔离。
⑨ 无能为力感与个人应对无效。
⑩ 生长发育异常。
⑪ 精神困扰。

The fifth step is“ evaluation”. It is the crucial step of determining whether the desired outcomes have been achieved. *It is the ongoing measurement of goal achievement in relation to the stated desired outcomes.*① The effects of interventions must be periodically evaluated and the plan modified accordingly. Evaluation is used to identify effective as well as ineffective strategies. It cannot be assumed that a plan is effective until this evaluation is completed. The reasons for unachieved goals are: 1) the problem is misidentified; 2) the goals are unrealistic; 3) the nursing care is ineffective; 4) additional problems have developed.

The steps of the nursing process are described separately in order to help you understand them. However, in practice they overlap a great deal. For example, while you are giving the injection to the patient (implementation), you may be observing his breath condition at the same time (assessment). When you observe some redness and swelling in his skin (assessment), you may conclude that the skin integrity goals have not been met (evaluation).

The nursing process is beneficial to the nurse and the client. Using the nursing process can increase the nurs菇 job satisfaction. Additionally, this method will save the nurs菇 time, energy, thereby increase the nurs菇 ability to find creative solutions to client care and increase the nurs菇 confidence in the nursing role. Nursing process provides continuity and coordination of care to the client. *It promotes individualized care and shows that careful assessment and problem intervention benefit clients by preventing complications and hastening recovery.*②

Daily Language

1. This patient has many problems. What should I consider first?
   这个病人有许多的护理问题，我应当首先考虑哪个？
2. According to the Maslow菇 hierarchy of needs, you should consider the first level needs such as breath, fluid and food needs.
   根据马斯洛的需要层次论，你应当先考虑第一层次的需要，比如呼吸、体液以及食物等需要。
3. You should also pay attention to the clien菇 security need, because he is unconscious.
   你也应当关注这个病人的安全问题，因为他意识不清。
4. The clien菇 health status may change with time. So it is necessary to revise the nursing plan and nursing interventions accordingly.
   病人的健康状况会随时间的变化而有所变化。所以护理计划和护理措施也必须进行相应的调整。

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① 评价是对与预期结果相关的行为目标的连续性测量过程。
② 护理程序可以促进个体化护理，(其具体) 体现为全面而仔细的护理评估和针对问题的护理措施可以预防并发症，并促进病人的早日康复，使病人受益。

Unit 5 Communication in Nursing

1. Understand the major functions of communication;
2. Distinguish the difference between the verbal communication and nonverbal communication;
3. Identify the criteria of effective communication.

Communication is the interchange of information between two or more people in order to influence others and to obtain information. In other words, communication is the exchange of ideas or thoughts. It is a basic component of human relationship and *serves four major functions*: *control*, *motivation*, *emotional expression*, *and information*.①

Communication is a two-way process involving the sending and receiving of a message. It is made up of seven parts: 1) the communication source, 2) encoding, 3) the message, 4) the channel, 5) decoding, 6) the receiver, and 7) feedback. *This process includes all the techniques by which an individual affects another*.② There are many factors influencing the communication process, including a person's social-cultural background, language, age, education, and the limitations and attributes of nonverbal communication.

Communication is an important aspect of nursing practice. Effective communication is essential for the establishment of a good nurse-patient relationship. Learning about communication is thus an important consideration for nurses to develop effective communication skills in the nursing practice. *Nursing practice involves three kinds of communication*: *social communication*, *structured communication*, *and therapeutic communication*.③ Social communication is often carried out in an informal setting. It is usually satisfying to all parties participating. *Structured communication refers to definite planned content*.④ The process in which a nurse teaches a diabetes patient to give an injection or discusses care with a pneumonia patient is a good example of structured communication. *Therapeutic communication is a process that helps clients overcome temporary stress*, *get along with other people*, *adjust to the unalterable*, *overcome psychological blocks and so on*.⑤ Both nurses and doctors use

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① (沟通) 主要有四种功能: 控制、激励、情感表达和信息传递。
② 这个过程包括个体影响他人的所有技巧。
③ 护理实践涉及三种类型的沟通: 社会沟通、结构沟通和治疗性沟通。
④ 结构性沟通指事前计划好内容的沟通。
⑤ 治疗性沟通是帮助病人克服暂时性压力、学会与人相处、适应不可变因素、克服心理障碍等情况的过程。

therapeutic communication in many settings and in many circumstances. For instance, it is used to support the anxious preoperative patient or to help the person with cancer to accept and cope with the diagnosis.

Communication is a natural activity that we all engage in unconsciously. It is generally carried out in two different modes: verbal and nonverbal. Verbal communication is an exchange of information by use of words, including both spoken and written words. Nonverbal communication is an exchange of information without use of words. It is also called body language. Touch, gestures, facial expressions, eye contact, physical appearance, and body movements are all examples of non-verbal communication. Although verbal and nonverbal communication occur concurrently, the majority of communication is nonverbal (about 80%～90%). This may be surprising to those who associate communication with only verbal expression. *Non-verbal communication tells others more about what a person is feeling than what is actually said, because nonverbal behavior is controlled less consciously than verbal communication.*① *It can either reinforce or contradict what is said verbally.*② In communication, nurses should learn to observe and interpret the patient's nonverbal behavior. On the other hand, the nurse's own nonverbal behavior is also under constant scrutiny by patient. Therefore, it is necessary for nurses to gain awareness of their actions and to learn to convey understanding, respect, and acceptance to patients.

In hospitals or clinical settings, nursing actions are often communicated in writing after they have been carried out. *It is very important for nurses to remember that nursing actions must not be recorded in advance because the nurse may determine to reassess the patient if the action should not or cannot be implemented.*③ Therefore, nursing activities are always recorded after they are completed, when the nurse can accurately record exactly what occurred. *Sometimes, it is important to record a nursing action immediately after this action is implemented. This is particularly true of medications, treatments, etc, because recorded data about a patient must be up to date, accurate, and available to other nurses and health care members.*④ Nursing actions are also communicated verbally for most of the time. For example, *when a patient's health status is changing rapidly, the charge nurse and the doctor may want to keep up to date with verbal reports.*⑤

In the clinical setting, when nurses communicate with patients or staff of the medical team, they need to consider several criteria of effective communication. These criteria include: 1) simplicity, 2) clarity, 3) timing and relevance, 4) adaptability, and 5)

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① 一个人的非语言行为所表达的内涵往往比他所说出来的要多些，因为非语言行为更少受到意识的控制。
② 非语言沟通可以强化语言所表达的内容，也可能与语言表达的内容相矛盾。
③ 由于护士会根据病人情况进行重新评估来确定某项操作是否应该执行或能否执行，所以护士必须牢记护理行为不能事前记录。
④ 有时，及时记录完成的护理行为是非常重要的，尤其是在给药或治疗后进行记录。因为关于病人的记录资料必须及时、准确并利于其他医护人员对资料的利用。
⑤ 当病人健康状况有较大变化时，主管护士和医生希望及时收到口头报告。

credibility. Simplicity means the use of commonly understood words. Many people have a tendency to over-communicate. Their messages are wordy, contain too many extraneous explanations, or use words that are highly academic, technical, or slangy. It is natural for some nurses to use complex technical terms in communication. However, patients can often misunderstand these terms. As a nurse, she needs to learn to select simple words intentionally. Simplicity is brevity. Brevity is particularly important in writing. However, *shortcuts for the sake of simplicity can sometimes lead to incomplete or unclear communication.*① For instance, initials or abbreviations such as bid, ICU, should be avoided unless the nurse is certain that the patient will understand them.

Clarity means saying exactly what is meant. The goal of clarity is to make listeners know the what, how, why, when, who, and where of any specific event. To ensure clarity in the communication, nurses need to speak slowly and enunciate words well.

No matter how clearly or simply words are stated or written, the timing needs to be appropriate to ensure that words are heard. Nurses need to be aware of both relevance and timing when communicating with patients.

*Spoken messages need to be altered in accordance with behavioral cues from the receiver. This adjustment is referred to as adaptability.*② Moods and behavior may change minute by minute, hour by hour, or from day to day. In this sense nurses need to avoid routine or automatic speech. What a nurse says and how it is said must be individualized and carefully considered. This requires astute assessment and sensitivity on the part of the nurse.

Credibility means trustworthiness and reliability. Credibility may be the most important criterion of an effective communication. To become credible, nurses need to be knowledgeable about the subject matter being discussed and to have accurate information. *Nurses also need to convey confidence and certainty in what they are saying.*③ However, they should not sound overconfident or authoritarian. They should state messages in a constructive way and focus on being helpful to patients. People value the nurse who acknowledges limitation and can say, "I don't know the answer to this, but I'll find someone who does."

Daily Language

1. Mrs. Liu has been admitted to hospital for three days.
   刘女士已经入院三天了。
2. You look so sad. There must be something bothering you. Maybe I can help.
   您看起来很伤心，一定有什么烦心的事。或许我能帮助您。

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① 但有时为了追求简单方便，缩写也可能导致沟通不完整或不清楚。
② 口头交流应该根据听者行为反应做出相应的调整，这种调整被称为适应。
③ 护士还应对自己所说的内容显示出自信和肯定。

Unit 6 Health Education

1. Identify the determinations of health;
2. Understand the purposes of health education;
3. Clarify the conditions of learning and principles of health teaching;
4. Describe the required conditions of health education.

Health education is one of the important tasks for nursing. To understand health education, nurses must understand the definition of health. One definition of health is presented by the World Health Organization (WHO): "Health is a state of complete physical, mental, and social well being, not merely the absence of disease or infirmity." (1958) *This definition of health is both qualitative and quantitative; health is measured both in numbers and by factors such as satisfaction with life, hope, and personal and social goals.*①

Determinations of health classified by the health field are as follows: 1) *human biology, including the physical and mental aspects of health developed within an individual as a result of an individual's basic biology and organic makeup.*② 2) Environment, including physical and other factors related to health that are external to the body and over which the individual has little control. Food, air and water quality, noise pollution, communicable disease, garbage and *sewage disposal*③, and much of the social environment just are few examples. 3) Health care organizations, usually referred to as the health care system, including providers, institutes and services of a medical nature; 4) Life-style, including the aggregate of decisions made by the individuals that affect their health and over which they more or less have control. *Personal desires and habits that are not good from a health point of view create self-imposed risks.*④ When these risks result in illness or death, the person's life-style can be said to have contributed to the illness. Health education is essential to an effort to promote healthy life-style. *It is a holistic process with intellectual, psychological, and social dimensions relating to activities that increase the abilities of people to make informed*

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① 这个定义既包含健康定性的方面又包含了其定量的方面；即健康既可用数字测量，又可用生活满意度、希望、个人和社会目标等因素进行测量。
② 在个体基础生物学和器官构成基础上发展起来的包括生理和心理两方面健康的人类生物学。
③ 污水处理。
④ 从健康角度来看，对健康不利的个人要求和习惯将危害自我健康。

*decisions that affect their personal, family, and community well-being.*①

*The purposes of health education are to reach a number of health goals, including individual and family development, rehabilitation, informed consent, coping, and self-care.*② Health education is a specialized communication process that is directed toward improving the learne蒂 health attitudes and behaviors. Thus, teaching and learning in health education processes are not separate activities. The Health education process consists of four steps: assessing learning needs, planning the objectives and goals of education, implementing teaching strategies and skills, and evaluating behavior changes and abilities of decision making of learners.

Nurses have the most direct contact with patients and their family members and are the largest resource for delivery of health education. In order to achieve the goals of health education, nurses must know conditions of learning as well as the principles of health education. The conditions of learning include: 1) The learners have a desire for learning. 2) *The learning environment is physical comfort, mutual trust and respect, mutual helpfulness, freedom of expression, and acceptance of differences.*③ 3) The learners perceive the goals of a learning experience to be their goals. 4) *The learners accept a share of the responsibility for planning and operating a learning experience and have a feeling of commitment toward it.*④ 5) The learning process is related to and makes use of the experience of the learners. 6) The learners have a sense of progress toward their health goals.

The principles of health education include: 1) Expose a learner to new possibilities for self-fulfillment. 2) Help each learner clarify his (her) own aspirations for improved behavior. 3) Help each learner identify his or her problems in life. 4) Accept each learner as a person of worth and respect his or her feelings and ideas. 5) Seek to build relationships of mutual trust and helpfulness among the learners by encouraging cooperative activities. 6) Expose his or her own feelings and contribute his or her resources as a co-learner in the spirit of mutual inquiry. 7) *Involve the learner in a mutual process of formulating learning objectives.*⑤ This process should include the needs of patients, of the hospitals, of the nurse, and of the society. 8) *Share his (her) thoughts about options available in the designing of learning experiences and the selection of materials and methods and involve the learner in deciding among these options jointly.*⑥ 9) Help the learner to organize himself to share responsibility in the process of mutual inquiry. 10) Help the learner exploit his own experiences as resources for learning through the use of such techniques as discussion, role

① 这是一个集知识、心理和社会等各方面因素为一体的过程，通过一系列的活动，以提高人们做出有利于他们个人、家庭和社区健康有关活动的决策能力。
② 健康教育的目的是实现许多健康目标，包括个体及家庭的发展、康复、知情同意、应对和自我护理。
③ 学习环境具备身体舒适、相互信任和尊重、互相帮助、自由表达和接受不同意见等特点。
④ 学习者接受共同承担制定和执行学习计划的责任，并建立起实现学习目标的责任感。
⑤ 让学习者参与制定学习目标的互动过程。
⑥ 分享学习者在学习设计和学习资料及方法选择方面可以采用的思想，并让学习者参与共同决定选择方案。

play, case methods, and so on. 11) Help the learner to apply new learning to his experience, thus to make the learning more meaningful and integrated. 12) Involve the learner in developing mutually acceptable criteria and methods for measuring progress toward the learning objectives. 13) Help the learner develop and apply procedures for self-evaluation according to the criteria.

In the assessment stage of health education, it is necessary for nurses to choose health education materials and methods carefully when working with learning populations. *The educators must know a learner's level of motivation through the health belief model or another assessment tool.*① In planning the health education programs for individual, families, or social people, there is a vast array of educational materials that can be used to support the learning goals. *The sources of health education include: professional organizations, hospitals, medical societies and community educational centers, volunteers and nonprofit organizations, and government sources.*② Health education can be implemented in many settings: face to face discussion, topic presentation, material reading, skill demonstration are just a few examples. All kinds of learning have cognitive, affective, and psychomotor components. Based on theses components, the results of health education can be evaluated by measuring the changes in values, attitudes, and health behaviors of the learners. *Health education is an activity that facilitates and stimulates the learning potential of the individual or aggregate.*③ The nursing educator thus serves as a resource person who respects and enhances each learner's unique cultural background, belief system, and life experience.

Vocabulary

1. professional [prə'feʃənl] *n*. 专业人员 *adj*. 专业的，职业的
2. protocol ['prəutəkɔl] *n*. 草案，协议
3. commitment [kə'mitmənt] *n*. 许诺，承担义务
4. partnership ['pɑːtnəʃip] *n*. 合伙，配对
5. empowerment [im'pauəmənt] *n*. 授权
6. integrate ['intigreit] *vt*. 使成整体，使一体化
7. qualitative ['kwɔlitətiv] *adj*. 性质上的，定性的
8. quantitative ['kwɔntitətiv] *adj*. 数量的，定量的
9. contributed [kən'tribjutəd] *v*. 捐助，捐献，贡献
10. psychological [ˌsaikə'lɔdʒikəl] *adj*. 心理（上）的
11. rehabilitation [ˌriːə(h)ˌbili'teiʃən] *n*. 复原，康复

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① 护理教育者必须运用健康信仰模式或其他评估工具了解学习者的学习动机水平。
② 健康教育资源包括：专业组织、医院、医学会及社区教育中心、志愿者及非盈利组织，以及政府机构。
③ 健康教育是一种促进和激励个人或群体学习潜能的活动。

Unit 7 Community Health Nursing

1. Define terms: community, community health, community health nursing, primary health care;
2. Compare community health nursing with primary health care;
3. Discuss the characteristics of community health nursing.

A community, broadly defined, is a *collection of people who share some important feature*① or features of their lives in common. It can be a society of people holding common rights and privileges. Communities may be identified by *geographic boundaries*②, such as a city, a county, or the neighborhood, or identified by its *common interest*③, such as a religious community, a group of migrant workers, or citizens concerned about air pollution, or defined by *people and agencies toward solving some health-related problem*④. The function of any community includes its member菸 sense of belonging and shared identity, values, norms, communication, and supporting behavior.

*Community health is much more than environmental programs and large-scale efforts to control communicable disease.*⑤ Community health practice involves six components: promotion of health, prevention of health problems, treatment of disorders, rehabilitation, evaluation, and research.

*Community health nursing is an art and a science that synthesizes knowledge from the professional nursing theories and public health sciences.*⑥ The major goal of community health nursing is to preserve the health of the community and its populations. The focuses of community health nursing are on health promotion and health maintenance of individuals, families, groups, and whole community. The community health nurses play an important role in identifying, assessing, and responding to the health needs of populations. They work in collaboration with communities, families, individuals, voluntary organizations, self-help groups, informal health care providers, governments, and the private sector.

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① (社区是) 共有一些重要特征的群体。
② 地理范围。
③ 共同的兴趣。
④ 解决健康相关问题的人群或机构。
⑤ 社区健康比环境规划的范围和大规模地控制传染病要广泛得多。
⑥ 社区健康护理是综合专业的护理理论知识和公共卫生学知识的一门艺术和科学。

Primary Health Care, defined as theoretical foundation for community health nursing, is approved at WHO conference in Alma Ata in 1978. *It is essential health care based on practical, scientifically sound and socially acceptable methods and technology made universally accessible to individuals and families in the community through their full participation and at a cost that the community and country can afford to maintain at every stage of their development in the spirit of self-reliance and self-determination.*① It forms an integral part both of the country's health system and of the overall social and economic development of the community. It is the first level of contact of individuals, the family and community with the *national health system*②. It brings health care as close as possible to where the people live and work, and constitutes *the first element of a continuing health care process*③. Several essential components of Primary Health Care have a great deal of significance to community health nurses, and are basic to community health nursing. These components include: essential health care services, acceptable and affordable care, universally accessible care, and full participation of the community.

Several characteristics of community health nursing, therefore, deserve special emphasis. Firstly, community health nursing is population-focused. It is concerned with the health status of people in the aggregate or groups. Secondly, in community health nursing, the promotion of health and prevention of illness are of first-order priority. Thirdly, community health nursing uses aggregate measurement and analysis. Finally, community health nursing uses principles from management and organization theory to provide effective administration of health care services.

Daily Language

1. One of characteristics of community heath is population-focused.
   社区健康的一个特点是关注群体。
2. The clients of community health nursing include not only individuals, but also families, aggregate, and community.
   社区健康护理的服务对象不仅包括个体，还包括家庭、群体和社区。
3. A family, an aggregate, or a community is composed of groups of people.
   一个家庭、群体或一个社区是由一群人组成的。
4. What is the difference of nursing intervention provided to individual, family, aggregate and community?

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① 初级卫生保健是一种基本的卫生保健。它依靠切实可行的、科学的、社会可以接受的方法和技术使社区中个人和家庭通过积极参与可以普遍享受，同时也是社区和国家在各个发展时期依靠自身力量和决心能够负担得起的卫生保健。
② 国家卫生系统。
③ 连续性健康照顾的第一要素。

Unit 8 Stress and Adaptation

1. Define the concept of stress;
2. List the types of stressors;
3. State the factors that influence the impact of stressor;
4. List the consequences of stress.

Stress is an inevitable aspect of life. From the moment we are born until our death, we constantly encounter stressful events in our daily lives. Stress is described as "a factor that induces bodily or mental tension" from the dictionary. And according to a famous stress theorist Hans Seyle, *stress is a nonspecific response of the body to any demand*①. It is a complex set of reactions that help to prepare the person's reaction to the stressor.

Stressors commonly exist in our daily life. They may originate within as well as outside of us. They may be physiological or physical in nature, or they may be social and psychological. Illnesses or diseases are the common *physiological stressors*②. Extreme heat or cold, loud noises, and bright lights are all physical causes of stress. Low self-esteem, loneliness, and hopelessness are examples of *psychological stressors*③. Conflict with colleagues and pressure from job requirements are common *social stressors*④.

Not all stressors have equal impact on *an individual's homeostatic balance*⑤. *Factors affecting the impact of stressor are its magnitude or strength, its duration, the person's prior experience with the stressor, the perceived significance to the individual, and the presence of additional stressors*.⑥

The consequences of stress are multidimensional. Physically, stress can threaten a *person's physiological homeostasis*⑦. *Emotionally, stress can produce negative or non-constructive feeling about self*.⑧ Intellectually, stress can alter a person's perceptual and problem-solving abilities. Socially, stress can alter a person's relationship with others.

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① 压力是机体对外在需求作出的非特异性反应。
② 物理性压力源。
③ 心理性压力源。
④ 社会性压力源。
⑤ 个体的恒定状态。
⑥ 影响压力源强度的因素有：压力源本身的强度、压力源的持续时间、个体应对压力的经验、个体对压力源重要性的感知，以及是否还存在其他的压力。
⑦ 机体的体内平衡。
⑧ 在情绪方面，压力可使机体产生自我感觉不良。

Spiritually, stress can change a person's general outlook on life.[①] Many illnesses, such as hypertension, gastric ulcers, bronchial asthma, and coronary heart disease have been linked to stress.

Stress initiates stress response. A successful response to the stressor is termed adaptation. *Adaptation would be any way the body tries to assist the individual to meet needs and maintain homeostasis.*[②]

Hans Selye, who was known as the "father of stress", proposed the general adaptation syndrome (GAS) to describe individual's biochemical mechanism of adapting to stress. He suggested that regardless of what caused the stress, the human reaction occurred in three distinct stages: *alarm reaction stage, stage of resistance or adaptation, and the stage of exhaustion*[③]. In the alarm reaction stage, the body becomes aware of the stressor and prepares to fight. The prominent bodily change at this time is the increase of the hormonal levels. In the stage of resistance or adaptation, the body attempts to repair any damage caused by the stress and the body's adaptation to the stressor takes place during this stage. When the body is successful in its fight against the threat, the GAS ends with the stage of resistance as homeostasis is restored. If the stage of resistance is not successful, the body may enter the stage of exhaustion. This stage, a prolonged state of preparedness, depletes the body's energy store and eventually its ability to maintain resistance to the threat. Without outside intervention, the outcome of the stage of exhaustion is illness or even death.

So far, researches on stress indicate that stress is essentially the adjustive demand placed on the body. When excessive, stress overtaxes the adaptive resources of the individual and eventually leads to a breakdown in functioning. However, a moderate level of stress can improve the performance and functional efficiency of the body and is necessary and desirable for optimal human functioning.

Daily Language

1. You look upset/ concerned/ worried.
   你看起来很心烦/ 焦虑/ 闷闷不乐。
2. What's the matter?/ What's wrong?
   怎么了？
3. I understand it's a really stressful situation.
   我明白这确实很让人紧张。
4. Take it easy. And relax yourself.

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① 在精神方面，压力可以改变个人的人生观。
② 适应是机体动用各种方法满足外在需求并维持体内环境稳定的过程。
③ 警告期、抵抗或适应期，以及衰竭期。

Unit 9 Rehabilitation

1. Define rehabilitation;
2. Describe the content of rehabilitation, and the degrees of disability;
3. Tell the nature of rehabilitation.

Rehabilitation is the process by which health professionals facilitate an individual to move toward health. It pays attention to a person's medical, psychological, and social problems so that *the ability to live more independently is enhanced*[①]. *Rehabilitation is less involved in treating disease than in promoting the person's ability to perform activities of daily living, get out into the community, enjoy leisure time, or return to work*.[②]

It is important to keep in mind certain definitions used to describe the situation of an individual with a disability, involving chronic illness, impairment, functional limitation or disability, and handicap. According to The Commission on Chronic Illness, *chronic illness is defined as "all impairments or deviations from normal which have one or more of the following characteristics: being permanent, leaving residual disability, requiring special training of the patient for rehabilitation, caused by a nonreversible pathological condition, expected to require a long period of supervision, observation, or care."*[③] Chronic illness may lead to mobility problems, and must receive priority for rehabilitation.

Impairment refers to the residual limitation resulting from disease, injury, or a congenital defect. *It is any alteration in physical or physiologic functions at the organ level*.[④] The impairment of physiologic function is extremely common in old age. *When the impairment causes an alteration in the person's daily activities, it is called a disability*.[⑤] That means disability is the inability to perform some key life function, so it is also called functional limitation. Functional limitation resulting from disability is *at personal level*[⑥]. When a person with a disability is prevented from *fully expressing his or her potential*[⑦] because of

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① 增强独立生活的能力。
② 与促进个体日常生活的能力、融入社区的能力、享受闲暇的能力或重返工作岗位的能力相比，康复较少涉及疾病的治疗。
③ 慢性病是指"任何具有以下一种或多种特征的损害或正常状态的偏移：永久性、会留下残疾、需要专门的康复训练、由不可逆转的病理改变所致、需要长期的的监护、观察或照料"。
④ 它是生理功能在器官水平的任何改变。
⑤ 当残损引起个人日常活动改变时，就称为残疾。
⑥ 在个体水平上。
⑦ 充分展示其潜能。

societal barriers, the person has been handicapped. Therefore, handicap refers to the interaction of a disabled person with the environment. Function limitation resulting from handicap is at social level. The handicapped persons lose the ability to fulfill usual social roles and normal activities.

*Rehabilitation is a dynamic process of planned adaptive change in lifestyle in response to unplanned change imposed on the individual by disease or traumatic incident.*① The focus is not on cure, but on *living with as much freedom and autonomy*② as possible at every stage and in *whichever direction the disability progresses*③. The difference between rehabilitation and many other medical systems is team orientation. Rehabilitation team members meet regularly to discuss patient problems and deficits of function, to make comprehensive therapeutic plan, to decide the patient rehabilitation goals, and to allow staff members and patient to bring up problems and seek solutions.

The primary members of the team are rehabilitation nurses, physical and occupational therapists, speech pathologists, and vocational counselors.

A rehabilitation nurse is the most crucial team member. The nurse provides important nursing care directly and ensures techniques learned from physical or occupational therapists are reinforced and practiced. Moreover, because of long contact with patients, a nurse plays a key role in supporting psychological adjustment to disabilities and in training family members.

Physical therapists assess strength, range of motion, endurance, and mobility skills. The focus of physical therapists is on promoting ambulation and transfers. Occupational therapists train patients in performing basic or instrumental activities of daily living, and in performing perceptual and sensory evaluations. Cognitive retraining is also within their purview. Speech pathologists assist the aphasic patient in achieving communication, and help with cognitive assessment and evaluation of swallowing function. Recreation therapists assist patients in improving ability to find leisure activities that are satisfying and enjoyable. This ability can significantly help patients with disability adjust to disability. Physicians provide medical expertise and often serve as team facilitators. Those with primary responsibility for the rehabilitation are specialists in physical medicine and rehabilitation.

Rehabilitation can be provided in any setting where the health care is delivered. Specialized rehabilitation centers that may be *freestanding hospital*④ or departments within hospitals play stronger role for rehabilitation interventions. The patient's home is a logical site for much rehabilitation care. *Appropriate patient for home rehabilitation is someone who may benefit from outpatient care but stay at home because of transportation problems and limited*

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① 康复是一个动态过程，它有计划地适应生活方式的改变，是对疾病或外伤事故所致的意外改变的反应。
② 自由和自主地生活。
③ 无论残疾往哪个方向发展。
④ 独立的医院。

endurance.[1] Nursing home is another important site for providing rehabilitation. The physical, occupational and speech therapies are available in nursing home. The patients who have medical limitations that prohibit them from tolerating therapy in an inpatient unit, and need both skilled nursing and physical and/or occupational therapy, may be admitted in nursing home.

Daily Language

1. He has a stroke.
   他中风了。
2. He has already been in paralysis.
   他已经瘫痪了。
3. It's a good beginning for your rehabilitation.
   这对你的康复是一个好的开端。
4. The therapeutic exercise will help you to stand up.
   治疗性运动能帮助你站起来。
5. This exercise is designed to prevent deformity and maintain normal muscle tone and function.
   设计这套练习的目的在于预防畸形和维护肌肉的弹性和功能。
6. The most commonly used form of isotonic exercise is range of joint motion exercises.
   最常见的等张运动是关节活动。
7. The next step is elbow movement.
   接下来做肘关节运动。

Vocabulary

1. rehabilitation [rɪˌhæbɪlɪˈteɪʃ(ə)n] *n*. 复原，康复
2. facilitate [fəˈsɪlɪteɪt] *vt*. (不以人作主语) 使容易，推动，帮助，促进
3. disable [dɪˈseɪb(ə)l] *n*. 使残废 *v*. 使残废，使失去能力
   disability *n*. 无力，无能，残疾
4. chronic [ˈkrɒnɪk] *adj*. 慢性的，延续很长的
   chronic illness 慢性病
5. impairment [ɪmˈpeəmənt] *n*. 故障；缺陷；损害，损伤
6. handicap [ˈhændɪkæp] *n*. 障碍，不利条件
7. deviation [ˌdiːviˈeɪʃ(ə)n] *n*. 背离；偏差，偏离

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① 适合接受家庭康复的病人是那些可以从门诊治疗获益（即不必住院治疗），但又因交通问题、体力有限不能到门诊治疗的患者。

Unit 10 Growth and Development

1. Define the terms: growth, development;
2. Interpret the principles of growth and development ①;
3. State the influencing factors of growth and development.

An individual experiences numerous changes throughout life. Theses changes include the continuous replacement of cells, tissues, and fluids. Some changes are not physical in nature; they are changes in emotion, behavior, and feelings.

*Growth and development is usually applied to as a unit to express the sum of the many changes that take place during the lifetime of an individual.*② However, They stand for different interrelated dimensions respectively. *Growth is an increase in number and size of cells as they divide and synthesize new proteins and the results in increased size and weight of the whole or any of its parts.*③ *Development, however, makes reference to the gradual process of change and differentiation; advancement from simple to more advanced levels of complexity; and the emerging and expanding of the individual's capacities through growth, maturation, and learning.*④ It is a sequential series of changes in function.

Principles of growth and development

In all dimensions of growth and development, *there is a definite, predictable sequence*⑤. *Growth and development proceed in regular directions that are head-to-tail, near-to-far, gross-to-fine, and simple-to-complex.*⑥ Infants gain control of the head before they have control of the *trunk and extremities*⑦. They hold their head up before their chest, hold their back erect before they stand, use their eyes before their hands, and achieve control of their hands before they have control of their feet. In infants, shoulder control precedes mastery of the hands, and the whole hand is used as a unit before the fingers can be manipulated. Children creep before they stand, and stand before they walk. The central

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① 生长发育。
② 生长发育通常作为一个整体来表示一个人一生中所发生的许多变化之和。
③ 生长是指细胞在分裂和合成新蛋白质时数量和大小的增长，并使整体或某部分的大小和重量增加。
④ 然而，发育是指分化与变化的渐进过程，从简单到复杂的进步，以及通过成长、成熟和学习，个人能力的展现和扩大。
⑤ 有一个特定的、可预知的顺序。
⑥ 生长发育遵循由上到下、由近到远、由粗到细，以及由简单到复杂的有规律的顺序。
⑦ 躯干和四肢。

nervous system develops more rapidly than the peripheral nervous system. Later facets of the personality are built on the early foundation of trust. The child babbles, then forms words and finally, sentences; writing emerges from scribbling.

However, growth and development do not progress at the same rate or pace, *characterized by spurts of growth and periods of rest*①. The younger the child is, the rapider the physical growth is. The fastest physical growth appears during the first 6 months of life, especially during the first 3 months, which is the first *growth spurt*②. The other spurt occurs in puberty.

Growth and development proceed *at a highly individualized rate*③. No two persons should be expected to react in the same manner to the same stimuli. Every child grows in his or her unique and personal way.

Last, there are *limited times*④ during the process of growth and development. Both physical maturation and psychological development have *critical periods*⑤ when an environmental event has maximal influence on them. For example, primary socialization occurs during the first year and the infant establishes a basic trust in the world, which is essential to a healthy personality.

Influencing factors of growth and development

Many factors such as heredity, gender, conditions of the mother, and others play important roles in the patterns of growth and development. They are interrelated, and their interactions result in the creation of a unique individual.

Heredity: Growth patterns appear to be genetically controlled. *At conception, the individual is endowed with a complex set of biological potentials involving height, skin, hair and eye color, and talents and interests.*⑥ Only identical twins have the same combinations of chromosomes (karyotype). In addition to these characteristics, there are many other potential traits that may be inherited. Certain abnormalities may be genetically transmitted. Examples of such inherited disorders include hemophilia, thalassemia, and Phenylketonuria.

Gender: Gender is an influential factor of growth and development. For example, the pubescent growth spurt may begin about 2 years earlier in girls than in boys. The development of language and motion is also earlier in girls than in boys. Therefore, gender differences should be taken into account when assessing children's growth and development.

Conditions of the pregnant mother: Development begins with conception or the union of sperm and ovum. Nutrition, emotion, and health status of the pregnant mother will affect

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① 具有生长高峰和间歇期的特征。
② 生长高峰。
③ 具高度的个体差异性。
④ 有限的时间。
⑤ 关键期。
⑥ 受孕时，个体就被赋予了一套复杂的生物学潜力，包括身高、肤色、头发和眼睛的颜色，以及天赋与兴趣。

fetal development . For example, malnutrition and viral infections such as *German measles*①, *herpes zoster*②, and cytomegalovirus infection may cause abortion, *premature birth*③, and developmental abnormalities in the fetus . Any substance, agent, or process that interferes with normal prenatal development needs to be avoided during pregnancy .

Finally, good environment and health facilities and adequate nutrition are essential for a child's growth and development . Illness will decelerate the growth and development .

Daily Language

1 . Please take your baby to the well-baby-clinic to have a check-up a month later .
请在一个月以后将你的孩子带到儿童保健门诊来检查。

2 . Did your baby have a check-up before ?
你孩子以前进行过儿保吗 ?

3 . What's his registration card number ?
请问他的儿保号是多少 ?

4 . I'm going to weigh him first .
我要先给他称体重 .

5 . Please put him on the platform balance .
请把他放在天平上。

6 . What's my baby's weight ?
我的孩子多重 ?

Vocabulary

1 . replacement [rɪˈpleɪsmənt] *n* . 代替，替换

2 . synthesize [ˈsɪnθəsaɪz] *vt* . 合成，综合，接合

3 . differentiation [ˌdɪfəˌrenʃiˈeɪʃən] *n* . 区别，鉴别；分化，分化作用

4 . maturation [ˌmætjuˈreɪʃən] *n* . 成熟

5 . peripheral [pəˈrɪfərəl] *adj* . 周围的，外围的，外部的，边缘的
the peripheral nervous system 周围神经系统

6 . babble [ˈbæbl] *v* . 牙牙学语，言语模糊或说话快而含糊不清 *n* . 儿语

7 . scribble [ˈskribl] *v* . 潦草地写，乱写，滥写

8 . puberty [ˈpjuːbəti] *n* . 青春期

9 . gender [ˈdʒendə] *n* . 性，性别

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① 风疹。
② 带状疱疹。
③ 早产。

Unit 11 Vital Signs

1. Interpret the terms: vital signs, body temperature, pulse, respiration, and blood pressure;
2. State the normal limits of vital signs and how to measure them;
3. Identify factors that influence each of the vital signs.

Body temperature, pulse, respiration, and blood pressure are the vital signs. 偏离正常值 *Variations from the normal limits*① in the temperature, pulse, respirations and blood pressure readings indicate a change in the health status of the person. So it is essential that the nurses be able to take vital signs, interpret data, *communicate them to others*②, and plan nursing intervention appropriately. Vital signs are always assessed during *a basic screening physical examination*③ and *upon admission to the hospital*④.

Body temperature is the balance between heat produced and heat lost by the body. *It is surprisingly consistent in healthy individuals.*⑤ To take the temperature, we use a thermometer. There are several kinds of thermometers: *glass thermometer*; *electronic thermometer*; *and disposable thermometer*⑥.

To take temperature, we put the thermometer in places that air can't reach. The three most common places to take temperature are the mouth, the rectum, and the axilla. Body temperature can be measured by two scales: either in Fahrenheit (F) or Celsius (C) degrees. The average normal adult oral temperature is 98.6℉ (37℃); the rectal temperature 99.6℉ (37.5℃); and the axillary temperature 98℉ (36.7℃).

Many factors, such as age, infection, temperature of the environment, amount of exercise, and emotional status, can affect a patient's temperature and should be considered when a temperature reading is evaluated. For example, the lowest temperature occurs between about 2 a.m and 6 a.m, and the highest between about 4 p.m and 8 p.m. In addition, studies of temperatures in healthy older adults indicate that the normal body

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① 偏离正常值。
② 传递数据。
③ 全面体检。
④ 入院时。
⑤ 健康人的体温常保持惊人的一致。
⑥ 玻璃温度计、电子温度计和可弃式温度计。

temperature decreases as age increases .

Selection of the appropriate method for taking the body temperature depends on several factors, such as age, consciousness level, and emotional status of the patient . For example, *the oral method is contraindicated for a small child, a confused or unconscious patient, or patients with pain, injury, or inflammation of the mouth .*①

Your pulse is your heart rate, or the number of times your heart beats in one minute . Many factors can influence the pulse rate: body temperature, age, anger, fear, exercise, pain, medications, and heart disease . Normally the pulse rate for an adult is between 60 to 100 times per minute . The term bradycardia describes an adult pulse rate below 60 beats/ min, tachycardia refers to an adult pulse rate above 100 . The *quality and rhythm of the pulse*② must also be observed and described . *The normal pulse rhythm is regular, with a steady length of time between each beat .*③ The most frequent site for taking the pulse is at the radial artery of the wrist because of its convenience . But other sites, such as carotid artery and apical pulse, may also be used . Measure the apical pulse with a stethoscope .

*Respiration is defined as the process of inhaling oxygen-bearing air into the lungs and exhaling carbon dioxide out of the lungs .*④ When a person inhales, his chest expands . When he exhales, his chest contracts . A rate of 16 to 20 is considered" normal" for the adult . *Normal respirations are regular, even and quiet .*⑤ But they can be affected by all the same factors that cause the pulse rate to vary . Different terms are used to describe the characteristics of respirations . For instance, the patient having difficulty in breathing is said to have dyspnea . Apnea describes the absence of respiration . Rapid respiration is termed hyperpnea . Count respirations by observing the number of expansions of the chest . While you are counting respirations, do not let the patient know, or he will probably not breathe naturally . *A good way is remaining your fingers on the patient's wrist after the pulse has been taken and counting the rise and fall of his chest for* 30 *seconds, then multiplying by* 2 *and recording .*⑥ During this procedure, *a watch displaying seconds*⑦ is needed .

*Blood pressure is a measurement of the force applied to the walls of the arteries as the heart pumps blood through the body .*⑧ There are two readings when you measure blood pressure, the higher one is called the systolic and the lower one the diastolic pressure . Normal blood pressure ranges from 110 to 140 millimeters (as the heart beats) over 60 to 90 millimeters (as the heart relaxes) . Blood pressure is usually written as 120/ 80 mmHg, for

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① 口腔测量法不适用于小孩、精神错乱及无意识的病人，以及有口腔疼痛、创伤或炎症的病人。
② 脉搏的强弱和节律。
③ 正常脉率跳动均匀规则，间隔时间相等。
④ 呼吸就是将含氧的空气吸入肺里、把二氧化碳排出的过程。
⑤ 正常的呼吸表现为有规律、均匀、平静。
⑥ 正确的方法就是在你测完脉搏之后仍然将手放在病人的手腕上，数病人的胸部起伏次数 30 秒，然后乘以 2，再做记录。
⑦ 秒表。
⑧ 血压是心脏喷出的血液在动脉血管壁上所呈现的压力。

example. A systolic blood pressure of over 160 or a diastolic pressure over 100 is termed hypertension; a systolic blood pressure below 100 is termed hypotension, which may not require medical treatment. Variations in blood pressure can be caused by several factors, such as age, exercise, emotion, and pain. To measure the blood pressure, you need a cuff and a sphygmomanometer.

Daily Language

1. Wha蒜 my temperature?
   我的体温多少度？
2. I蒜 39 degrees Celsius.
   摄氏 39 度。
3. Please roll up your sleeves. 蒜l take your blood pressure.
   请把袖子卷起来，我要给你测血压。
4. How about my blood pressure?
   我的血压多高？
5. I蒜 120 over 80.
   收缩压 120，舒张压 80。
6. Is it normal?
   正常吗？

Vocabulary

1. pulse [pʌls] *n*. 脉搏
2. respiration [ˌrespəˈreɪʃn] *n*. 呼吸
3. vital [ˈvaɪtl] *adj*. 生命延续所必需的，极重要的
   vital signs 生命体征
4. intervention [ˌɪntəˈvenʃn] *n*. 干涉，干预
   nursing intervention 护理措施
5. thermometer [θəˈmɒmɪtə(r)] *n*. 温度计，体温计
6. disposable [dɪˈspəʊzəbl] *a*. 一次性的；用后即可丢弃的
7. rectum [ˈrektəm] *n*. 直肠
   rectal [ˈrekt(ə)l] *adj*. 直肠的
8. axilla [ˈæksɪlə] *n*. 胳肢窝，腋窝
   axillary [ˈæksɪləri] *adj*. 腋窝的
9. scale [skeɪl] *n*. 刻度，标度
10. Fahrenheit [ˈfærənhaɪt, ˈfɑːr-] *adj*. 华氏温度计的 *n*. 华氏温度计

Unit 12 Health Assessment

1. Describe the purposes of health assessment;
2. State the sources of information for the assessment;
3. Describe the means that ensure a successful assessment.

Health assessment is the first step of the nursing process. It forms the basis of nursing care. Whether the patient is young or old, male or female, well or ill, assessment is an ongoing process of evaluating the whole person as a physical, psychosocial, and functional being. Hence, *when we assess the person, the holistic assessment should include physical and psychosocial aspects.*①

Health assessment is the orderly collection of information regarding the patient's health state. *It aims to recognize the patient's current health state, actual and potential health problems, and areas for health promotion.*② The sources of the information for the assessment include the health history, physical assessment, and laboratory and diagnostic data.

The health history interview is a way of gathering subjective data, often from the patient. *However, in some instances, such as trauma, coma, infant, and toddler, the historian may be someone other than the patient.*③ Note the name of the historian as well as the relationship between the historian and the patient. In addition, assess the reliability of the historian. The health history is considered subjective because nurses cannot confirm all information. A complete health history includes patient's profile (providing demographics that may be linked to health status), reason for seeking health care and *chief complaint* (*CC*)④, history of the present illness, past health history (medical, surgical, medications, communicable diseases, allergies, injuries/accidents, disabilities/handicaps, blood transfusions, childhood illness, and immunizations), family health history, social history (alcohol use, tobacco use, drug use, sexual practice, travel history, work environment, home environment, hobbies, stress, education, economic status, etc.), health maintenance

① 当我们评估这个人（这里指病人）时，整体的评估应该包括生理和心理社会等方面。
② 它（健康评估）的目的在于认识病人当前的健康状态、现存的和潜在的健康问题，以及需要进行健康促进的方面。
③ 然而，在一些情况下，如（病人）外伤、昏迷或是婴幼儿，其病史的提供者则可能是其他人。
④ 主诉，缩写为 CC。

activities, such as sleep, exercise, stress management, and health check-ups.

The CC is the sign (objective finding) or symptom (subjective finding) that causes the patient *to seek health care*① . *The cause for seeking health care and CC should be recorded as direct quotes from the patient*.② The duration of the signs/symptoms should also be included in the CC. *The history of the present illness (HPI) is a chronological account of the patient's CC and the events surrounding it*.③ Nine characteristics of each CC can be ascertained for a complete HPI. These characteristics include location, radiation, quality, quantity (the severity, volume, number, or extent of the CC), associated manifestations, aggravating factors, alleviating factors, setting, and timing. The patient may refer to the CC using terms such as minor, moderate, or severe.

Taking a whole health history from a patient may require 30 to 60 minutes. So, tell the patient the amount of time that will be required before the interview starts.

The health history usually represents the first step of patient assessment and is usually performed previous to the *physical assessment*④ . *Analysis of the information gathered from patients reveals areas needing notice in the physical assessment*.⑤

Physical assessment findings form a second source of information in the assessment. *These findings compose the objective data, or information that is observable, measurable, and verifiable by more than one person*.⑥ These data are also recognized as physical signs of illness or wellness. The physical assessment data can be obtained through a body-system or head-to-toe approach. Inspection, palpation, percussion, and auscultation are the techniques used by the nurse to assess patients during a physical examination. Physical assessment of a patient serves many purposes:

1. Confirmation of the complaints that cause the patient to seek health care.

2. Screening of general well-being. The findings will serve baseline information for further assessments.

3. Monitoring of current health problems.

4. Formation of diagnoses and treatments.

The final source of information in the assessment phase is laboratory and diagnostic data. *Results of serum, urine, pus, or other body-fluid studies, cultures, radiographic studies, and diagnostic procedures constitute objective data about the patient's status*.⑦

Usually, the nurse is the first person from the health care group to interact with the patient. She will often assume the role of mediator between the patient and the large health

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① 求医。
② 寻求健康照顾的原因和主诉应该直接引用病人的原话。
③ 现病史是按时间顺序将病人的主诉和相关问题进行的记录。
④ 体格检查。
⑤ 分析从病人处收集的信息，可以提示我们在体检中需要关注之处。
⑥ (体格检查) 结果得到的数据或资料是客观的，它们可经由多人观察、测量和证实。
⑦ 血清、尿液、脓液或其他体液检查、细菌培养、放射线检查和其他诊断方法检查的结果，组成了病人身体状况的客观资料。

care system. *She can foster an atmosphere of safety, harmony, and comfort by approaching each patient with an acceptable, respectful, and nonjudgmental attitude.*[①] The patient is an active and equal participant in the interview process and should feel free to openly communicate feelings, thoughts, perceptions, and factual information. When the nurse begins to approach the patient, she should follow the instructions for a successful assessment:

1. Provide the patient with privacy for physical examination and redressing.

2. Sit opposite the patient who is sitting or lying on the bed.

3. Start the interview with a pleasant introduction.

4. Offer the patient with an explanation of what is to follow and a proper time frame for the interview.

5. Tell the patient that you will be jotting down notes during the interview to help discussion and ensure accuracy of information.

6. Advise the patient about potential discomfort or sensations prior to performing procedures.

7. Have the patient make as few position changes as possible.

8. Ensure that all needed equipments are available and arranged in a logical manner.

9. Ask the patient about pain or other sensations potentially associated with certain assessment.

10. Offer cleaning wipes or tissues to the patient following certain examination (e.g., the pelvic examination).

11. Collect all available information of the patients.

12. Respect the patient's modesty.

The assessment phase is fluid and dynamic.[②] Nurses are constantly compiling, validating, interpreting, and recording the data. With this information, the nurse is prepared for the next phase: nursing diagnosis.

Daily Language

1. What brought you here today? / What made you ask for health care today?
   今天您哪儿不舒服？
2. What is the current treatment for this problem?
   针对这个问题目前的治疗是什么？
3. Are you allergic to any medications?
   你对药物过敏吗？

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① 通过易接受的、有礼貌的和无偏见的态度去接近病人，可营造一个安全、融洽、舒适的氛围。
② 评估是一个动态的阶段。

Unit 13 Isolation and Infection Control

1. State the isolation category;
2. Describe the main measures of each category of isolation;
3. State the aims of hand-washing, wearing gowns, and gloves.

Isolation practices are an attempt to prevent the spread of infection by breaking the chain of infection at the source, at the host, or in the method of transmission. *The isolation practices utilized today incorporate the knowledge of the chain of infection, the cost of isolation practices, and the psychosocial effects on patients and their families.*①

In 1970 the *Centers for Disease Control and Prevention (CDC)*② published the first Guidelines for Isolation Techniques. In 1983, as diseases and pathogens became more virulent, the isolation system was revised. The 1983 Isolation Guideline was a basis for hospitals and other institutions developing their own policies for isolation. *Based on the category system for isolation, it provided strict, respiratory, enteric, contact, AFB*③ *tuberculosis, drainage/ secretion, and blood/ body fluid precaution.*④

The aim of the strict isolation is to prevent extremely contagious infections transmitted by air or by contact. Possible diagnoses include diphtheria, varicella (chickenpox), and herpes zoster. The patient should have a private room, preferably a *negative-ventilation room*⑤. The healthcare workers should wear a mask, gown, and gloves, and must wash hands thoroughly, and handle infected materials carefully.

The purpose of respiratory isolation is to prevent transmission in short distances in the air. The possible diagnoses are measles, epidemic cerebrospinal meningitis, and pneumonia. The patient should have a private room. The healthcare workers should wear a mask for close contact, wash hands thoroughly, and handle infected materials carefully.

In order to prevent transmission of infections by contact with fecal matter, when nurses take care of patients with viral hepatitis A, gastroenteritis of various causes, clostridium

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① 今天所实施的隔离原则综合考虑了感染的传播过程的有关知识、隔离所产生的费用，以及隔离对传染病人及其家庭社会心理的影响。
② 美国疾病预防和控制中心，缩写为 CDC。
③ acid-fast bacill 的缩写，即抗酸杆菌。
④ 按照隔离的类别分类系统，隔离分为严密隔离、呼吸道隔离、消化道隔离、接触隔离、AFB 结核菌隔离、脓汁/ 分泌物隔离，以及血液/ 体液隔离。
⑤ 负压通气的房间。

*difficile, and enterovirus-caused meningitis or endocarditis, they can use enteric precautions.*① *A private room is required only if the patient has poor hygiene.*② If patient care involves risk of soiling, the healthcare worker should wear a gown and gloves, wash hands thoroughly, and handle infected materials carefully.

Contact isolation is used to prevent transmission from direct or close contact but not requiring strict isolation. *Possible diagnoses are most respiratory infections in children and infants, multiresistant bacteria either in infection or colonization, burns, or draining skin or wounds.*③ The patient should have a private room. The healthcare worker should wear a gown and gloves if patient care involves contact with infected materials, wash hands thoroughly and handle infected materials carefully.

The aim of AFB tuberculosis isolation is to prevent transmission of AFB droplet nuclei. Possible diagnosis is laryngeal or pulmonary tuberculosis. The patient should have a private room with negative ventilation and keep the door closed. The healthcare worker should wear a gown and gloves if patient care involves contact with infected materials, handle infected materials carefully, and wash hands thoroughly.

When the healthcare workers contact with skin lesions, oral secretions, or wounds, they can use drainage/secretion precaution to prevent transmission from *infected draining site*④. Possible diagnoses include decubitus ulcers, burns, wounds, or abscesses. The healthcare worker should wear a gown and gloves if patient care involves contact with infected materials, wash hands thoroughly, and handle infected materials carefully.

Blood/body fluid precautions are used to prevent transmission of infections by contact with infected blood or body fluid, including prevention of *mucous membrane*⑤ and eye exposure by use of a mask and *eye shields*⑥. Possible diagnoses are AIDS, viral hepatitis B, malaria, and syphilis. A private room is required only if the patient has poor hygiene. The healthcare worker should wear a gown and gloves if patient care involves contact with infected materials; wash hands thoroughly (especially after any exposure to body fluids), avoid needlestick injuries, handle all contaminated articles carefully, and clean up blood spills *with a* 1∶10 5.25% *sodium hypochlorite solution and water*⑦. Most institutions combine universal precautions and blood/body fluid precaution for all patients. Universal precautions are used to prevent patient-to-staff transmission of disease and mainly emphasize precaution to prevent transmission through blood and body fluid (fluids that are known to transmit HIV and

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① 为了预防经由接触排泄物所致的疾病的传播，当护士护理甲型病毒性肝炎病人、各种原因引起的肠胃炎病人、艰难梭菌和肠道病毒所致的脑膜炎或心内膜炎的病人时，他们应当使用消化道隔离。
② 只有当病人的个人卫生非常糟糕时才可提供单人房间。
③ 可能的诊断包括儿童和婴幼儿的大多数呼吸道感染、多重耐药的细菌感染或在定植阶段、烧伤和皮肤有分泌物或有伤口时。
④ 有感染性分泌物的部位。
⑤ 粘膜。
⑥ 眼罩。
⑦ 将 5.25% 漂白水与水以 1∶10 的比例溶解。

hepatitis B). *Universal precautions also address prevention of sharp injury, barrier precautions (gowns, gloves, masks, eyeshields, and ventilation devices for resuscitation), and hand-washing.*①

Measurements of isolation are handwashing, gowns, masks, and gloves. Proper handwashing requirements include mechanical action, use of cleanser and water, rinsing with water from high level down to hands, and drying with clean towel. Healthcare workers must wash hands between patients, after contacting secretions or excretions (even with gloves on), before performing any invasive procedure, wound care, or care of high-risk susceptible patient, especially in the intensive care unit or neonatal intensive unit.

Wearing gowns can prevent contamination of clothing, thereby decreasing transmission of organisms between patients. If a patient has diarrhea with incontinence, wearing gowns can also prevent soiling. In *burn unit*② and nurseries, wearing gowns could offer protective isolation. To prevent gown contamination, health workers should wash hands before gowning, always tie securely in the back, pull gloves up over the gown cuffs, and have a wall clock in the room (so a wristwatch is not needed).

Masks can prevent airborne transmission, direct contact with mucous membranes and inhalation of infectious microorganisms in either large or small droplets. Gloves not only protect the health workers from infection, but also protect the patients from endogenous flora on healthcare workers' hands, and protect other patients from transient colonization on healthcare workers' hands.

Vocabulary

1. pathogen [ˈpæθəˌdʒən] *n*. 病原体（物）
2. enteric [enˈterik] *adj*. 肠的
3. tuberculosis [tjuˌbəːkjuˈləusis] *n*. 肺结核
4. diphtheria [difˈθiəriə, dip-] *n*. 白喉
5. varicella [vˌærɪˈselə] *n*. 水痘
6. herpes [ˈhəːpiːz] *n*. 疱疹
7. zoster [ˈzɔstə] *n*. 带，带状疹子
8. measles [ˈmiːzlz] *n*. 麻疹，风疹，包虫病，痧子
9. cerebrospinal [ˌseribrəuˈspainl] *adj*. 脑脊髓的
10. meningitis [ˌmeninˈdʒaitis] *n*. 脑膜炎

    epidemic cerebrospinal meningitis 流行性脑脊髓膜炎

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① 普遍性的预防还强调锐器刺伤的预防，防护物的使用（包括手套、口罩、护目镜和能保持正常通气的通风设施）和洗手。

② 烧伤病房。

Unit 14 Medication

1. State the significance of medication administration;
2. List the routes for medication administration;
3. Describe the contents of nursing assessment for medication administration;
4. Speak out the key points of nursing care in regard to medication administration for discharged patients.

Medication treatment is a very important medical means, through which many diseases can be cured or stopped when the administered medications *take effect*① . Since administration of medication is one part of nursing care, the nurse is expected to have a knowledge basis of drugs, including drug name, classifications, preparation, and effects and *adverse effects*② of drug, as well as physiologic factors that affect drug action.

The route to be used when administering a medication should be stated clearly, because while some drugs can be given in more than one route, others may be used safely only in one way. Medications are administered in different ways. The common routes used in clinic nursing are oral administration, inhalation, administration by injection, vaginal administration, rectal administration and so on. Among them, the most common route is oral, that is, given by mouth. Any medication given by mouth may be considered an oral medication, even if it is absorbed in the mouth and not in the stomach or intestine. Oral medications are available in two forms: solid preparations and liquid preparations. Solid preparations include tablets, capsules, and pills. Liquid preparations include elixirs, spirits, suspensions, and syrups.

Parenteral medications are given through injection with a needle. They are the most rapidly absorbed, because they are administered directly into or close to the circulation or into their sites of action. Their common routes of administering include *intradermal injection*③, *subcutaneous injection*④, *intramuscular injection*⑤, and *intravenous injection*⑥.

Nurses should apply the nursing process to the administration of medication, and assess

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① 生效，见效。
② 副作用。
③ 皮内注射。
④ 皮下注射。
⑤ 肌肉注射。
⑥ 静脉注射。

the client's response both during and after drug therapy. The outcomes of the treatment should also be evaluated after accomplishing the plan of care. Assessment begins with taking a medication history in addition to the total nursing history. The *medication history*① helps nurses get very specific information about the use of medication. Therefore, questions should be focused on the person's use and need for medications, as well as on previous responses to therapy. Specialized clinics such as oncology or diabetes clinics usually adopt the basic medication history to include even more specific questions. Do not administer any medication until these data have been gathered. Communicate the information you have gathered to other members of the health care team by transcribing it on the person's permanent medical record. Communicate critical information such as allergies or previous *adverse reactions*② to medications by noting it on the person's *arm band*③ and in a conspicuous place on the cover of the chart. Also, document this information in the nursing care plan and notify the pharmacy.

Basic considerations for a physiologic assessment are: age, weight, sex, and condition of the person. A person's age, weight, and sex are factors that affect the pharmacokinetics of medication. For instance, because of reduced circulation, absorption of medications in a 70-year-old person is generally slower than that in the body of a 20-year-old person. Males and females have different body structures, tissue mass, and reproductive and hormonal systems. Some medications, such as oral contraceptives, work only in the body of a female. Age and weight are often the determining factors in the selection of medication dosages or routes of administration. For example, an infant may be given a small dose of antibiotics in a liquid form to simplify administration, but a school-age child can take a larger dose in chewable tablet form. The reduced muscle mass of a thin person may be a consideration for administering intramuscular injections in certain sites. The alterations in a person's condition or function also influence the movement of medications through the body. Absorption and distribution of medications may be changed by heart diseases; liver diseases alter biotransformation; kidney diseases may lead to slowed excretion of drug metabolites. Other functions including nutritional and fluid status, gastrointestinal functions and changes in activity level should be assessed. Deficits in hearing or vision, integumentary, circulatory, gastrointestinal, and musculoskeletal systems, ability to communicate, and level of consciousness should be noted.

Psychosocial factors should be always considered in regard to both the administration of medications and client's *adherence to*④ the *medication regimen*⑤. Factors you may need to assess include motivation, person's attitude toward therapy, *drug dependence and misuse*⑥, person's understanding and knowledge level of a medication regimen, and support from

① 药物史。
② 不良反应，副反应。
③ 腕带。
④ 坚持，遵守。
⑤ 药物治疗方案。
⑥ 药物依赖与滥用。

*significant others*① . These factors often determine *whether the client is compliant or noncompliant with*② the *therapeutic regimen*③ .

In many cases, clients continue a prescribed medication at home after discharge from hospital . The nurse should help the client establish a medication schedule that best fits the prescribed frequency of mediations and the client's *life style*④ . The client should know the drug name, dosage, route of administration, and frequency . He should understand the desired effects of medications and their possible adverse effects . The nurse should stress the importance of *taking medicines as prescribed*⑤ and for as long as prescribed . Drugs used for maintaining health, such as those to control *high blood pressure*⑥, need to be taken continuously as ordered to avoid *recurrence of symptoms*⑦ . Nurses have a teaching responsibility to society in relation to the abuse of any drug .

Daily Language

1 . Please take 2 pills at a time and three times a day .
请每次服两片，每天服三次。

2 . Why should I take these medicines ?
我为什么需要服这些药？

3 . We give them to you for preoperative preparation .
我们给你这些药是用于作术前准备。

4 . When and for how long should I take them ?
我应该什么时候服药？需要服药多久？

5 . You should take them after meals for 3 days .
应该饭后服用，并且需要服三天。

6 . Is there any discomfort after taking them ?
服药后会有不舒服的感觉吗？

7 . Please pay attention to your reactions after taking them . If you have any discomfort, don't hesitate to inform us as soon as possible .
请注意你服药后的反应。如果感觉有任何不适，不要犹豫，马上告诉我们。

Vocabulary

1 . medication [ˌmedɪˈkeɪʃn] *n* . 药物，药物治疗，药物处理

① 重要关系人。
② 是否依从、顺从。
③ 治疗方案。
④ 生活方式。
⑤ 按医嘱服药。
⑥ 高血压。
⑦ 症状复发。

Unit 15 Injections

1. List the major nursing diagnoses for patients who are receiving injection;
2. List the common ways of giving injections;
3. Describe the purposes of each injection and the requirements for selecting sites for each of them;
4. State the common procedures of injection except for intravenous injection.

The safe preparation and administration of intradermal, subcutaneous, intramuscular, and intravenous medications is a routine nursing responsibility that requires dexterity, sterile technique, a knowledge of the actions, usual dosage, desired effects, and potential side effects of the drug being given; and a knowledge of how and where to give the drug. The latter requirement necessitates knowledge of human anatomy. Because drugs given by these routes not only are absorbed more quickly than by mouth but also are irretrievable once injected, a firm mathematics foundation and routine practice of the *three checks and seven confirmations*① are mandatory.

The major nursing diagnosis to keep in mind when giving injection of medications is High Risk for Injury. Patients can be injured by medications given in the wrong dosage, at the wrong time, by an incorrect route, or in an incorrect site. They can also be injured by the omission of essential medications or the administration of an incorrect medication. Although this nursing diagnosis will not appear in the nursing care plan, it applies to every situation in which a patient is being given medications. Another nursing diagnosis when administering medications is Knowledge Deficit. In this case the knowledge deficit would be related to some aspect of the medication regimen, for example, the need to learn how to safely administer injections.

To give any injection, a syringe, a needle, a swab and disinfectant to clean the skin, and a medication are needed.

Intradermal injections can be used for diagnostic purposes, such as *allergy testing*② and *tuberculin testing*③, and some vaccinations. Intradermal injections are placed just below the

① 三查七对。
② 过敏试验。
③ 结核菌素试验。

epidermis. Sites commonly used are the *inner surface of the forearm*①, *dorsal aspect of the upper arm*②, and the *upper back*③. The dosage given intradermally is very small, usually less than 0.2 ml.

Subcutaneous injection of medications is to ensure slow absorption, as the fatty tissue has less blood flow. Medications given subcutaneously include epinephrine, growth hormone, insulin, erythropoietin, and some vaccines. Because there is subcutaneous tissue all over the body, various sites can be used for subcutaneous injections. These sites include *outer aspect of the upper arm*④, *anterior aspects of the thigh*⑤, *lower abdominal wall*⑥, and upper back. Usually, no more than 1 ml of solution is given subcutaneously. Larger amounts increase the clien蒜 discomfort, and may cause poor absorption.

Intramuscular injection of medications is for quick absorption. It is used when larger doses are needed than can be given subcutaneously, when a drug is too irritating to be given subcutaneously, or when a drug cannot be taken orally. A key point is the selection of a safe site, which should be away from large nerves, bones, and blood vessels. The sites for intramuscular injection should be rotated when therapy requires repeated injections. The amount of 3 ml is considered the maximum for one site. Several sites are available for intramuscular injections, such as dorsogluteal site, deltoid muscle site, femoral quadriceps site and so on. *The dorsogluteal site for administering an intramuscular injection is lateral and slightly superior, on the one third point of line drawn from the coccyx to the iliac spine.*⑦ Another method of locating the dorsogluteal site is to divide the buttock into 4 quadrants, and the injection is given in the upper outer quadrant. *The deltoid muscle site for intramuscular injection is located by palpating the lower edge of the acromion. At the midpoint, in line with the axilla on the lateral aspect of the upper arm, a triangle is formed.*⑧ Intramuscular injections into the deltoid muscle should be limited to 1 ml of solution and used only for adults. The femoral quadriceps site is used to infants and small children because it is the largest and most well developed muscle mass in the infants.

The percutaneous insertion of a plastic or metal needle into a vein is used for withdrawing blood, giving direct intravenous medication, and administering intravenous infusion of fluid, nutrient, or blood. The common sites for venipuncture and peripheral intravenous therapy include the dorsum of the hands and the inner sides of the elbows and arms. The considerations in choosing a site for venipuncture or cannulation include: 1) Is the patient in shock or

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① 前臂内侧。
② 上臂背面。
③ 上背部。
④ 上臂外侧。
⑤ 大腿前面。
⑥ 下腹壁。
⑦ 进行肌肉注射的臀大肌注射点是尾骨与髂脊连线的外侧稍偏上 1/3 处。
⑧ 肌肉注射的三角肌注射点可通过触摸肩峰下缘进行定位。在上臂外侧面，与腋窝成一直线的中点处形成一块三角形肌肉。

circulatory collapse? 2) What kind of medication is to be administered? 3) What is the planned duration of therapy? 4) In what conditions are the patien荫 veins? 5) What are the patien荫 psychological responses to the therapy? The intravenous site should be changed every two or three days to reduce the risk of infection and damage to vessel. To avoid it, the nurse must use *aseptic technique*① strictly during venipuncture, and use sterile technique for all equipment, as well as change intravenous tube, fluid, and dressing according to standard requirements.

The common procedures of injections (except for intravenous injection) include:

1. Confirm the physician荫 order.
2. Assemble equipment (such as choosing the appropriate needles and syringes).
3. Identify the patient.
4. Explain the procedure to the patient and place him/her in appropriate position.
5. Wash your hands.
6. Follow the principles of aseptic technique.
7. Draw the medication solution into the syringe and eliminate the air from it.
8. Choose the appropriate site without excessive pigment, nodules, lesions or hair.
9. Sterilize the area.
10. Dart needle into tissue quickly at a required angle.
11. Aspirate by pulling back on barrel, assessing for blood return.
12. Inject the drug into the body after judging no blood.
13. Remove the needle quickly.
14. Press the site with a swab.
15. Assist the patient to a comfortable position.
16. Wash your hands.
17. Record the administration of the medication.
18. Observe and evaluate the response of the client to the medication.

Daily Language

1. Before that, I have to take an allergy testing for you.
   用药之前我必须先给你做一个皮试。
2. Please don荫 touch the sterilized area.
   请不要碰到消毒的部位。
3. Please pay attention to your reactions after administering the intradermal injection, and don荫 go out within 15 minutes.

① 无菌技术。

25 . cannulation [ˌkænjuˈleɪʃən] *n* . 管子，套管插入术

26 . dressing [ˈdresɪŋ] *n* . 敷料

27 . nodule [ˈnɑːdjuːl] *n* . 节结

28 . lesion [ˈliːʒən (ˈleʃ-)] *n* . 损害，身体上的伤害

29 . dart [dɑːt] *v* . 飞奔，投掷 *n* . 飞快的动作

30 . barrel [ˈbærəl] *n* . 桶，筒，活塞筒

Questions on Reading

1 . What are the major nursing diagnoses for patients who are receiving injection ?

2 . What are the common ways for giving injections ?

3 . Compare the differences in purposes of intradermal injection, subcutaneous injection, intramuscular injection, and intravenous injection .

4 . How do you select the sites for administering each of the injection ?

5 . What are the common procedures of injections except for intravenous injection ?

Unit 16 Blood Transfusion

1. Explain the purposes for blood transfusion;
2. State the characteristics of whole blood and blood product;
3. State what a nurse should pay attention to when administrating blood and blood products.

*Administration of blood*① and *blood products*② is an effective treatment *to restore blood volume, replace clotting factors, and improve oxygen carrying capacity*③. Whatever the strongpoint is, potential complications must be concerned. Nurses must recognize these complications and try to prevent them as well as be able to intervene appropriately.

Whole blood may be ordered as a part of patient therapy or, because blood can be divided into its component parts, only the special components needed in a patient's therapy may be used. The followings discuss the indications and contraindications for administrating whole blood and some of its components.

*Whole blood transfusion*④ is most commonly used in instance of acute massive blood loss or for *total blood exchange in neonates*⑤. Whole blood is not indicated when blood volume is normal or increased. *Because blood products give the only needed elements to the recipient, this practice makes it possible to provide more needs with fewer donations and to decrease the risk of complication, such as blood carried disease and circulatory overload, to the recipient.*⑥ Common blood products are as the following:

*Packed red blood cells*⑦ are the blood product remaining after up to 80% of the plasma is removed from the whole blood. *Because the red blood cells provide the same oxygen-carrying capacity as whole blood but in smaller volume, they may be used in situations in which the patient is at risk for circulatory overload but has the need of hemoglobin for its oxygen-*

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① 输血。
② 成分血。
③ 恢复血容量、补充凝血因子以及提高机体携氧能力。
④ 输注全血。
⑤ 新生儿全血置换。
⑥ 由于成分血只向受血者提供需要的血液成分，这就使得较少的献血者能满足更多人的需求，同时降低受血者由于输血引发的并发症，如血液传染性疾病和循环负荷过重。
⑦ 压积红细胞。

carrying capacity.① Typing and crossmatching is needed before the use of red blood cells.

Platelets are used in cases of thrombocytopenia caused by lack of platelet production, resulting in postponed bleeding time.② They are only indicated for treatment of life-threatening hemorrhage. Platelets should be type-compatible, but RH antigens are not found on platelets. The risk for acquiring blood-carried disease is the same as for whole blood.

Plasma is the fluid portion of the blood remaining after the red blood cells, platelets, and leukocytes have been removed. Fresh frozen plasma can be stored for 12 months and provides clotting factors, proteins, and fluid volume. Plasma protein fraction is the portion of the plasma remaining after fibrinogen and globulin have been removed. This component is used when the replacement of intravascular volume is necessary. The risk of hepatitis is eliminated because of a pasteurizing process.

*Human serum albumin*③ increases the *colloidal osmotic pressure of the blood*④ and is usually recommended as a treatment for shock, burns, and hypoproteinemia. Typing and crossmatching is not required.

Cryoprecipitate is prepared from fresh frozen plasma and contains clotting factor Ⅷ— the factor lacking in hemophiliacs. It holds a very small quantity of red cells, which makes cross-matching unnecessary. But it is usually advisable to give ABO compatible if the total volume is more than 100 ml. It may be stored frozen for 12 months, and the risk for blood-carried disease is still present.

The followings are general guidelines for administration of blood and blood products:

1. The transfusion should be started within 30 minutes after the blood is removed from refrigerated storage in the facility蒜 blood bank. Blood or blood product should not be stored in the refrigerators on the nursing unit, because refrigerators used for blood storage are monitored to keep the right temperature.

2. Nurses should stay with the patient for the first 15 minutes or first 50 ml of the transfusion to observe for reactions and complications. Blood should be infused slowly during this time. Factors including the blood product being used, the patien蒜 age, and clinical condition determine the rate of administration.

3. Because red cells settle to the bottom and plasma rises to the top, gentle mixing of whole blood is necessary before and during the transfusion. Squeeze the bag gently every 20～30 minutes to prevent red cells settling.

4. If blood hangs at room temperature longer than 4～6 hours, abandon it immediately in order to avoid the danger of *bacteria proliferation and red blood cells hemolysis*⑤.

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① 因为红细胞的携氧能力与全血一样，而容量较小，故可用于有循环负荷过重危险但又需要血红蛋白来提高机体携氧能力的病人。
② 血小板制品用于因血小板生成障碍导致的血小板减少症，该症会导致出血时间延长。
③ 人血白蛋白。
④ 血浆胶体渗透压。
⑤ 细菌繁殖和红细胞溶血。

5 . Do not add medications directly to blood or a blood product . If the tubing being used to administer blood must be used to give medications, rinse it with saline before and after .

6 . *An IV of normal saline may be piggybacked into the blood tubing to maintain access to the line in case it is necessary to discontinue the blood .*①

7 . Before massive transfusion, blood should be warmed in order to prevent a severe temperature drop . Immersing a blood-warming coil in a water bath at a temperature of 37℃ is the method usually taken to achieve this purpose . Blood should not be warmed unless it is going to be given immediately . Blood that has been warmed should not be returned to the blood bank or refrigerator for future use .

8 . A blood pump may be used to increase flow rate when blood volume and oxygen-carrying capacity need to be increased right away .

9 . Infectious agents may still be present although all blood is tested for hepatitis B virus and *AIDS*② antibodies . Needles should be removed with gloved hands and placed in the sharp container after being used . All bags and transfusion equipment should be disposed in the special plastic-lined containers kept in the soiled utility room and never in the wastebasket in the patient's room .

Daily Language

1 . How are you feeling today ?
你今天感觉如何？

2 . I feel frail and I still get edema .
我感觉软弱无力，而且仍然有水肿。

3 . The blood test shows that your serum albumin is low . That is why you still get edema .
血液检查显示你的血清白蛋白低，所以你仍然有水肿。

4 . Administration of human serum albumin will alleviate your symptom . It's one of the blood products .
输入人体白蛋白会减轻你的症状。它是一种成分血（血液制品）。

5 . Administration of blood products needs typing and cross-matching .
输血液制品需要做血型测定和交叉配血。

6 . Not all the blood products need typing and cross-matching .
并不是输所有血液制品都需要血型测定和交叉配血。

① 输入生理盐水的静脉通道应与输血袋的管道相连，以便在不需要输血时可以继续输液。
② 爱滋病（获得性免疫缺陷综合症）。

Unit 17 Wound Care

1. Interpret the term: wound;
2. State the process of wound healing;
3. Identify factors that influence wound healing;
4. Describe the basic wound caring methods.

The skin encloses the body structures and the body fluids, which protect the inner body from external assault. When skin barrier is pierced, *the inflammatory process of the individua荫 immune response*① is enacted to get rid of the foreign material, if possible, and to prepare the injured body area for healing. This injured body area, whether internal or external, is called a wound. *Wounds are often described according to how they are acquired: incised, contused, abraded, punctured, or lacerated.*② Wounds can also be further described according to the likelihood and degree of wound contamination. Clean wounds are uninfected wounds in which there is no inflammation and the respiratory, alimentary, genital, and urinary tracts are not entered. Clean wounds are mainly closed wounds; or, if necessary, they are drained with closed drainage. *Clean-contaminated wounds*③ are surgical wounds in which the respiratory, alimentary, genital, or urinary tracts have been entered. Such wounds reveal no evidence of infection. Contaminated wounds consist of open, fresh, accidental wounds. *Some surgical wounds, involving a major break in sterile technique or a large amount of spillage from the gastrointestinal tract, are also called contaminated wounds.*④ Contaminated wounds show evidence of inflammation. Dirty or infected wounds include old, accidental wounds containing dead tissue and wounds with evidence of a clinical infection.

Following injury, the process of healing takes place. Only certain tissues of the body are able to regenerate. Healing takes place by primary or secondary intention. *Primary intention healing*⑤ occurs when the tissue surfaces are or have been approximatey closed and there is least or no tissue loss; it is characterized by the formation of minimal granulation tissue and

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① 机体免疫应答产生的炎症反应。
② 伤口通常根据它的形成被描述为：切割伤、挫伤、摩擦伤、穿刺伤，或撕裂伤。
③ 半污染伤口。
④ 一些无菌技术不完备或有大量胃肠道溢出物的外科伤口也被称为污染伤口。
⑤ 一期愈合。

scarring. Primary intention healing includes three stages: *the defensive stage*, *the reconstructive stage*, *and the mature stage*① . *Secondary intention healing*② occurs when the wound is extensive and has large tissue loss. Secondary intention healing differs from primary intention healing in three ways: longer repair time; greater scar; and more prone to be infected.

Both internal and external factors can affect wound healing, either positively or negatively. External factors include preoperative stays, preoperative preparation, and intraoperative elements. Internal factors include vasculature, immune status, nutrition, obesity, and stress. Some ways are recommended to control the external factors that affect wound healing, which include treating bacterial infections before surgery, shortening preoperative hospital stay, administering enteral or parenteral nutrition preoperatively to the malnourished clients if the surgery is not urgent, and bathing with an antimicrobial soap the night before surgery when clients having *elective surgery*③ . Hair near the operative site should not be removed unless absolutely necessary; If hair must be removed, it should be clipped or removed with a depilatory, rather than shaved.

There are three main complications of wound healing:

Hemorrhage. Some bleeding from a wound is normal intraoperatively and postoperatively. Hemorrhage, however, is abnormal. It may be caused by a dislodged clot, a slipped ligature, or erosion of a blood vessel.

Dehiscence with possible evisceration. Dehiscence is the partial or total rupturing of a wound. Some factors can result in evisceration, including obesity, poor nutrition, multiple trauma, failure of suturing, excessive coughing, vomiting, and dehiscence.

Infection. Infection of a surgical wound may become apparent 2 to 11 days postoperatively. Drainage from a suspected infected wound should be cultured.

Just as there are many types of wounds, there are many ways of caring for wounds. In general, there are four basic measures to take to accelerate the wounding healing.

*The Center for Disease Control*④ recommends the following wound care practices for preventing wound infection. 1) Wash hand before and after touching surgical wound. 2) Touch an open or fresh surgical wound only when wearing sterile gloves or using sterile forceps. After the wound is sealed, sterile gloves are no longer required. 3) Remove or change the dressing over open and closed wounds when it becomes wet. 4) Take a specimen of any drainage from the wound that is suspected of being infected. Send the specimen to the laboratory for culture and *Gram stain*⑤ .

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① 组织防御期、组织重建期和组织成熟期。
② 二期愈合。
③ 择期手术。
④ 疾病控制中心。
⑤ 革兰氏染色。

Promoting Healing and Preventing Complication . Adequate nutrition is crucial for wound healing . Clients unable to eat a diet that provide sufficient calories and nutrients for wound healing may require a central line for total parenteral nutrition or an enteral *feeding tube*① . Protein, vitamins, and trace metals play a major role in the healing process . Prevention of stress on a wound, vomiting, abdominal distention, or strenuous coughing, is also important during the wound healing process . *Vomiting is often prevented by withholding food and oral fluid until there is no nausea* .② Abdominal distention often can be prevented by frequent position changes and ambulation . Ambulation improves respiration and blood circulation, helping the delivery of nutrients and oxygen to the wound area . Changing Dressings is another important way to promote the healing of wounds . For the primary intention wounds with minimal drainage and no tissue loss, sterile dry dressings are used . Surgical dressings are changed to prevent the growth of microorganisms therefore helping the surgical wounds healing .

Monitoring Wound Drains: *Surgical drains are inserted to permit the drainage of excessive serosanguineous fluid and purulent material and to promote healing of underlying tissue* .③ These drains may be inserted and sutured through the incision line, but they are most commonly inserted through stab wounds several centimeters away from the incision line therefore the incision itself may be kept dry . In order to keep effective draining, holding the drain securely in place and assessing the amount and character of drainage, including odor, thickness, and color, are necessary for nurses .

Daily Language

1 . What should I do for this wound ?
对这个伤口，我该注意什么？

2 . Keep the dressing from wet . Take some antibiotics, if necessary .
保持敷料的干燥，必要的时候口服一些抗生素。

3 . When shall I come to take the stitches out ?
我什么时候来拆线？

4 . Please come to me in seven days .
请七天后来拆线。

5 . How were you injured ?
你是怎么受伤的？

6 . The wound is rather large, so I shall stitch it .

① 管喂。
② 通过限制食物和水的摄入来防止呕吐，直到病人没有恶心为止。
③ 插入外科引流管将额外的血性液体以及脓性物质引流出来，从而促进深部组织的愈合。

Unit 18 Intubation

1. State the indications for intubation;
2. Identify the types of intubation;
3. State the complications of intubation.

Intubation is a very important step to set up effective ventilation for patients. *Muscle relaxants are used to facilitate surgery thereby necessitates the use of mechanical ventilation.*① *Patients with a full stomach, protecting against aspiration of regurgitated gastric contents, patients with prone position, making maintenance of the airway difficult, and patient in whom the airway cannot be satisfactorily maintained by any other technique may undergo intubation.*② *During cardiopulmonary resuscitation, intubation allows: ventilation with* 100% *oxygen without leaks, suction clearance of inhaled debris, and a route for the administration of drugs.*③

A variety of equipments exist and which is chosen will be determined by the circumstances. *Laryngoscope with* a *curved blade and functioning light*④ is the essential device for intubation. Tracheal tubes with working *cuff*⑤ are in a variety of sizes. The length is expressed in centimeters and the internal diameter in millimeters. They should be lightly lubricated. For males: 22~24 cm in length, 8.0~9.0 mm internal diameter; and for females: 20~22 cm in length, 7.5~8.5 mm internal diameter. Inflate the cuff by syringe once the tube is in place. The tube is connected to the anesthetic system or ventilator tubing by *catheter mounts or*‘ *elbow*’⑥. And suction should be switched on and immediately handed in case the patient vomits or regurgitates. In addition, a *semi-rigid introducer*⑦ is to help mould the tube to a particular shape; and the bandage or tape is to secure the tube.

Oral intubation requires *abolition of the laryngeal reflexes*⑧ and appropriate monitoring

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① 如果手术使用了肌松剂，就需要对病人实施机械通气。
② 对处于以下三种情况的病人可能需要插管：饱食的病人，为了防止其吸入返流的胃内容物；处于俯卧位而使通气困难的病人；使用其他技术而不能达到满意通气效果的病人。
③ 在进行心肺复苏的过程中，插管可以确保以100%的纯氧进行密闭通气，吸出误吸物。插管还是一种给药的途径。
④ 有可靠光源的弯咽喉镜（作者注：儿童插管可用直镜片，用光导纤维作为光源更为可靠）。
⑤ 气管导管前端的气囊。
⑥ 导管的接口或肘形接口。
⑦ 具可塑性（有一定韧性并具备一定强度）的金属丝。
⑧ 终止病人的喉反射。

of the patient . The patient should be positioned with the neck flexed and the head extended at the atlantooccipital joint . This is the so-called‘ sniffing the morning air’ position . The patient蒜 mouth is fully opened by the index finger and thumb of the right hand *in a scissors action*① . The laryngoscope is always held in the doctor蒜 left hand and *the blade is introduced into the mouth along his right-hand side of the tongue, displacing it to the left*② . The blade is advanced until the tip lies in the gap between the base of the tongue and the epiglottis . *Force is applied in the direction in which the handle of the laryngoscope is pointing,*③ the effort coming from the upper arm of the doctor not from the wrist, to lift the tongue and epiglottis to expose the larynx . *A triangular opening can be seen, with the apex anterior and the whitish colored true cords lateral .*④ *The tracheal tube is introduced into the right side of the mouth, advanced and seen to pass through the cords until the cuff lies just below the cords .*⑤ The tube is then held firmly by the doctor蒜 fingers of the right hand and *the laryngoscope is carefully removed*⑥ . The cuff is then inflated sufficiently to prevent leak during ventilation . *Finally the position of the tube is confirmed by listening for breath sounds in both axillae and then it is secured in place .*⑦

*For nasotracheal intubation, a well-lubricated tube is introduced usually through the right nostril along the floor of the nose with the bevel pointing medially to avoid damage to the turbinates .*⑧ It is advanced into the oropharynx, where it is usually visualized using a laryngoscope in the manner described above . It can be advanced directly into the larynx by pushing on the proximal end .

Hypoxia, trauma, and reflex activity are the common complications in intubation .

Hypoxia happening during intubation can deteriorate the impaired respiratory conditions . *Measuring the carbon dioxide in expired gas is the best way to detect hypoxia in oesophageal intubation;*⑨ low carbon dioxide indicates oesophageal intubation . *Less reliable signs are ‘ burping’ sounds as gas escapes, diminished breath sounds on auscultation, decreased chest movement on ventilation, and gurgling sounds over the epigastrium .*瘢 *Abnormal anatomy or airway pathology are the most common reasons for failed intubation and inability to ventilate .*瘢 In fact many cases can be predictable at the preoperative assessment . Possible causes include the tube becoming kinked, disconnected, or inserted too far, and passing into

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① 以剪刀的手势。
② 沿着病人舌头的右侧将喉镜前端镜片插入病人口中，将舌头推向左侧。
③ 用力的方向为喉镜把手指向的方向。
④ 这样可以见到一个三角形的开口，三角形顶点向前，两侧为发白的声带。
⑤ 气管导管沿着病人嘴的右侧插入，一直往前通过声带，直到气囊刚好进到声带下面。
⑥ 小心地取出喉镜。
⑦ 最后通过在两侧腋部听呼吸音而确认气管导管的位置，并加以固定。
⑧ 进行鼻 - 气管插管时，将润滑好的导管从右鼻孔沿着鼻底插入。为避免损伤鼻甲，应将导管的洞斜面居中。
⑨ 若气管导管插入了食管，最好的检测方法是检测呼末二氧化碳。
瘢 其他一些不太可靠的评估方法有：气体逸出时有“暖气”声，听诊时呼吸音减弱，机械通气时胸廓运动减弱，并且在上腹部可听到“咕咕”声。
瘢 解剖异常或气道的病理性改变是最常造成插管和通气失败的原因。

one main bronchus . Severe bronchospasm and *tension pneumothorax*① will result in failed ventilation after intubation . *And hypoxia is also induced by regurgitated gastric contents which can cause blockage of the airways directly or secondary to laryngeal spasm and bronchospasm* .②

Direct trauma occurs during laryngoscopy and insertion of the tube to lips, teeth, tongue, pharynx, larynx, trachea, nose, and nasopharynx during nasal intubation, which will cause soft tissue swelling or bleeding . Indirect trauma includes the injury to the mandible (dislocation), and the cervical spine and cord, especially where there is *pre-existing degenerative disease*③ or trauma .

*Reflex activities during intubation*④ include three aspects . Firstly, hypertension and arrythmias occur in response to intubation and may hurt patients with coronary artery disease, and aortic or intracranial aneurysms . *In patients at risk, specific action is taken to attenuate the response, for example, pretreatment with β-blockers, analgesics, or intravenous lignocaine* .⑤ Secondly, vomiting is also caused by intubation . This may be stimulated when laryngoscopy is attempted in patients who are inadequately anesthetized . It is more frequent when there is material in the stomach . For example, in emergencies when the patient is not starved, in patients with *intestinal obstruction*⑥, or when gastric emptying is delayed, as after opiate analgesics or following trauma . Finally, laryngeal spasm is a fatal complication . Stimulation of the epiglottis or larynx will result in *adduction of the vocal cords*⑦ which blocks the airway up and endangers the patient .

Daily Language

1 . I am so sorry to tell you that your husband's respiratory condition is not very good .
我很遗憾的告诉您，您丈夫的呼吸状况不很好。

2 . His lungs cannot supply enough oxygen for his body so we must use the ventilator for him . Now we are going to intubate him .
他的肺不能供给机体足够的氧气，所以我们必须给他用呼吸机。现在我们准备给他插管。

3 . What is a ventilator for ?
呼吸机是做什么用的？

4 . To put it simple, the ventilator is used to supply oxygen and help the patient to exhale

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① 张力性气胸。
② 反流的胃内容物堵塞气道、喉痉挛或气管痉挛也可能引发低氧血症。
③ 先前存在的退行性疾病。
④ 插管而引起的机体的反射。
⑤ 对于危重的病人，应该采取一些特殊的措施来减弱病人的反射，例如预先给β-受体阻滞剂，止痛剂或静脉使用利多卡因。
⑥ 肠梗阻。
⑦ 声带的内收。

Unit 19 Suction

1. State the routes of suction;
2. State the procedure of suction;
3. Remember the cautions for suction.

In some facilities, a doctor's order is needed for suctioning. This order is usually written as a *prn*① order. Gradually, suctioning becomes a nursing decision. It is used for patients with critical oxygenation level. *Carefully assess the patient before you proceed with suctioning unless the patency of the patient's airway is threatened.*② Assess by listening to chest sounds and sounds of the higher respiratory tract. You can sometimes hear gurgling sounds from the back of the throat. *A patient with severely compromised respirations from the presence of considerable secretions may appear cyanotic and have labored breathing.*③ It is important for you to weigh the suction decision, because the irritation of the catheter may intensify the buildup of secretions. Also, during suctioning the patient cannot breathe in oxygen. Suctioning performed too frequently can increase the accumulation of secretions and cause a degree of hypoxia. *Identify whether the patients needs a short period of hyperventilation with a high concentration of oxygen before suctioning.*④ Be familiar with the suction equipment available and the details of the procedure.

Wash your hands for infection control and plan for any needed assistance from another colleague. Choose the appropriate equipment. Gather the following: sterile catheter of appropriate size, sterile gloves, sterile container or basin, and sterile saline. *Be sure wall suction connecter or portable suction machine is available.*⑤ *Tongue depressor*⑥ is used for oral suction. Goggles provide front and side protection, but most people who wear eyeglasses believe they can provide adequate protection. Mask, if desired, should be prepared. Suction trap also should be available when a sputum specimen is needed.

First of all, identify the patient to be sure that you are performing the procedure on the

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① 需要时使用。
② 除非病人气道的开放性受到威胁（而需要立即进行吸痰）外，吸痰前请仔细评估病人。
③ 一个由于呼吸道有大量的痰液而呼吸严重受损的病人可能会出现紫绀和呼吸困难。
④ 在吸痰之前应确认病人是否需要一短时间的高浓度氧气的通气。
⑤ 确保中心负压接头或可移动的负压机可以正常使用。
⑥ 压舌板。

correct patient . Explain what you are going to do . Suctioning can be irritating to the alert patients . Our cough reflex, the basis for our protective response, makes it natural to resist foreign objects that enter the respiratory tract . Explaining the necessity and process of the procedure can alleviate patient's fear and attain increased cooperation . Tell the patient that you will insert a catheter gently, and ask the patient to relax as much as possible . Inform the patient that coughing may be induced, but explain that it is all right in that it may help to raise the secretions to a level where they can be suctioned . Secure adequate room lighting so you can suction safely or have an assistant prepare a gooseneck lamp or flashlight for you . Place the patient in *semi-Flowler's position*① with the head turned slightly towards you for oropharyngeal suctioning . *For nasopharyngeal suction it is helpful to have the patient's neck hyperextended to promote smooth insertion of the suction catheter .*② Place a clean towel across the patient's chest to protect the gown . If prior oxygenation is needed, do so at this time . It is good practice to routinely have the patient take two to three deep breaths before you start . Open the suction kit, maintaining the sterile technique . Use the wrapper as a *sterile field*③ . If the sterile solution and container are not in the suction kit, set up the sterile container and open and pour the sterile solution into it at this time . Turn on the wall suction mechanism or the suction machine . Put on sterile gloves . Most kits contain two sterile gloves so that both hands are protected from contact with respiratory secretions and the patient is protected from microorganisms . Reserve the gloves on your dominant hand for contact with the suction catheter, and use the other hand for any other needs . If the sterile solution and container are in the kit, open the container and fill it with solution . *Pick up the catheter with your dominant hand and attach the connector end to the suction tubing, which is held in you clean, gloved nondominant hand .*④ Take care not to touch the tubing with the dominant hand to avoid contamination . Test the wall unit or the machine by suctioning water once through the tubing and catheter . (The water also serves to lubricate the catheter .)

Nurses can insert the catheter either through the mouth, the nostrils, or the tracheostomy . Through the mouth, if necessary *use a tongue blade, to hold the tongue aside for visibility*⑤ . Slide the catheter along the side of the mouth to the oropharynx . Through the nostrils, slide the catheter gently along the floor of an unobstructed nostril to the nasopharynx . *Never apply suction while you are inserting the catheter because you could damage the mucous membranes and further deprive the patient of oxygen .*⑥ Use a catheter only for a single route . If you must change the route, obtain a new sterile catheter . For example, if you have attempted to suction a patient orally and have met with resistance, and

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① 半坐卧位（将处于仰卧位的病人头部抬高，角度小于 45 度）。
② 经鼻咽部吸痰时，应使病人的颈部处于过伸位，以便吸痰管顺利地插入。
③ 无菌区。
④ 用你的优势手持吸痰管，用带着手套的、干净的非优势手将吸痰管连接在吸引管上。
⑤ （使用压舌板）将舌头推向一侧，（使咽部）得到暴露。
⑥ 在插入吸痰管的时候不要开启负压，以免损伤黏膜，使病人吸不到氧气。

then decide to proceed with the nasopharyngeal approach, discard the first catheter and obtain a second one. If you are not using a kit, open separate sterile packs and proceed. Some facilities allow using the same catheter for oral suction if the oral suctioning is done last. Holding your thumb over the opening in the catheter, apply suction for no longer than 15 seconds. *It is good practice as you begin to hold your breath during the suction period to remind you that patient is not receiving oxygen or inspiring while you suction.*① When suctioning orally, suction carefully in the cheeks, where secretions are easy to pool. Withdraw the catheter under suction with a rotating motion which aspirates the secretions protruding from the catheter's tip. Flush the catheter with sterile water to remove secretions.

Also, you can suction through the tracheostomy. Switch the nondominant, clean-gloved hand to the suction tubing and control button. Using the dominant, sterile-gloved hand on the catheter, proceed to suction. Insert the catheter 4～5 inches into the tracheostomy *without occluding the port*② on the suction catheter, so that suction is not being applied. Suction at this time would further deprive the patient of oxygen.

The patient may cough in response to the irritation caused by the catheter, which helps to raise secretions. Persistent coughing may indicate tracheal spasm, in which case you should stop the procedure. Using the nondominant hand, briefly turn off the suction to listen to the patient's breath sounds and assess the need for repeated suctioning. If the patient's breathing is not clear, repeat suction, allowing the patient to rest 20～30 seconds between suctioning periods. When the patient's breathing sounds become clear, stop suctioning. You should never apply suction more than three times. Sometimes you will stop because you have not been successful in reaching the secretions. At other times you may be forced to stop because the patient is actively resistant to the procedure. Knowing when to stop suctioning requires good nursing judgment. Remember this procedure can be painful and frightening for the patient.

Finally, with your nondominant hand, grasp the cuff of the sterile glove and pull it downward over the used catheter in you gloved hand. This method neatly encloses the used catheter in the glove, making disposal more sanitary. Detach the catheter from the tubing and dispose of it safely in a receptacle. Reposition the patient for comfort, wash your hands, and offer oral hygiene.

Record the procedure on the patient's chart. Include date, time, and the amount and character of secretions removed. The patient's tolerance of the procedure should also be noted.

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① 开始吸痰时屏住你的呼吸是一种很好的操作习惯，这样做可以提醒你病人在吸痰时是没有吸到氧气或吸气的。

② 不能堵住吸痰管上的小孔。

Unit 20 Cardiopulmonary Resuscitation

1. Interpret the term "CPR";
2. State the steps of basic CPR;
3. Identify the methods of carrying out basic CPR.

*Cardiac arrest is the sudden loss of cardiac output, which is potentially reversible with prompt restoration of circulation and oxygen delivery.*① *Strictly defined, basic life support (BLS) is the initial phase of emergency cardiac care, including recognition of cardiac arrest, delivery of rescue breathing (ventilation), and chest compressions (circulation).*② *Advanced cardiac life support (ACLS) includes BLS, ECG monitoring, rhythm identification, and restoration of hemodynamic stability through intubation, defibrillation, and pharmacologic therapy.*③ *Modern CPR began in* 1960 *with the landmark study by Kouwenhoven et al,*④ which reported combining closed chest compression, mouth-to-mouth breathing, and external defibrillation. As they explained it, 正如他们解释的那样，他们的步骤非常容易执行："任何地方任何人都可以进行心脏复苏程序，所需要的只是两只手。" their algorithm was remarkably easy to perform: "Anyone, anywhere, can now initiate cardiac resuscitative procedures. All that is needed are two hands."

护士应有效地执行心肺复苏术(CPR)5，无论是在街上还是在医疗机构

Nurses are expected to carry out *Cardiopulmonary Resuscitation (CPR)*⑤ efficiently, 护士必须能够作为团队成员或单独对所有年龄段的人执行这些救生程序 whether on the street or in a health care facility. Nurses must be able to perform these life-saving procedures as a member of a team or alone on individuals of all ages. Cardiopulmonary 心肺复苏有效地恢复了血流动力学的稳定，恢复了自主循环。 resuscitation effectively restores hemodynamic stability, and return of spontaneous circulation. *Prolonged survival is significantly lower because of underlying illness and the postresuscitation syndrome, specifically central nervous system injury and left ventricular stunning after resuscitation.*⑥

Basic CPR is easy to remember as ABC: Airway, Breathing, and Circulation.

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① 心脏骤停指心输出量的突然终止。如果循环和氧供得到迅速恢复，这一过程可能是可逆的。
② 严格地说，基本生命支持（即基本心肺复苏）是针对心脏急症实施抢救的第一步，包括识别心脏骤停、救治呼吸（通气）以及胸部按压（即恢复循环）。
③ 进一步生命支持（即高级心肺复苏）包括基本生命支持，心电图监测，心律的确认，通过插管、除颤和药物治疗而恢复血液动力学的稳定。
④ 现代心肺复苏以 Kouwenhoven 等人 20 世纪 60 年代开始的研究为标志。
⑤ 心肺复苏。
⑥ 由于基础疾病和复苏后综合征的存在，病人长期生存率很低，尤其是病人复苏后出现了中枢神经系统受损及左室顿挫时更是如此。

Airway

Firstly, determine unresponsiveness. When you find a person in a state of collapse, call out for help and when someone responds, send that person *to activate the emergency medical system*① in health care facility or, if you are outside the health care facility, in your community.

请派该人通知急救系统
如果您不在医疗机构之外，请在您的社区中启动紧急医疗系统

Secondly, position the person on the back on a firm, flat surface. If you must roll the person over, *try to move the entire body as a single unit*②, avoiding worsening any injury to the neck, back, or long bones.

其次，将人仰卧在稳固、平坦的表面上。如果您必须将人翻身，请尝试将整个身体作为一个整体移动2，避免加重对颈部、背部或长骨的任何伤害

*Thirdly, tilt the patient's head back by placing the palm of one hand on the forehead and applying firm backward pressure.*③ Place the fingers of the other hand under the bony part of the lower jaw near the chin, and lift to bring the chin forward. This will clear the tongue out of the airway.

将另一只手的手指放在靠近下巴的下颌骨质部分下方，
然后抬起以使下巴向前。这会将舌头从气道中清除。

Breathing

Firstly, determine breathlessness. Place your ear close to the patient's mouth and do three things: LOOK at the chest and the stomach for movement; LISTEN for breathing sounds; and FEEL for air against your cheek.

Secondly, keeping the airway open by using *the head-tilt/chin-lift maneuver*④, gently pinch the nose closed using the thumb and index finger of the hand on the forehead. This will prevent air from escaping through the patient's nose.

用手的拇指和食指在额头上轻轻捏住鼻子。
这将防止空气通过患者的鼻子逸出

Finally, place your mouth over the patient's mouth; make an airtight seal, and give two initial breaths of 1～1.5 *second each*.⑤ Be sure to allow time for deflation of the patient's lungs between breaths. If the patient's chest rises and falls, it shows that air has entered. If mouth-to-mouth breathing is not desirable or possible (for example, in the presence of vomiting or injury to the mouth or jaw), you can close the mouth with your palm and breathe into the nose. The position of the head is the same as for mouth-to-mouth breathing. Mouth-to-stoma breathing is possible if the patient has a permanent tracheostomy. In such a case it would not be necessary to tilt the head back to open the airway.

确保在两次呼吸之间留出时间让患者的肺部放气

Circulation

Firstly, kneel at the level of the patient's shoulders. You will then be in a position to perform both rescue breathing and chest compression without moving you knees. The patient should be on a hard surface. In a health care facility, slip a *cardiac board*⑥ under the patient.

Secondly, locate the lower margin of the patient's rib cage on the side nearest you. *Run the fingers of the hand nearest the patient's legs up along the rib cage to the indentation where*

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① 通知急救系统。
② 尽力将病人作为一个整体来搬动。
③ 第三，放在病人前额上的手掌向后施力，使病人头部后仰。
④ “压前额-抬下巴”的方法。
⑤ 最后，用你的嘴唇包绕病人的嘴，做到不漏气，并给予两次每次持续 1～1.5 秒的初始人工呼吸。
⑥ 心脏按压板。

the ribs meet the sternum.①

Thirdly, place the heel of your other hand just above those fingers, at right angles to the sternum. This will keep the main force of compression on the sternum and decrease the chance of rib fracture.

Fourthly, remove your fingers from the indentation and place that hand on the top of the one already in position. Your hand should be parallel. Your fingers may be either extended or interlaced, but they must be kept off the chest.

Fifthly, with your shoulder directly above the patient's chest, compress downward, keeping your arms straight. You should move the sternum of an adult 1.5～2 inches with each compress. You should release compression pressure after each compression to allow blood to flow into the heart. The time allowed for release should be equal to the time required for compression. Therefore, your compression should be a rhythmical 50% down and 50% back up. Your hands should not be lifted from the chest or their position changed in any way so that your hands will not lose correct position.

If you are the only rescuer present, you are responsible for both rescue breathing and cardiac compression. *The proper ratio is* 15 *compressions to* 2 *breaths at a rate of* 100 *compressions/min*.② If two rescuers are present, it is best to position yourselves on opposite sides of the patient. Rescuer 1 should be positioned at the patient's side and compress the sternum at a rate of 100 compressions/min. Rescuer 2, positioned at the patient's head, maintains an open airway, and ventilates the patient twice after every fifteen compression.

Daily Language

1. You should call for help first. (Help!)
   先求救。(救命!)
2. Tilt his head, lift his chin, and check breathing.
   压额抬颏，检查病人呼吸。
3. Give two breaths.
   吹两口气。
4. Position hands in the center of the chest and firmly push down two inches on the chest 15 times.
   稳定地按压胸腔 2 寸，并反复 15 次。
5. Continue 2 breaths and 15 pumps until help arrives.
   持续地以按压 15 次、吹气两次的速度循环，直到救援到来。
6. The patient's pupil diameter is about 0.3 cm, and light reflex disappears. Blood

① 用你靠近病人腿侧的手，沿胸廓（即肋缘）向上滑行到肋骨与胸骨交接的切迹处。
② 恰当的节律是每分钟 100 次心脏按压，并且每 15 次心脏按压对应两次人工呼吸。

Unit 21 Nutrition

1. State the kinds of essential nutrients for a person;
2. State three kinds of diseases relevant to nutrition;
3. Identify the meaning of nutrition.

*Nutrition is the sum of all the interactions between an organism and the food it consumes.*① People require food or essential nutrients for the growth and maintenance of all body tissues and the normal functioning of all body processes. *Nutrients are the organic and inorganic ingredients of foods and required for proper body functioning.*② Providing structural material for body tissues, providing energy for body processes and movement, and regulating body processes are three basic functions of nutrients.

An adequate food intake consists of a balance of essential nutrients: water, carbohydrates, proteins, fats, vitamins, and minerals. The most basic nutrient is water. Carbohydrates, fats, and proteins are the energy-providing nutrients. Foods differ in their nutritive value, and no one food provides all essential nutrients.

The amount of energy that nutrients supply to the body is their *caloric value*③. A calorie is a unit in heat energy. *A small calorie is the amount of heat required to raise the temperature of* 1 *g water* 1℃.④ A large calorie (Calorie, kilocalorie) is the amount of heat required to raise the temperature of 1 kg water 1℃ and is the unit used in nutrition. *The energy liberated from each gram of carbohydrate or protein after it is metabolized is about* 4 *kcal*;⑤ from each gram of fat, about 9 kcal. The average North American receives approximately 45% of energy from carbohydrates, 40% from fat, and 15% from protein. In most other parts of the world, people derive far more energy from carbohydrates than from fats, or proteins.

Carbohydrates are composed of carbon, hydrogen, and oxygen. There are two basic kinds of carbohydrates: glucose (simple carbohydrates) and starch (complex carbohydrates). Once in the cell, glucose is either used for immediate release of energy or stored in the form of

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① 营养是有机体与其消耗的食物之间相互作用的总和。
② 营养素是指食物所含的有机和无机成分，是维持机体的各项功能所必需的。
③ 卡价。
④ 1 小卡是指可以将 1 克水温度升高 1 摄氏度所需的热量。
⑤ 每 1 克碳水化合物或蛋白质代谢后所释放的能量为 4 千卡。

glycogen . Although all cells of the body are capable of storing some glycogen, certain cells (liver and muscle) can store large amounts . Glucagon from the alpha cells of the pancreas, and epinephrine from the adrenal medulla activate glycogenolysis . When the body恭 stores of carbohydrates fall below normal, certain quantities of glucose are formed from protein and fat . Up to 60% of the amino acids of the body恭 protein can be converted into glucose . *During starvation, the body depletes first its fat and later its protein reserves .*① Glucose undergoes a series of reaction to produce energy for the body恭 varied demands .

Proteins are organic substances that upon hydrolysis or digestion yield amino acids . Proteins are composed of carbon, hydrogen, and oxygen, just like carbohydrates, but protein also contains nitrogen . Every cell in the body contains some protein, and about 3/ 4 of body solids are proteins . Urine and bile are the only body substances normally lacking protein . Amino acids are categorized as essential or nonessential . *Essential amino acids are those that cannot be manufactured in the body and must be supplied .*② Nonessential amino acids are those that the body can manufacture . *Positive nitrogen balance indicates nitrogen input exceeds output .*③ A state of positive nitrogen balance often exists under the following conditions: childhood and adolescence, pregnancy, and phases of physical exercise, when there are increases in muscle growth . Eating more protein than necessary does not result in an increase in positive nitrogen balance but can lead to weight gain . Negative nitrogen balance exists when nitrogen output exceeds intake . This state usually occurs when a person consumes a diet inadequate in essential amino acids and calories or exposed to unusual stress, for example, trauma .

Fats are groups of organic substances (fats, oils, waxes, and related compounds) that are greasy and insoluble in water but soluble in alcohol or ether . These organic substances are lipids . Fats have the same element (carbon, hydrogen, and oxygen) as carbohydrates, but the hydrogen content is higher . Cholesterol is a fatlike steroid alcohol, mostly synthesized in the liver . Some cholesterol is absorbed from the diet . *High blood levels of one or more of the lipids, particularly cholesterol or triglycerides, is called hyperlipidemia .*④

*Vitamin, an organic compound that cannot be manufactured by the body, is needed in small quantities to catalyze metabolic process .*⑤ Vitamins are generally classified as fat soluble or water soluble . Fat-soluble vitamins include A, D, E, and K . Water-soluble vitamins include C and the B-complex vitamins: B_1, B_3, B_6, B_9, B_{12}, etc . The body cannot store water-soluble vitamins, thus people must get a daily supply in the diet . The body can store fat-soluble vitamin, so a daily supply of fat-soluble vitamin is therefore not absolutely

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① 当人处于饥饿状态时，机体首先消耗其储存的脂肪，然后将蛋白质消耗掉。
② 必需氨基酸是机体不能自己产生、必须靠外界供给的氨基酸。
③ 正氮平衡指氮的摄入量超过了排出量的状况。
④ 血中一种或多种脂质的水平升高，尤其是当胆固醇或甘油三酯升高时，即称为高脂血症。
⑤ 维生素，一种不能由机体自行合成的有机化合物，所需量很小，在代谢的过程中起酶的作用。

necessary. Microminerals are those that people require daily in amounts less than 100 mg. They include iron, zinc, manganese, iodine, fluoride, copper, cobalt, chromium, and selenium.

*Clinical nutrition is concerned with the diagnosis and treatment of diseases that affect the intake, absorption, and metabolism of dietary constituents and with the promotion of health through the prevention of diet related diseases.*① Adult diseases of clinical nutrition encompass the most common causes of mortality in the developed world. They include obesity, hypertension, diabetes, dyslipidemias, increased risks of cardiovascular disease, some cancers, and pulmonary failure, intestinal disorders related to inadequate nutrient absorption, eating disorders, and malnutrition associated with chronic illness and surgical trauma. Scientific advances on the relationship between dietary substances and the cellular mechanisms of disease occur with regularity and frequency. Yet, despite the prevalence of nutritional disorders in clinical medicine and despite increasing scientific evidence on the significance of dietary modification to disease prevention, present day practitioners of medicine are typically untrained in the relationship of diet to health and disease. In the absence of reliable medical advice on nutrition, patients increasingly turn to herbal dietary supplements, costly diet schemes for weight reduction, and other unproved and potentially harmful remedies. Standardization of curricula for nutrition education of medical students is increasingly relevant to the cost-effective integration of nutritional concepts into medical practice.

Daily Language

1. Do you know what you must pay attention to in your diet?
   您知道在饮食上有哪些问题是您必须注意的吗？
2. You should control the quantities of energy you take in, especially the amount of the fats.
   您应该控制您摄入能量的总量，尤其对摄入脂肪的量应该严加控制。
3. You should eat more fish, vegetable, and fruit. Also you should do exercises for weight reduction.
   您应该多吃鱼、蔬菜和水果。同时您还应该进行体育锻炼以减轻体重。

Vocabulary

1. nutrient [ˈnjuːtriənt] *adj*. 营养的，滋养的 *n*. 营养素，滋养物
2. organic [ɔːˈɡænik] *adj*. 器官的；有机的；器质的

① 临床营养学是关于诊断和治疗影响饮食成分的摄入、吸收和代谢的疾病，并且通过预防饮食相关疾病而促进健康的学科。

Unit 22 Shock

1. State the definition of shock;
2. State the pathophysiology of shock;
3. List the complications of shock;
4. List the classes of shock and their causes.

Shock is a life-threatening condition that occurs when cardiovascular system is unable to meet the oxygen and nutritional needs of the body荫 cells and is usually reflected by a sudden drop in blood pressure. Insufficient blood pressure may be due to a variety of factors. Common causes include inadequate *cardiac output*① due to *heart failure*②, a sudden loss of blood volume due to hemorrhage, or a sudden decrease in *peripheral vascular resistance*③ due to anaphylaxis.

Adult respiratory distress syndrome (*ARDS*)④, *disseminated intravascular coagulation* (*DIC*)⑤, *and multisystem organ failure* (*MSOF*)⑥ are three especially serious conditions that may follow a prolonged episode of shock. Clients with ARDS will usually develop *pulmonary failure*⑦ despite high levels of supplemental oxygen and *mechanical ventilation*⑧. DIC results from ischemic damage to the endothelial lining of blood vessels and may precipitate shock and death. MSOF is a major cause of mortality following shock. It usually begins with respiratory failure, followed by failure of the kidneys, heart, liver, and finally cerebral and gastrointestinal functioning.

Different forms of shock are categorized and named by the cardiovascular characteristics that occur. The four shock categories are: (1) *hypovolemic shock*⑨ caused by a decrease in the circulating blood volume. Hypovolaemia normally results from a haemorrhage, which can be either internal, such as bleeding from the spleen, or external, for example, from trauma;

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① 心输出量。
② 心脏衰竭。
③ 外周血管阻力。
④ 成人呼吸窘迫综合征。
⑤ 弥散性血管内凝血。
⑥ 多系统器官功能衰竭。
⑦ 肺衰竭。
⑧ 机械通气。
⑨ 低血容量性休克。

(2) *cardiogenic shock*① caused by cardiac failure; (3) *extracardiac obstructive shock*② caused by a blockage of blood flow in the cardiovascular circuit outside the heart; and (4) *distributive shock*③ caused by excessive dilation of the venules and arterioles. Most cases of clinical shock show some components of each of these categories. However, this classification system is helpful in understanding the causes of shock.

When in shock, the person蒜 blood pressure is very low. Depending on the specific cause and type of shock, symptoms will include one or more of the following:

瞌Anxiety or agitation

瞌Confusion

瞌Pale, cool, clammy skin

瞌Low or no *urine output*④

瞌Bluish lips and fingernails

瞌Dizziness, light-headedness, or faintness

瞌Profuse sweating, moist skin

瞌Rapid but weak pulse

瞌Shallow and rapid breathing

瞌Chest pain

瞌Unconsciousness

Assessment and treatment of the patient will depend on the type of shock, as signs and symptoms differ for each. The aim of the care is to re-establish perfusion of the vital organs and to prevent the shock from progressing. Regular observations are essential; the frequency depends on the patien蒜 condition, but at a bare minimum they should be recorded half hourly. The nurse should check for any fall in blood pressure and changes in heart rate, as well as the strength of the pulse, as a weak and thready pulse is indicative of shock. The patient should ideally be monitored for *cardiac rate and rhythm*⑤, blood pressure, and *pulse oximetry*⑥. The nurse should regularly assess the patien蒜 conscious level and assessment should be used in conjunction with the *Glasgow coma score*⑦, with any deterioration reported immediately. The patien蒜 respiratory rate should be measured, and the depth and pattern noted. High flow oxygen should be given to any patient in shock, so pulse oximetry and *blood gases*⑧ are taken to provide an assessment of the patien蒜 respiratory status. Nurses should observe for tachypnea and respiratory tiredness, as these indicate that the patient might require mechanical ventilation.

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① 心源性休克。
② 心外阻塞性休克。
③ 分布性休克。
④ 尿量。
⑤ 心率和心律。
⑥ 脉搏氧含量。
⑦ 格拉斯哥评分。
⑧ 血气分析。

Patients in shock should be kept warm, but should not be warmed too quickly, as this will cause peripheral vasodilation and could cause hypotention. A *central venous catheter*① should be inserted to measure the fluid status of the patient. The *central venous pressure* (*CVP*)② should be measured hourly to monitor the circulating fluid level of the patient. *Strict asepsis*③ must be maintained at all times to prevent infection.

Strict fluid balance is essential and any *losses and inputs*④ need to be recorded. If large amounts of fluids are to be administered, they should be warmed prior to transfusion, as this will reduce the incidence of hypothermia. A *urinary catheter*⑤ should be inserted and hourly urine measurements recorded. The nurse must ensure that if the patient is *actively bleeding*⑥, blood has been cross-matched and intravenous fluids are being administered as quickly as possible, especially if the patient is bleeding excessively externally. The volume of blood lost should ideally be placed. Any patient in shock requires a minimum of two wide bore cannulas for rapid intravenous fluid and drug administration.

The patient should *lie in flat*⑦ in the acute stage of shock and regular positional changes should be given to monitor pressure areas and to prevent fluid from accumulating in the lungs. It is essential when nursing a patient in shock to explain all the procedures to the patient and relatives. Time should be taken to keep relatives informed of the situation, and it is imperative that the patient is reassured and is allowed to ask questions regarding his or her care.

Daily Language

1. He is anxious and agitated.
   他有些烦躁不安。
2. His vital signs are not stable.
   他的生命体征不稳定。
3. His heart rate is about 135 times per minute, and the pulse is weak and thready.
   他的心率在每分钟 135 次左右，脉搏细弱。
4. His blood pressure keeps around 70/56 mmHg.
   他的血压一直在 70/56 mmHg 左右波动。
5. His respiratory rate is about 35 times per minute, and it is shallow and rapid.
   他的呼吸浅快，次数在每分钟 35 次左右。
6. He looks pale, and the skin is cool and clammy.

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① 中心静脉插管。
② 中心静脉压，缩写为 CVP。
③ 严格无菌（技术操作）。
④ 出入量。
⑤ 尿管。
⑥ 活跃性出血。
⑦ 平卧。

Uuit 23 Pain

1 . State the concept of pain and the types of pain;
2 . Describe the characteristics of different types of pain;
3 . Sum up the approaches to pain management;
4 . Discuss what nurses should do to relieve pain .

Pain is a universal experience . *How we feel pain is a complicated process that is carried out by the nervous system* .① It is any sensation caused by stimuli that injure or destroy tissue . Pain can be caused by temperature, pressure, or chemicals released by damaged tissues . The process of pain is not fully understood . *It is currently thought that pain is mediated by the stimulation or inhibition of gating mechanisms located in the spinal cord, the central control system of the cortical and limbic areas of the pain, and a central biasing system located in the brainstem . This theory is called the gate control theory of pain and was developed by Melzack and Wall* .② Although this is not a completely validated theory, it does provide a comprehensive view of what we presently know about pain .

Pain serves a useful purpose in many instances . It provides us with a warning that our body is at risk of, or is, being damaged . This gives us time to react and remove the damaging stimulus, e.g . when you burn your hand on the oven . *Sometimes though, pain comes from tissue damage that we cannot move away from such as inflammation in joints, or a tumor pressing on organs in the body* .③ *Pain can also warn of emotional or stress related problems* .④ Pain is a subjective symptom . *It is exactly what you say it is and hurts when you say it hurts* .⑤ No one can fully appreciate the pain of another . In other words, no two persons feel the same degree of pain in the same way . There is no direct relation between stimulus and individual response .

There are two main types of pain: acute and chronic pain . Acute pain generally occurs suddenly, is temporary, and *eases of its own accord*⑥, or after successful treatment of the

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① 我们对疼痛的感受是一个由神经系统传导的复杂的过程。
② 目前普遍认为疼痛是通过刺激或抑制位于脊髓段的门控机制、大脑皮层和边缘的疼痛感受区和位于脑干的边缘系统来调节的。该学说被称为“疼痛门控理论”，是由 Melzack 和 Wall 创立的。
③ 有时候，疼痛来自我们无法回避的组织损伤，如关节炎症、体内压迫器官的肿瘤等。
④ 疼痛也可提示与情绪或焦虑相关的健康问题。
⑤ 当一个人述说疼痛存在时它就存在。
⑥ 自动地减轻。

underlying problem. *Chronic pain persists, or recurs for prolonged and often indefinite periods of time.*① There are two aspects to feeling pain: the actual physical sensation and the emotional response. Sensitivity and reaction to pain are influenced by previous experience with pain, age, sex, emotional factors, and cultural background. *Fear and anxiety about pain decrease an individua蒜 threshold and tolerance for pain, which reduces his/her ability to cope with the pain.*②

*An awareness of the clien蒜 usual response to painful stimuli enables the nurse to better evaluate the level of pain and facilitates the development of effective strategies to minimize the pain experience.*③ The pain control nurse may assess the following: the site of pain; the nature of pain (how long it lasts, how quickly it starts, whether it is intermittent or constant); what the pain feels like; what makes the clien蒜 pain feel better or worse; how bad the pain is; and how the clien蒜 pain affects his/her daily life.

Major types of physiologic pain are superficial pain, visceral pain, referred pain, radiating pain, phantom pain, etc. Superficial pain is of cutaneous origin, usually due to trauma, which causes strong stimulation of a large number of nociceptors. *It is well localized and may have a bright, pricking quality or be dull and burning.*④ Both types are usually of short duration. Visceral pain originates in the deeper structures of the body, which are poorly localized. It is often associated with, or manifested solely by, referred pain and tends to have an aching quality. Nausea and vomiting are frequently present. *Referred pain can most often be experienced at a focus of the body separated and removed from the organ where the noxious stimulus is occurring. Usually the pain is initiated in one of the visceral organs and referred to an area of the body surface.*⑤ Also, pain may originate in a viscus and be referred to another deep area of the body not exactly coincident with the location of the viscus producing the pain. The knowledge of these different types of referred pain is extremely important in diagnosis because many visceral ailments cause no other signs except referred pain. Radiating pain is a sensation that pain is extending from the initial site to another point. *This can be possibly due to recruitment of neurons in adjacent cord segments.*⑥ *Phantom pain is a sensation in a body part that has been amputated.*⑦ Even though the nerves supplying the amputated part were severed, the remaining neurons may continue to send impulses to the same area of the brain as before.

Nurses may use a variety of tools to assess client蒜 pain. They may use a pain scale to

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① 慢性疼痛持续时间长，或常常不定时的反复发作。
② 疼痛引起的恐惧和焦虑会降低个人的疼痛阈值和疼痛忍耐力，从而降低他/她应对疼痛的能力。
③ 了解病人对疼痛刺激的反应可帮助护士更好地评价病人的疼痛程度，制定出有效的护理策略来减轻病人的疼痛。
④ 它（皮肤痛）定位准确，表现为锐痛、刺痛（或钝痛）或烧灼痛。
⑤ 牵涉痛大多发生于人体的某一部位，而该部位与受到有害刺激的器官并不相关。通常痛觉产生于某内脏器官，而牵涉到身体表面。
⑥ （放射痛）可能与相邻脊髓节段的神经元的聚集现象有关。
⑦ 幻觉痛是源于身体已被截除部位的一种感觉。

assess how bad the pain is . A simple scale consists of a line where one end is marked "no pain" and the other end is marked "worst pain" (the patient would be asked to mark on the line where he thinks his pain level is) . More complicated pain scales can also be used . For example, the face scale is useful for assessing pain in children . A nurse may also use a simple drawing of the body to record the site of the pain . After assessing the pain, the nursing plan for pain control should be carried out . Pharmacological approaches, nonpharmacological approaches and nursing care are three ways commonly used to relieve pain .

Drugs have been used in the relief of pain for many centuries, *but their mode of action remains obscure*① . Aspirin is the most frequently used drug that relieves pain and also reduces inflammation and fever . It is taken by mouth alone or in combination with caffeine, or codeine for the relief of the less severe types of pain, such as headache, toothache, neuralgias, and the pain of rheumatoid arthritis . *Narcotic analgesics produce analgesia by acting directly on the central nervous system . It is the drug that induces stupor and insensibility and relieves pain .*② The term is used particularly for morphine and other derivatives of opium but is also applied to other drugs that depress brain function (e.g . general anesthetics and hypnotics) . *Because of their ability to cause dependence and tolerance, sleeping drugs have largely replaced narcotics; but they are still used for relief of severe pain .*③

Postoperative clients need analgesics for 24 to 72 hours after surgery and perhaps longer, depending on the procedure . To manage the pain, the nurse requires knowing the type of surgical procedure that has been performed as well as patient characteristics that may influence the choice of analgesic . Frequent assessment of pain severity using techniques such as visual analog scales can help optimize pain control . *Nonsteroidal anti-inflammatory agents provide good analgesia after most minor surgical procedures and can decrease the amount of opioid analgesics needed after more extensive procedures .*④ Intravenous and epidural administered opioids are useful; however, better analgesia may be achieved when local anesthetics are administered by infiltration, peripheral nerve block or continuous epidural infusion . The use of two or more analgesic techniques together produces better pain relief than a single medication or administration route . A team approach can lessen the amount of postoperative pain . Family physicians should aggressively treat postoperative pain and actively support hospital postoperative pain treatment program .

The nurse is the care provider with the most responsibility for evaluating the client's need for analgesic medications . Nurses often worry that a client will become addicted . Clients and families also frequently express the same concern . Addiction can be a problem in some instances, but when responsible health care providers are working with clients in pain, they

① 但是它（药物止痛）的作用方式依然不清楚。
② 麻醉止痛药通过直接作用于中枢神经使痛觉消失，它是一种引起昏睡、感觉迟钝和缓解疼痛的药物。
③ 由于麻醉药容易产生依赖性和耐药性，大部分已被安眠药所替代，但是仍然用来缓解剧烈的疼痛。
④ 非甾类抗炎药物对于多数小手术后疼痛止痛效果好，还可以减少大手术后的麻醉药的用量。

continually evaluate the client's response, and adjust the medication schedule accordingly. The age and weight of the client, the severity of the health problem, other extenuating factors, and history of response to analgesic medication are considerations in deciding adequate medication management.

There should be less fear of addiction in terminal ill clients in excruciating pain. It is difficult to justify an attitude that encourages living with unbearable pain while dying.[①] Nurses often feel helpless in working with clients with relentless pain. However, all efforts will be made to make a client as comfortable as possible.

Daily Language

1. Do you still feel pain?
   你还感觉痛吗？
2. I'll give you some pills. They're useful to relieve the pain.
   我给你一些药片，它们能减轻你的疼痛。
3. This patient needs some medicine to relieve his pain.
   这个病人需要一些麻醉药来缓解疼痛。
4. That boy's arm is in rupture. He has a sharp pain.
   这个男孩的手臂断了，他感到疼痛剧烈。
5. The pain seems relieved.
   疼痛似乎减轻了。
6. Can anything be done for the pain?
   对于疼痛有什么办法吗？
7. I have a bad headache. May I have a pain killer?
   我头痛厉害，请给我一片止痛药好吗？

Vocabulary

1. receptor [rɪˈseptə(r)] *n*. 接受器，感受器，受体
2. stimulus [ˈstɪmjʊləs] *n*. 刺激物，促进因素，刺激
3. inhibition [ˌɪnhɪˈbɪʃn] *n*. 禁止，阻止，压抑
4. brainstem [ˈbreɪnstem] *n*. 脑干
5. perceptual [pəˈseptʃuəl] *adj*. 知觉的，有知觉的
6. phenomenon [fəˈnɒmɪnən] *n*. 现象
7. cutaneous [kjuːˈteɪniəs] *adj*. 皮肤的，影响皮肤的

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① 对于遭受极度疼痛的临终病人来说，不必太担心药物的成瘾。我们难以证明鼓励病人在临终时还要忍耐疼痛的态度是正确的。

Unit 24 Anxiety

1. Identify the concept of anxiety;
2. Distinguish anxiety from fear;
3. Describe the characteristics of anxiety;
4. State different levels of anxiety;
5. Identify the nursing interventions and nursing goals for patients with anxiety.

*Anxiety is defined as an uncomfortable feeling of apprehension or dread that occurs in response to internal or external stimuli and can result in physical, emotional, cognitive, and behavioral symptoms.*① The concept of anxiety can be better understood by exploring some of its characteristics. Anxiety is an emotion and a subjective individual experience. It is an energy and therefore cannot be observed directly. A nurse can only infer that a patient is anxious based on certain behaviors and she needs to validate this with the patient. Also, anxiety is an emotion without a specific object. It is provoked by the unknown and precedes all new experiences such as entering school, starting a new job, or giving birth to a child.

The characteristics of anxiety differentiate it from fear. *Fear is an individual ideation that has a specific source or object.*② The person can identify and describe the source of fear. *Fear involves the intellectual appraisal of a threatening stimulus;*③ *anxiety involves the emotional response to that appraisal*④. Fear is caused by physical or psychological exposure to a threatening situation. Fear produces anxiety. Thus, fear is the appraisal of danger; anxiety is the unpleasant emotion evoked by fear.

Another characteristic of anxiety is that it is communicated interpersonally. This is significant to the nurse. If a nurse is talking with a patient who is anxious, within a short time the nurse will also experience feelings of anxiety. Similarly, if a nurse is anxious in a particular situation, it will readily be communicated to the patient. The contagious nature of anxiety can therefore have positive and negative effects on the therapeutic relationship. *A nurse must diligently monitor these effects.*⑤

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① 焦虑是一种不舒适的感受，是对内外刺激的一种恐惧反应，可导致身体、情绪、认知和行为的症状出现。
② 恐惧是个体对特定的物体或场境的害怕。
③ 恐惧是对危险刺激的理性评价。
④ 焦虑是对评价的一种情绪反应。
⑤ 护士须随时观察病人的这些反应。

Although a specific etiology for anxiety is still uncertain, sources indicate that a combination of biologic, psychosocial, and environmental factors seems evident. A lot of researches find that anxiety is related with psychoanalytic factors, such as unconscious conflict, repressed wishes, drives, and desires; *biologic factors, such as genetic predisposition, vulnerability inherited, neuroanatomic sites, and potential for pathology*①; neurophysiologic factors, such as excessively responsive autonomic nervous system, and neurotransmitter disruption; interpersonal factors, *such as early interactions with parental figure*②; behavioral factors, such as internal stimuli; *existential factors*③, such as loss of meaning in life; and environmental factors, such as disasters, rape, assault, and continual traumas/stressors.

*Peplau*④ (1963) identified four levels of anxiety and described their effects.

Mild anxiety is associated with the tension of day-to-day living. During this stage the person is alert and the perceptual field is increased. The person sees, hears, and grasps more than before. This kind of anxiety can motivate learning and produce growth and creativity.

Moderate anxiety, in which the person focuses only on *immediate concerns*⑤, involves the narrowing of the perceptual field as the person sees, hears, and grasps less. The person *blocks selected areas*⑥ but can attend to more if directed to do so.

Severe anxiety is marked by a significant reduction in the perceptual field. The person tends to focus on a specific detail and not think about anything else. All behavior is aimed at relieving anxiety, and much direction is needed to focus on another area.

Panic is associated with awe, dread, and terror. *At this stage details are blown out of proportion*.⑦ Because of a complete loss of control, the person is unable to do things even with direction. Panic involves *the disorganization of the personality*⑧. There may occur increased motor activity, decreased ability to relate to others, *distorted perceptions*⑨ and loss of rational thought. Panic is frightening and paralyzing experience. A person in panic is unable to communicate or function effectively. A prolonged period of panic would result in exhaustion and death.

Hildegard Peplau suggested that the nurse works with the client, first, to identify the level of anxiety; next, to understand the behaviors associated with it; and finally, to better cope. The nurse plays an important role in helping the client overcome his anxiety. The basic goals of intervention are to decrease the client's anxiety and help totally deliver him from it so

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① 遗传素质、先天易感性、神经类型、潜在的病理变化等生物因素。
② 早期与父母的交流。
③ 生存因素。
④ 佩普洛（美国心理学家）。
⑤ 直接相关事件。
⑥ 阻断特定领域事件。
⑦ 在此阶段，个体的感知被不合理地放大。
⑧ 人格紊乱或人格混乱。
⑨ 知觉歪曲。

as to restore him to a normal level of functioning .

All types of people may participate in assisting individuals in anxiety . They can be psychologists, teachers, welfare workers, clergy, police, social workers as well as nurses . The nurse may act as an educator, an adviser, and a model . Her ability to be aware of client's feelings and thoughts is crucial in coping with a client in anxiety . During the initial interview, the client first of all needs to gain a feeling of safety . For this end, the nurse must listen carefully and with patience .

Speak in a soft voice to present a calm, peaceful, and acceptable atmosphere . *It is usually hard to encourage discussion on any topic other than the symptoms* .① However, it is better neither to minimize the symptoms nor focus excessively on them . As the client and the nurse begin to know each other better, there will be a firmer foundation for further *mutual trust*② . Then proceed to work with the client to establish the nurse-client relationship, which comes slowly but eventually will allow him greater comfort in *giving vent to*③ true feelings . *With this end in view*④, a nurse should examine her own moods from time to time so as to avoid any negative influence upon the client . Tell the patient explicitly that the key to the problem will surely be found . Avoid argument or angry talk . Don't get involved in a quarrel or a power struggle . Moreover, it is also important to watch out for silence . It should be studied and not maintained for a long period . If the client has been keeping silent for a long while, the nurse may ask him what he is thinking about .

Once the patient can recognize his anxiety, the communication should be aimed at revealing the effect and reality of conflicted areas . That is to say, the nurse helps the client see which of his values are being threatened . She can instruct and direct the client . Correct his abnormal behavior . Direct him to use relaxing treatment . If the symptoms are evident, she can advise the doctor to give the client suitable medicine .

In evaluating the nursing care, the nurse can review and look into each phase of the nursing process . If the intervention has been successful, it means that the person's degree of anxiety and his ability to function have returned to a normal level .

Daily Language

1 . Would you like to talk about your job ?
你愿意谈谈您的工作吗？

2 . What happened recently ?
最近发生什么事了？

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① 通常，很难要求病人去讨论其他话题而不讨论其症状。
② 相互信任。
③ 发泄。
④ 由于这个原因。

Unit 25 Fever

1. Identify the definition of fever;
2. Explain the cause of fever and the signs of fever;
3. Assess, plan, and implement the care for a person with fever.

Body temperature is the result of the balance between heat production and heat loss. Normal oral body temperature in the adult ranges from 36.7℃ to 37.6℃ (98.1℉ to 99.6℉). A healthy person蒜 body temperature slightly increases during the day since the morning (from 6:00 a.m.). The peak body temperature is reached at 6:00 to 10:00 p.m. and the lowest is between 2:00 and 6:00 a.m.

An elevation of temperature above normal is referred to as fever or pyrexia. According to the degrees of the temperature, *fever is classified as low-grade fever, mild fever, moderate fever, and high fever*①. Temperate between 37.6℃~38.2℃ (99.8℉~100.8℉) is considered a low-grade fever, between 38.3℃~38.9℃ (101℉~102℉) is considered a mild fever, and between 38.9℃~39.4℃ (102℉~103℉) is considered a moderate fever. Anything around 39.4℃ (104℉) or above is considered a high fever, and delirium or convulsions may occur.

Fever is part of our body蒜 *defense mechanism*②. It is our body蒜 natural response to fighting germs. When our bodies are fighting infections, the body temperature rises. The cause of the fever is quite a complicated process. Our blood and lymphatic systems produce white blood cells to fight off infection. As white blood cells increase in number, our bodies are heated up, causing fever or a rise in body temperature. Although fever is usually an indication of infection or illness, it is believed that fever helps the body fight infection by stimulating natural defense mechanisms. That is why it is not good to try to decrease a fever too much or too fast. When body temperature is above 105 degrees, harm can occur in the nervous system. In some people, usually children, seizures can occur when the body temperature is above 38.9℃. Therefore, if a fever goes too high, measures must be taken to bring the fever down. Antipyretics are medications usually used to lower fever, such as Tylenol, aspirin,

① 发热分为低热、微热、中热和高热。
② 防御机制。

and so on . Don't give a child aspirin products when they have *chicken pox*① or the flu, which could result in *Reye's syndrome*② that is potentially lethal .

Fevers are primarily caused by viral or bacterial infections . However, other conditions may also induce a fever, such as allergic reactions, autoimmune diseases, trauma, cancer, excessive exposure to the sun, intense exercise, hormonal imbalances, certain drugs, and damage to the hypothalamus . When an infection occurs, fever-inducing agents called pyrogens are released, triggering the resetting of the thermostat . In other circumstances, the immune system may overreact (allergic reactions) or become damaged (autoimmune diseases), leading to the uncontrolled release of pyrogens . A stroke or tumor can damage the hypothalamus, causing the body's thermostat to malfunction . Excessive exposure to the sun or intensely exercising in hot weather can result in heat stroke, a condition in which the body's cooling mechanisms fail . *Malignant hyperthermia is a rare, inherited condition in which a person develops a very high fever when given certain anesthetics or muscle relaxants in preparation for surgery .*③

Symptoms of fever include chills, aching everywhere, listlessness, red cheeks, feeling hot or cold, etc ., which vary depending on the stage of the fever .

During the onset of a fever, the major signs are chills, shivering, pallid skin, increased pulse rate, and rising rectal temperature even though skin reading is cool .

In the course of a fever, the patient may not complain of feeling hot or cold . Instead, he may complain of headache, irritability or restlessness, or weakness .

During the termination of a fever, the signs include diaphoresis, reddening of the skin, and possible dehydration .

When one has fever, even though the fever may work to help him, it still makes him feel uncomfortable . Nursing interventions are as follows . During the period of chills, *provide additional bedclothes*④ and note the onset and duration of the chill . During the stage of diaphoresis, increase circulation of air in the room to promote heat loss by vaporization . Because bedclothes may become wet with diaphoresis, it is important they be changed as necessary . *It is necessary to encourage the patient to increase fluid intake to prevent potential dehydration due to excessive fluid loss from diaphoresis .*⑤ The nurse should measure and record the patient's fluid intake and output . In addition to monitoring the patient's temperature, the nurse needs to assess the patient's pulse and respirations . *The increase in the pulse rate and respiratory rate is directly associated with the increase in the basal metabolic*

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① 水痘。
② 雷若综合征。
③ 恶性的过高热是一种很少见的遗传性疾病。在准备手术时使用麻醉药或使用肌肉松弛药后，可能导致恶性高热。
④ 增加被褥保暖。
⑤ 必须鼓励病人多饮水，以避免因出汗过多导致脱水。

rate and demands for oxygen that accompany a temperature rise.[1] The pulse rate can increase 10 to 15 beats per minute with a rise of each 1 degree Centigrade in temperature. The physician should be notified of the patient's vital and clinical signs for appropriate diagnostic procedures.

Daily Language

1. He has got a low-grade fever/ high fever.
   他发低烧/ 高烧。
2. You should take his temperature every two hours.
   你应该每两小时给他测一次体温。
3. If he has a low-grade temperature, you don't have to give him anything to lower the fever.
   如果他只是发低烧，不必给他吃退烧药。
4. Offer him plenty of fluid to bring his body temperature down.
   让他多喝点水，来降低体温。
5. If the temperature is over 38.5℃ or if the child is uncomfortable, you can give him some medicine to bring the fever down.
   如果体温超过 38.5℃，或孩子感觉不舒服，可以给他吃点退烧药。
6. You can lower the temperature by giving him a lukewarm bath. Do not use ice water or rubbing alcohol to reduce a child's temperature.
   你可以给他洗个温水浴来降温，但不要用冰水或酒精。

Vocabulary

1. pyrexia [paɪˈreksɪə] *adj*. 热病的
2. delirium [dɪˈlɪriəm] *n*. (暂时的) 精神狂乱，精神错乱，精神极度兴奋
3. convulsion [kənˈvʌlʃən] *n*. 惊厥，痉挛，抽搐
4. mechanism [ˈmekənɪzəm] *n*. 机构，机制
5. lymphatic [lɪmˈfætɪk] *adj*. 含淋巴的，淋巴腺的
6. antipyretic [ˌæntɪpaɪˈretɪk] *n*. 退热药，解热药
7. Tylenol [ˈtaɪlənɔl] *n*. 羟苯基乙酰胺 (一种醋氨酚牌子的商标)
8. autoimmune [ˌɔːtəʊɪˈmjuːn] *adj*. 自体免疫的
9. hypothalamus [ˌhaɪpəˈθæləməs] *n*. 视丘下部
10. pyrogen [ˈpaɪrəʊdʒən] *n*. 热原 (质)

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① 呼吸和脉搏增加与由于体温升高而引起的基础代谢增加和氧需要量增加直接相关。

Unit 26 Diarrhea

1. Interpret the term: diarrhea, dehydration, acute diarrhea, and chronic diarrhea;
2. Describe the common causes for diarrhea;
3. List the types of diarrhea which indicate medical supervision;
4. Describe the prevention measures for diarrhea.

The definition of diarrhea depends on what is normal for you. For some, diarrhea can be as slight as one loose stool per day. Others may have two or more than three daily bowel movements usually and not be considered as diarrhea. *So the best explanation of diarrhea is "an abnormal increase in the frequency and liquidity of one's stools. Sometimes the loose stools mix with blood, pus, and mucus."*①

Diarrhea may be considered as a symptom of more serious diseases, injury or food poisoning and can also be indicative of a chronic syndrome such as *Crohn's disease*②. It is also a result of severe radiation sickness. *The basic physiopathology of diarrhea may be divided into the general classification of decreased fluid absorption, increased fluid secretion, motility disturbance, or a combination of these.*③

Common causes of diarrhea are bacterial infections, viral infections, parasites infections, food intolerances, reaction to medicines, intestinal diseases, and functional bowel disorders. Some people develop diarrhea after stomach surgery or removal of the gallbladder. *The reason may be a change in how quickly food moves through the digestive system after stomach surgery or an increase in bile in the colon that can occur after gallbladder surgery.*④ In many cases, the cause of diarrhea cannot be found. Usually, as long as diarrhea *goes away by itself*⑤, the cause often doesn't matter or a broad search for the cause is not necessary.

According to its duration, diarrhea can be divided into two groups: acute diarrhea and chronic diarrhea. Acute diarrhea may be defined as diarrhea that lasts less than 2 weeks. *This can nearly always be supposed to be infective although it is only confirmed in a minority of*

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① 因此，对腹泻的最好解释是：大便次数和通畅性增加。有时稀便中混有血液、脓液和黏液。
② 克隆重病。
③ 腹泻的基本病理生理学改变包括液体吸收减少、分泌增多、肠蠕动紊乱和以上各因素的综合反应。
④ 其原因可能与胃部手术后食物经过消化道的速度改变，或是因为胆囊术后结肠内的胆汁增多有关。
⑤ 自愈。

cases.① It is usually accompanied by abdominal pain, nausea, and vomiting. When diarrhea persists for at least 2 weeks or when it subsides and returns more than 2 to 4 weeks after the initial episode, it is considered chronic. Such diarrhea is usually painless and persistent. Malabsorption and malnutrition are also sequelae of chronic diarrhea.

Severe diarrhea may be debilitating and life threatening. Diarrhea can cause dehydration, which means the body lacks enough fluid to function well. Dehydration is particularly dangerous in children and the elderly, and it must be treated quickly to avoid severe health problems. Common signs of dehydration include: thirst, less frequent urination, dry skin, fatigue, light-headedness, and dark colored urine.

The following types of diarrhea usually indicate medical supervision is necessary:

1. Diarrhea in infants.
2. Moderate or severe diarrhea in young children.
3. Diarrhea associated with bleeding.
4. Diarrhea that persists for more than 2 weeks.
5. Diarrhea that is related to more common illnesses such as non-cramping abdominal pain, fever, weight loss, etc.
6. Diarrhea in homosexual males (tends to be more severe and may be associated with AIDS).
7. Diarrhea in travelers (more likely to have foreign infections such as parasites).
8. Diarrhea in food handlers (possible to infect other persons).

Most types of diarrhea are harmless and will get away by themselves within hours to days. If people don't get dehydrated, there's little to worry about. *The treatment of diarrhea is based on the reason and is aimed at replacement of liquid and electrolytes*.② It includes the patient consuming adequate amounts of water to replace that loss, preferably mixed with electrolytes to provide necessary salt and some amount of nutrients. Oral solutions containing glucose and electrolytes may be sufficient to replace losses from mild diarrhea. For many people, further treatment and formal medical suggestion is unnecessary. Medicines that stop diarrhea may be useful in some cases, but they are not suggested for people whose diarrhea is from a bacterial infection or parasite — stopping the diarrhea traps the organism in the intestines, prolonging the problem. Instead, doctors usually prescribe antibiotics. Viral causes are either treated with medication or left to run their course, depending on the severity of diarrhea and the type of the virus. In situations of severe diarrhea, parenteral administration of fluids, electrolytes, vitamins, and potentially, nutrition is required.

For the most part diarrheas are infections, spread primarily via the fecal-oral route.③

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① (急性腹泻)几乎总是被认为是传染性的，尽管这只在少数病例中得到证实。
② 应针对腹泻的原因治疗，同时要补充丢失的液体和电解质。
③ 大多数腹泻主要由粪—口途径传染。

Invisible organisms such as microscopic viruses, bacteria, or parasites from the stool that hide on hands, toys, foods, and surfaces might be taken into the mouth.[①] Then these microbes flourish in your intestines, causing damage and diarrhea. Prevention is a matter of good hygiene. The measures mainly include washing your hands before preparing foods and eating clean food, washing your hands after using the bathroom, keeping your hands away from your mouth in general, washing after shaking hands with a number of people, and other measures that keep your hands clean. Prevention of non-infectious diarrhea is directed at merely avoiding the triggering events — such as taking a lactose-absorption enzyme with dairy products, or avoiding certain foods or spices, which are likely to cause loose stools. Trying to stay away from stressful situation is simple advice to provide, but hard to follow in real life.

Daily Language

1. I have got bad abdominal pain and several loose stools today. It seems that I have diarrhea.
   我有严重的腹痛和几次稀便。看来我腹泻了。
2. When did these symptoms start?
   这些症状是什么时候开始的？
3. How many times do you have bowel movements a day?
   一天解了几次大便？
4. Is there any blood or mucus in the stools?
   大便中有血或者黏液吗？
5. What is the shape of the stool?
   大便呈什么形状？
6. Just loose stool.
   稀便。
7. Have you got nausea and vomiting?
   有恶心和呕吐症状吗？

Vocabulary

1. diarrhea [daɪəˈrɪə] *n*. 痢疾，腹泻
2. stool [stuːl] *n*. 大便，厕所
3. pus [pʌs] *n*. 脓，脓汁
4. mucus [ˈmjuːkəs] *n*. 黏液，胶

① 看不见的生物体，如从大便中来的微小病毒、细菌，或藏在双手、玩具、食物和物体表面的寄生虫可能被吃进口里。

Unit 27 Constipation

1. Interpret the term: Constipation;
2. Identify the factors that lead to constipation;
3. State the complications of the constipation;
4. State the treatment of constipation.

*Constipation is passage of small quantity of hard, dry stool.*① Other symptoms of constipation include feeling bloated, uncomfortable, and sluggish. People who are constipated may find it difficult and painful to have a bowel movement. Common causes of constipation are as the following:

Not enough fiber in the diet. The most common cause of constipation is that people eat too much low fiber food, such as cheese, eggs, and meats. People who eat lots of *high-fiber foods*② such as vegetables, fruits, and whole grain are less likely to become constipated. Fiber, both soluble and insoluble, is the part of fruits, vegetables, and grains that the body cannot digest. Insoluble fiber passes through the intestines almost unchanged. The bulk and soft texture of fiber help prevent hard, dry stools.

Not enough liquids. Liquid like water and juice adds fluid to the colon and bulk to stools, making bowel movements softer and easier to pass. People with constipation must drink enough of these liquids each day. Liquids that contain caffeine, like coffee and cola drinks, and alcohol have a *dehydrating effect*③. Drink as less as possible if possible.

Lack of exercise can result in constipation, although doctors do not know the reason precisely. For example, constipation often takes place after an accident or during an illness when one must stay in bed and cannot exercise.

Medications such as pain medications (especially narcotics), antacids that contain aluminum and calcium, blood pressure medications (calcium channel blockers), antiparkinson drugs, antispasmodics, antidepressants, iron supplements, diuretics, and anticonvulsants may lead to constipation.

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① 便秘就是排除的粪便量少且坚硬干燥。
② 富含纤维的食物。
③ 脱水作用。

Irritable Bowel Syndrome (*IBS*).① People with Irritable Bowel Syndrome, also known as spastic colon, have spasms in the colon that influence bowel movements. Constipation and diarrhea are its common symptoms, usually alternating. Abdominal cramping, gassiness, and bloating are other common complaints.

Changes in life or routine. *During pregnancy, women may be constipated because of hormonal changes or because the heavy uterus compresses the intestine.*② Aging may also affect bowel regularity because a slower metabolism results in less intestinal movement and muscle tone. In addition, people are prone to be constipated when traveling because their normal diet and daily routines are changed.

*Abuse of laxatives.*③ Rumors about constipation may lead to a serious abuse of laxatives. It is popular among people who are preoccupied with having a daily bowel movement. Laxatives usually are not necessary and can be habit-forming. The colon begins to count on laxatives to bring on bowel movements. Ultimately, laxatives can damage nerve cells in the colon and disturb the colon蒜 natural ability to contract. For the same reason, regular use of enemas can also lead to a loss of normal bowel function.

*Ignoring the urge to have a bowel movement.*④ People who ignore the urge to have a bowel movement may finally feel no urge, which can lead to constipation. Some people delay having a bowel movement because they do not want to use toilets in comfort station. Others ignore the urge because of emotional stress or because they are too busy. Children may postpone having a bowel movement because they do not want to be interrupted when they are playing.

Specific diseases. Neurological disorders, metabolic or endocrine disorders, and systemic conditions that affect organ systems are usually causes of the constipation.

Problems with the colon and rectum. Intestinal obstruction, scar tissue (adhesions), tumors, colorectal stricture, *Hirschsprung蒜 disease*⑤, or cancer can compress, squash, or narrow the intestine and rectum, therefore, resulting in constipation.

*Constipation can result in complications including hemorrhoids caused by exerting to have a bowel movement or anal fissures caused when hard stool hurts the sphincter muscle.*⑥ Treatment for hemorrhoids may include warm tub baths, ice packs, and application of a special cream to the affected area. Treatment for *anal fissure*⑦ may include stretching the sphincter muscle or surgical removal of tissue or skin in the affected area. *Sometimes straining causes a small amount of intestinal lining to push out from the anal opening.*⑧ This condition, also called rectal prolapse, may lead to secretion of mucus from the anus.

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① 肠激惹综合症，缩写为 IBS。
② 妇女怀孕期间，由于激素水平改变或由于增大的子宫压迫肠道可能导致便秘。
③ 滥用泻药。
④ 忽略排便冲动。
⑤ 先天性巨结肠。
⑥ 便秘会导致并发症，包括用力大便引起的痔疮和坚硬粪便损伤肛门括约肌引起的肛裂。
⑦ 肛裂。
⑧ 有时用力大便会导致一部分直肠内壁从肛门口脱出。

Constipation may also result in hard stool which crams the intestine and rectum so tightly that the normal pushing movement of the colon is not enough to expel the stool. This condition, called fecal impaction, occurs most commonly in children and the elderly. Impaction stool can be made softer with mineral oil taken by mouth and by an enema. After softening the impaction, the doctor may break up part of the hardened stool and insert one or two fingers into the anus to remove them.

*Although constipation treatment depends on the cause, severity, and duration, in most cases, dietary and lifestyle changes will help relieve symptoms of constipation and help prevent it.*① The followings are the common measures to take when treating constipation.

Diet with enough fiber (20 to 35 grams each day) helps form soft, bulky stool. People inclining to constipation need to limit foods that have little or no fiber, such as ice cream, cheese, meat, and processed foods.

Certain lifestyle changes can help treat and prevent constipation, including drinking enough water and other liquid such as fruit and vegetable juice, taking part in daily exercise, and reserving adequate time to stimulate a bowel movement. In addition, the urge to have a bowel movement should not be ignored.

Laxatives are not necessary for the mildly constipation people. *But for those who have made diet and lifestyle changes but still keep constipating, doctors may prescribe laxatives or enemas for a restricted time.*② These treatments can help restore a chronically sluggish bowel. For children, short-term treatment with laxatives, along with retraining to build regular bowel habits, helps a lot.

Other treatments may be directed at a specific cause. For example, the doctor may suggest discontinuing medication or performing surgery to correct an anorectal problem such as rectal prolapse.

Daily Language

1. I haven't had a bowl movement for three days and I am feeling bloated and uncomfortable.
   我已经三天没有解大便了。我现在感觉腹胀，很不舒服。
2. Sometimes I bleed when I relieve the bowels.
   当我大便时有时要出血。
3. Use the suppositories and laxative. They will help bowl movement and relieve pain.
   使用栓剂和缓泻药。它们有助于通便并同时缓解疼痛。

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① 尽管便秘的治疗要根据便秘原因、严重度和持续时间，但在大多数情况下，饮食和生活方式的改变将有助于减轻便秘的症状和预防便秘。
② 对那些已经改变饮食及生活习惯却仍然便秘的人，医生可能会在一段时间内给予缓泻剂和灌肠剂治疗。

Unit 28 Pneumonia

1. List the types of pneumonia;
2. Explain the supportive treatments for pneumonia;
3. State the nursing diagnosis and nursing goals for pneumonia patients;
4. State the nursing care of pneumonia.

Pneumonia is an infection of the lung that affects 1 out of 100 people annually. It was the No. 1 cause of death until 1936. Since then, the use of antibiotics brought it under control. In China, there are approximately 2.5 million cases of pneumonia per year and around 125 thousand persons die of the disease, *ranking as the fifth leading cause of death*①. Pneumonia outbreaks tend to occur in spring and winter.

Pneumonia is not a single disease. It can have over thirty different causes. The most common ones include bacterial pneumonia, viral pneumonia, mycoplasma pneumonia, and pneumonia associated with aspiration of *foreign substances*②. 支原体肺炎和吸入异物相关的肺炎

Pneumococcus is the most common cause of bacterial pneumonia. Bacterial pneumonia can attack anyone from infants through the very old. However, those who may be at higher risk for pneumococcal disease are usually those who are 65 years of age or older, children more than 2 years of age and adults who have chronic illnesses, such as heart, lung, and kidney disease, cancer, diabetes, alcoholism, HIV infection, or *other conditions associated with an impaired immune system*③. 与免疫系统受损相关的疾病 The onset of pneumococcal pneumonia can vary from gradual to sudden. Signs and symptoms include: chills, chattering, chest pain, cough, fever, *rapid and shallow respiration*④, *nasal flaring*⑤, retractions, and dyspnea. 鼻翼煽动、回缩和呼吸困难

Most cases of viral pneumonia are mild and last a short time, but some cases are more serious and require hospitalization. People who are at risk for more serious viral pneumonia typically have impaired immune systems such as young children (especially those with heart defects), the elderly, people with HIV, and people taking medications to inhibit their immune systems in the treatment of autoimmune disorders. Symptoms of viral pneumonia

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① 在疾病死亡原因中列第五位。
② 异物。
③ 其他与免疫系统功能损伤有关的疾病。
④ 呼吸浅、快。
⑤ 鼻煽。

include: fever, a dry cough, headache, muscle pain, weakness, shortness of breath, sore throat, and sweating.

Mycoplasma pneumonia is an infection of the lungs caused by mycoplasma. Mycoplasmas are the smallest free-living agents of disease in humankind, *unclassified as bacteria or viruses, but having characteristics of both*.① Mycoplasma pneumoniae usually affects people younger than 40. The symptoms are generally mild. But in some people, they may progress to more severe symptoms. Common symptoms include the following: headache, fever, chills, excessive sweating, dry cough, chest pain, and sore throat.

Other kinds of pneumonia: Various special pneumonias are caused by the inhalation of food, liquid, gases or dust, and by fungi. *Pneumocystis carinii pneumonia* (*PCP*)② is caused by an organism believed to be a fungus. In many patients, PCP is the first sign of AIDS.

Antibiotics as a form of pneumonia treatment can cure bacterial and mycoplasma pneumonia most of the time, but there is no effective treatment yet for viral pneumonia. Antibiotics can lower body temperature within a day or two and produce a dramatic recovery. After temperature drops, medication must be continued for 4 or 5 days *according to physician's instructions*③, otherwise the pneumonia may recur. Recurrence can be more serious than the first attack. Besides antibiotics, patients can also be given *supportive treatment*④: proper diet, oxygen to relieve dyspnea or cyanosis, and medication to ease chest pain. Adequate rest is important to maintain progress toward full recovery and to avoid relapse.

氧气以缓解呼吸困难或发绀

The nursing diagnosis for pneumonia includes impaired gas exchange, ineffective airway clearance, hyperthermia, ineffective breathing pattern, risk for fluid volume deficit, and activity intolerance.⑤

Nursing goals for pneumonia are: promote rest, maintain patent airway, control fever, prevent dehydration, ease breathing difficulty, provide nutrition, monitor respiratory efforts, reduce anxiety and detect complications early.

Nursing care of all types of pneumonia is basically the same. The client needs rest to prevent exhaustion. The nurse must organize her work so that she does not disturb the client unnecessarily. *The patient must be turned and repositioned frequently to avoid pooling of secretion*.⑥ Remove heavy clothing and blankets and give *sponge bathing*⑦ when the client is flushed with fever. Encourage the client to take more fluids to prevent dehydration. Vital

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① 它（支原体）既不是细菌也不是病毒，但却具有两者的特点。
② 卡氏肺囊虫肺炎，缩写为 PCP。
③ 遵医嘱。
④ 支持疗法。
⑤ 肺炎的护理诊断包括：气体交换受损、清理呼吸道无效、体温过高、低效型呼吸形态、有体液不足的危险、活动无耐力等。
⑥ （护士应）经常为病人翻身并改变体位以防止分泌物的淤积。
⑦ 海绵擦浴。

signs and chest sounds are monitored to assess the progress of the disease and to detect early signs of complications .

Daily Language

1 . Your daughter must take these medicines in time . Even after the temperature returns to normal, she may need to continue to take the medication for 4 to 5 days according to the doctor's prescription, otherwise the temperature may shoot up again .
您女儿应按时服药。即使在体温恢复正常以后，她还要继续服药 4 至 5 天，否则温度可能再次升高。

2 . Besides, what your daughter needs most now is adequate rest and a balanced diet . So, it is helpful to let your daughter have a good sleep and eat well .
另外，您女儿现在最需要的是好好休息和均衡的营养。所以，您应该让她睡好觉，多吃点。

Vocabulary

1 . annually [ˈænjuəli] *adv* . 一年一次，每年
2 . outbreak [ˈaʊtbreɪk] *n* . (战争的) 爆发，(疾病的) 发作
3 . mycoplasma [ˌmaɪkəʊˈplæzmə] *n* . 支原体，支原菌
4 . aspiration [ˌæspəˈreɪʃn] *n* . 吸气
5 . pneumococcus [ˌnjuːməˌkɒkəs] *n* . 肺炎球菌
6 . alcoholism [ˈælkəhɒlɪzəm] *n* . 酒精中毒
7 . chattering [ˈtʃætərɪŋ] 咔哒声
8 . nasal [ˈneɪz(ə)l] *adj* . 鼻子的；鼻中的
9 . retraction [rɪˈtrækʃn] *n* . 收回
10 . hospitalization [ˌhɒspɪtəlaɪˈzeɪʃn] *n* . 医院收容，住院治疗
11 . defect [ˈdiːfekt] *n* . 缺陷
12 . fungus [ˈfʌŋɡəs] (*pl* . fungi) *n* . 真菌
13 . organism [ˈɔːɡənɪzəm] *n* . 生物体，有机体
14 . recur [rɪˈkɜːr] *v* . 复发
recurrence [rɪˈkʌrəns] *n* . 复发
15 . cyanosis [ˌsaɪəˈnəʊsɪs] *n* . 发绀，青紫
16 . relapse [rɪˈlæps] *n* . 复发 *vi* . 旧病复发
17 . sponge [spʌndʒ] *n* . (外科用的) 棉球，纱布

Unit 29 Hypertension

1. Define hypertension;
2. List at least four potential complications that can develop from uncontrolled hypertension;
3. Discuss the nursing management for a client with hypertension.

Hypertension refers to a condition characterized by sustained elevation of systolic arterial pressure of 140 *mmHg or higher, or a diastolic arterial pressure of* 90 *mmHg or greater, or both.*① Hypertension is divided into two categories: *essential (or primary) hypertension*② and secondary hypertension. Essential hypertension is a sustained elevation of blood pressure without any known cause for it. *Secondary hypertension describes a variety of conditions in which elevation of blood pressure results from or is secondary to some other disorder.*③ Many hypertensive individuals are asymptomatic or have mild symptoms like headache, fatigue, insomnia, and nervousness. However, all have an elevated systolic or diastolic blood pressure, or both. *If blood pressure is not controlled, it can lead to stroke, congestive heart failure, myocardial infarction, dissecting or rupture of an aortic aneurysm, and renal failure.*④ Initial medical management of hypertension depends on the degree of pressure elevation but generally non-pharmacologic interventions are used first. For mild elevations, weight reduction, decreased sodium intake, moderate exercise, and reduction of other contributing factors, such as smoking and alcohol use, may return the blood pressure to normal levels. If cholesterol and triglyceride levels are increased, a diet low in saturated fats is recommended. Depending on the client's response to non-pharmacologic therapy, pharmacologic therapy may be subsequently used until blood pressure is controlled. For some patients, pharmacologic therapy should be used first, and non-pharmacologic therapy may only be a *helpful adjunct*⑤.

原发性高血压是没有任何已知原因的血压持续升高

继发性高血压描述了血压升高由或继发于某种其他疾病的各种情况

许多高血压患者没有症状或有轻微的症状，如头痛、疲劳、失眠和紧张。然而，所有人的收缩压或舒张压都升高，或两者都有

高血压的初始医疗管理取决于压力升高的程度，但通常首先使用非药物干预

胆固醇 甘油三酸酯

根据客户对非药物治疗的反应，可以随后使用药物治疗直到血压得到控制。

The nurse should measure the blood pressure accurately and consistently in the same

根据客户对非药物治疗的反应

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① 高血压是指以持续性的收缩压（动脉血压）≥140 mmHg 和（或）舒张压（动脉血压）≥90 mmHg 为特点的临床综合征。
② 原发性高血压。
③ 继发性高血压是由于其他疾病引起的血压升高。
④ 如果血压未能得到控制，可导致中风、充血性心力衰竭、心肌梗塞、动脉瘤破裂和肾衰。
⑤ 辅助治疗。

患者的心输出量充足；疼痛会减轻或减轻；客户将在没有不适的情况下进行日常生活活动

manner . The major goals of nursing management are that the client's cardiac output will be adequate; pain will be reduced or relieved; the client will perform activities of daily living without discomfort; the client will understand the purposes and expected outcomes of treatment; the client will verbalize or demonstrate sufficient knowledge for safe self-care; and the client's anxiety will be reduced to a manageable level . *To make these goals measurable, more specific criteria must be added* .① The following are common nursing problems and corresponding nursing measurements accompanying patients with hypertention .

预期结果 可控水平 以下是高血压患者常见的护理问题及相应的护理措施。

Cardiac Output inadequate . Limit activity and provide rest to maintain or restore the heart rate to less than 100 bpm . Restrict caffeine and nicotine, which cause vasoconstriction . Relieve a fever, if it is present, since this increases the *workload of the heart*② . Demonstrate a calm, self-confident manner when providing care to avoid creating anxiety in the client and stimulating the sympathetic nervous system

血管收缩 刺激交感神经系统

Potential Possibility of Injury . *Some antihypertensive drugs cause postural hypotension* .③ The client may feel weak, faint, or dizzy when changing positions . To lessen the potential for injury, instruct the client to rise slowly from a *sitting or lying position*④ . In the morning, have the client first sit on the edge of the bed for a few moments and then stand . *Accompany the client when ambulating* .⑤ Identify the signal to use when needing assistance . Help the client to sit or, preferably, to lie down promptly upon feeling faint .

确定需要帮助时使用的信号。帮助服务对象坐下，最好是在感到昏厥时立即躺下

Uncomfortable Pain . Provide rest immediately if chest pain occurs and give oxygen . For the client with a severe headache and elevated intracranial pressure, raise the head slightly and avoid jarring the bed . Control sensory stimulation, like bright lights and noise, within the environment . Administer analgesics and vasodilators that have been prescribed *on a prn basis*⑥ .

出现胸痛应立即休息并吸氧。 对于头痛剧烈、颅内压升高的患者，头稍抬高，避免震动床 控制环境中的感官刺激，例如明亮的灯光和噪音。管理已在 prn 基础上开具的镇痛药和血管扩张剂

如果服务对象的高血压非常严重，则只进行必要的活动。例如，进行部分沐浴，而不是完整沐浴

Activity Intolerance .⑦ Carry out only essential activities if the client is extremely hypertensive . For example, *give a partial bath, rather than a complete bath*⑧, or delay hygiene indefinitely if the client is unstable . Provide periods of rest between activities . Assist the client if self-performed activities create dyspnea, chest pain, or tachycardia . Resume or increase activity gradually based upon the client's response .

如果自我进行的活动引起呼吸困难、胸痛或心动过速，请协助服务对象。根据客户的反应逐渐恢复或增加活动

Anxiety . Allow clients to express their feelings . Respond promptly to the client's signal for assistance . Stay with the client during periods of discomfort . Reassure the client that he or she will not be left alone . *Speak in a calm, well-modulated voice* .⑨ Encourage relaxation

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① 为了使这些目标可测量，必须有更具体的标准。
② 心脏的负荷。
③ 一些抗高血压药可引起体位性低血压。
④ 坐位或卧位。
⑤ 当病人走动时应陪伴在病人身边。
⑥ 长期备用医嘱。
⑦ 活动无耐力。
⑧ 给予部分擦浴，而不是全身沐浴。
⑨ 用镇静的、平和的声音说话。

使用沉默和适当的触摸作为治疗性沟通技巧

techniques, such as slow, controlled breathing or imagery. Give brief explanations. Use silence and appropriate touching as therapeutic communication techniques. When possible, allow at least one supportive family member or friend to be with the client. If a visitor contributes to client's anxiety, limit the interaction tactfully.

Knowledge Deficit. Reinforce or re-explain information provided by the physician.

在施用药物或其他治疗措施之前，确定预期结果和监测效果的方式。

Before administering drugs or other treatment measures, identify the expected outcomes and the manner in which the effects will be monitored. Inform the client of data indicating progress, such as loss of weight, lowering of blood pressure, and so forth. *Many clients fail to adhere to their treatment program*① because they have few, if any, symptoms and feel well. Stress that hypertension is a chronic condition that requires lifelong attention and treatment. The following points should be involved in the clients and family teaching planning:

— How to take the blood pressure.

— *Keep a log of blood pressure measurements for follow-up visits.*②

— Measures to reduce the risks from hypertension: annual physical examination; correction of obesity by eating a well-balanced and nutritious diet and engaging in a moderate program of physical exercise; moderation of salt intake; coping effectively with or avoiding problems or situations that produce stress whenever possible; improvement of general health habits by obtaining adequate sleep and rest.

除非事先与医生协商，否则切勿增加、减少或省略处方药

— *Follow the directions for medications*;③ never increase, decrease, or omit a prescribed drug unless first conferring with the physician.

— Report adverse effects from medications. 报告药物的不良反应

— Get medical approval before taking *nonprescription drugs*④.

— Inform all physicians and even the dentist of medications that are being taken.

否则避免烟草和含有咖啡因或酒精的饮料

— Avoid tobacco and beverages containing caffeine or alcohol, unless permitted by the physician.

Daily Language

1. Good afternoon, Paul. What is troubling you?

   下午好，Paul。有什么问题吗？

2. Have you ever had swollen ankles?

   您过去出现过踝关节肿大的现象吗？

3. Have you seen a doctor before?

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① 许多病人不能坚持治疗计划。
② 记录测量的血压以便追踪观察。
③ 遵照用药说明（书）。
④ 非处方药。

Unit 30 Diabetes Mellitus

1. Define diabetes mellitus;
2. Name the typical symptoms of diabetes mellitus;
3. List four complications of diabetes mellitus;
4. Plan a teaching program for the client with diabetes mellitus and his family.

多糖症,高血糖症

Diabetes mellitus is a metabolic disorder characterized by chronic hyperglycemia and caused by various pathogeny.① Two major groups of diabetes mellitus are *type* Ⅰ, *insulin-dependent diabetes*② and *type* Ⅱ, *non-insulin-dependent diabetes mellitus*③. A secondary form of diabetes mellitus may also be seen and is due to factors such as pregnancy, certain drugs, adrenal disorders (pheochromocytoma, *Cushing's disease*④), and certain pancreatic disorders, such as pancreatitis. Often, the signs and symptoms of diabetes are only temporary and disappear once the disorder is controlled or the cause eliminated. The three typical symptoms of diabetes mellitus are polyuria, polydipsia, and polyphagia. Additional symptoms include weight loss, weakness, thirst, fatigue, and dehydration. These signs and symptoms may *have an abrupt onset*⑤ in those with type Ⅰ diabetes. Those with type Ⅱ diabetes may *have a more gradual onset of symptoms*⑥. The complications of diabetes mellitus include diabetic ketoacidosis, hyperosmolar, *hyperglycemic nonketotic syndrome* (*HNKS*)⑦, hypoglycemia, vascular disturbances, and neuropathies. The treatment of diabetes mellitus may involve one or more of the following: diet, exercise, insulin, and weight control.

肾上腺

As diabetes mellitus is a chronic disease, one or more family members may take some or all of the responsibility of the *treatment regimen*⑧. To allow the client and family members time to understand material, nurses should plan a teaching program that presents material in small steps. Begin teaching by explaining diabetes: what it is, why treatment is necessary, and the various methods of treatment. Use audiovisual materials (video tapes, booklets,

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① 糖尿病是由多种病因引起、以慢性高血糖为特征的代谢紊乱。
② Ⅰ型糖尿病，即胰岛素依赖型糖尿病。
③ Ⅱ型糖尿病，即非胰岛素依赖型糖尿病。
④ 柯兴氏病。
⑤ 突然发作。
⑥ 逐渐出现的症状。
⑦ 高渗性非酮症性综合症，缩写为 HNKS。
⑧ 治疗方案。

drawings) to enhance learning. *Emphasize that the treatment of diabetes is highly individualized and that the treatment of one person cannot be compared with that of another.*① If the client requires insulin, use a chart to explain how to *rotate insulin injection sites*②. If the physician has recommended the use of a glucometer to monitor blood glucose levels, allow time for the client to use the glucometer and monitor his or her own blood glucose levels.

Develop a teaching plan to include one or more of the following general areas that apply to all diabetics:

— Signs and symptoms of hyperglycemia and hypoglycemia;

— The importance of weight reduction (if necessary);

— Methods of treatment of hypoglycemia: orange juice, two to three teaspoons of honey, hard candy, and glucose tablets;

— Problems that require contacting the physician: skin infections, pain in the extremities, visual problems, change in color or temperature of the skin of the extremities, frequent episodes of hypoglycemia, prolonged nausea and vomiting, and illness (especially a severe illness). Additional problems may be included, depending on the health of the client.

— Follow the exercise regimen suggested by the physician. During exercise, carry some food or other physician-approved form of glucose, which can be eaten if symptoms of hypoglycemia occur. (Many physicians recommend that diabetic clients always carry some form of easy-to-eat carbohydrate in case hypoglycemia occurs suddenly.) This is especially important for those taking insulin or those subject to episodes of hypoglycemia while taking an oral hypoglycemic agent.

— When using foods containing an artificial sweetener, the fat, protein, and carbohydrate in these foods still have to be counted within the framework of the diet.

— Read food labels because the sugar, fat, and protein contents must be counted in the diet. *Products labeled as "low calorie" and "dietetic" are not synonymous with "no sugar."*③ These products may contain sugar.

— Drink adequate amount of fluid especially in warm weather, when perspiring, or exercising.

— Take special care of the feet. Keep the toenails short and cut straight across. Bathe the feet daily and pat dry. *Dry the area between the toes thoroughly before putting on shoes, socks, or stockings.*④ Avoid tight fitting shoes. Never walk barefoot. Inspect the feet daily for cuts, bruises, corns, callouses, and swelling. Do not use commercial corn or callous removers or a razor or abrasive to remove corns or callouses. Consult a podiatrist if unable to

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① 强调糖尿病的治疗具有很大的个体差异，所以一个病人的治疗不能和另一个病人的治疗相比。
② 变换胰岛素注射部位。
③ 标有"低碳水化合物"和"食物疗法"的产品并不意味着"无糖"。
④ 在穿鞋、袜之前彻底擦干脚趾间缝隙。

cut the toenails correctly or if corns and callouses become a problem .

— Consult the physician regarding insulin or oral hypoglycemic agent dosage adjustments should illness occur .

Develop a teaching plan as the following for the diabetic requiring insulin for the control of diabetes:

— Strenuous exercise is only carried out with approval of the physician because the diet and the dosage of insulin may require major adjustments .

— *Do not switch brands of insulin without physician approval .*[①]

— Rotate insulin injection sites . Use a chart to record the site used until thoroughly familiar with a plan of rotation .

Develop a teaching plan as the following for the diabetic requiring an oral hypoglycemic agent for the control of diabetes:

— Do not increase or decrease the dose of the prescribed drug unless a dosage change is recommended by the physician .

— Take the drug at the prescribed time of day . If unable to take the drug because of nausea and vomiting, contact the physician .

— Tell your client that hypoglycemia can occur with the use of these agents .

Daily Language

1 . out-patient department 门诊部
in-patient department 住院部

2 . I've been feeling very thirsty and passing a lot of urine recently .
我近段时间常感到口渴，尿多。

3 . How is your appetite ?
您的胃口怎么样？

4 . Diabetes is a chronic disease . Right now, it can't be cured , but it can be controlled .
糖尿病是一种慢性疾病，目前不能治愈，但能控制。

5 . You should learn as much as you can about diabetes, have your urine tested regularly, stick to your diet, take your medicine on time, and have some regular exercise .
您应该多了解糖尿病的有关知识，定期进行尿检，坚持饮食疗法，按时服药，进行有规律的体育锻炼。

5 . You should see the doctor if you have skin infections, pain in the extremities, visual problems, change in color or temperature of the skin of the extremities, frequent episodes of hypoglycemia, or prolonged nausea and vomiting .

① 没有医生的许可，不能改变胰岛素的类型。

Unit 31 Hepatitis

1. State the transmission routes of viral hepatitis A, B, C and D.
2. Identify the main signs and symptoms of viral hepatitis.
3. Identify the prevention measurements for hepatitis A and B.

Hepatitis is an inflammation of the liver. It is caused by viruses, bacteria, alcohol, *drug abuse*①, some medicines, or serious harm to the liver. Hepatitis is a very serious health problem in parts of Asia, Africa, and the Caribbean. There are several different viruses which can cause viral hepatitis. They are hepatitis A, B, C, D, and E viruses. All of these viruses can cause acute, or short-term, viral hepatitis. Hepatitis B, C, and D viruses can also cause chronic hepatitis, in which the infection is prolonged, sometimes lifelong. *Other viruses may also cause hepatitis, but they have yet to be discovered and they are obviously rare causes of the disease.*②

People get viral hepatitis in many ways. Usually, people get hepatitis A or hepatitis E by eating food and drinking water infected with hepatitis A virus (HAV) or hepatitis E virus (HEV). People get hepatitis B or C by having unprotected sex (having sex without using a condom) with someone who has hepatitis B or C, being transfused with blood that is infected with the virus, sharing needles with someone infected with the virus, being accidentally stuck by an infected needle or using tools that are infected with the virus for tattoos and body piercing, and a mother passing the virus to her child *during the birthing process*③. People get hepatitis D only if they already have hepatitis B.

Hepatitis, in its early stages, may cause *flu-like symptoms*④. These symptoms may include malaise (a general ill feeling), fever, muscle aches, loss of appetite, nausea, vomiting, diarrhea, and jaundice. Some people may have no symptoms at all and may not even know they're infected. *If hepatitis progresses, its symptoms begin to point to the liver as the source of illness.*⑤ Chemicals normally secreted by the liver begin to build up in the blood. This causes jaundice (a yellowing of the skin and the eyes), foul breath, and a bitter

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① 药物滥用。
② 其他的一些病毒也可能引起肝炎，但至今还没被完全弄清楚，但显然这些病毒引起的肝炎较少见。
③ 在分娩过程中。
④ 流感样的症状。
⑤ 如果肝炎进一步发展，它的症状以肝脏（病灶）为中心。

taste in the mouth. Urine turns dark or "tea-colored," and stools become white, light, or "clay-colored." *There can also be abdominal pain, which may be centered below the right ribs (over a tender, swollen liver) or below the left ribs (over a tender spleen).*① Viral hepatitis symptoms are similar no matter which type of hepatitis a person has. Very rarely, a new case (acute) of viral hepatitis can cause *liver failure*② and death. It is important to know that symptoms are less common in children than adults, and people who have HCV infection are less likely to experience symptoms.

All of hepatitis patients should avoid alcohol, as it can worsen liver disease. There is no treatment for hepatitis A. Hepatitis A usually *resolves on its own*③ over several weeks. Acute hepatitis B usually resolves on its own, but very severe cases can be treated with lamivudine. HBV-infected persons should have a medical evaluation for liver disease every 6～12 months. Alpha-interferon, lamivudine, and adefovir are the three drugs currently licensed for the treatment of persons with chronic hepatitis B. These drugs are effective in up to 40% of patients. *Liver transplant*④ is the last resort, but livers are not always available. HCV-positive persons should also have a medical evaluation for liver disease every 6～12 months. Interferon, pegylated interferon and *ribavirin*⑤ are the only drugs licensed for the treatment of persons with chronic hepatitis C. Interferon can be taken alone or in combination with ribavirin, and *combination therapy is currently the treatment of choice*.⑥ Chronic hepatitis D can be treated with alpha-interferon. Hepatitis E usually resolves on its own over several weeks to months.

Hepatitis A vaccine is the best protection for hepatitis A. It is recommended for people more than two years of age who are in risk groups for HAV infection. It is recommended as a routine vaccination for children living in certain geographic areas where hepatitis A occurs at consistently higher rates than average. For a recent exposure to someone with HAV or if travel is imminent (leaving in less than 4 weeks) to an area of the world where hepatitis A is common, people should see doctor about his need for a dose of immune globulin (IG). A good way to prevent hepatitis A is washing hands with soap and water after using the toilet, changing a diaper, and before preparing and eating food. Hepatitis B vaccine is the best protection for hepatitis B. Routine vaccination is recommended for all persons 0～18 years of age, and for persons of all ages who are in risk groups for HBV infection. For optimal protection, all babies should be given their first dose of hepatitis B vaccine at birth before leaving the hospital. A pregnant woman should be tested for hepatitis B. *Infants born to*

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① 也有可能出现腹痛，腹痛以右肋下（右肋下可触到质地柔软肿大的肝脏）或左肋下（左肋下可触到质地柔软的脾脏）为中心。
② 肝衰竭。
③ 自愈。
④ 肝移植。
⑤ 利巴韦林（抗病毒药）。
⑥ 联合用药是目前的治疗方法之一。

*HBV-infected mothers should be given HBIG (hepatitis B immune globulin) and vaccine within 12 hours of birth.*① There is no vaccine to prevent hepatitis C. Many other ways to help one prevent hepatitis B and C are as follows: Don't share personal care items, such as razors, toothbrushes, and washcloth. Consider the risks before getting a tattoo or body piercing. Avoid being stuck by sharps and needles. Use condoms properly when having sex with someone who has hepatitis B or C. As for health care workers, it is important for them to stick to routine barrier precautions and safely handle needles and other sharps. In addition, they should be vaccinated against hepatitis B.

Daily Language

1. You'd better have a liver function test.
   您最好做一个肝功检查。
2. Please come to our hospital tomorrow morning before breakfast with an empty stomach to have a blood test. Don't eat or drink anything.
   请明天早上空腹来医院查血。记住什么都不要吃。
3. Your liver is enlarged and tender to the touch.
   您的肝脏肿大，有触痛感。
4. Take the medicine I prescribe. You can't drink any alcohol and should have a good rest. There is no special treatment. It will take you some time to recover, so you have to be patient.
   按我开的处方服药。您不能喝酒，注意休息。该病没有什么特殊的治疗方法。您需要一段时间才能康复，所以您要有耐心。

Vocabulary

1. hepatitis [ˌhepəˈtaɪtɪs] *n*. 肝炎
2. condom [ˈkɒndəm] *n*. 避孕套
3. tattoo [təˈtuː, tæˈtuː] *v*. 刺花样 *n*. 文身
4. malaise [mæˈleɪz] *n*. 不舒服
5. jaundice [ˈdʒɔːndɪs] *n*. 黄疸 *vt*. 使患黄疸
6. stool [stuːl] *n*. 肠内排出的粪便
7. interferon [ˌɪntəˈfɪərɒn] *n*. 干扰素
   alpha-interferon α-干扰素
   pegylated interferon 聚二醇化干扰素

① 感染了乙型肝炎病毒的母亲所生育的婴儿应当在出生12小时内注射乙肝免疫球蛋白和疫苗。

Unit 32 Care of the Client with Surgery

1. Briefly state the category and purposes of the surgery;
2. Describe the nurse's tasks during the 3 phases of surgery;
3. State the goal of the recovery room nurse;
4. Describe the main nursing care during postoperative phase.

手术是在手术过程中使用仪器治疗伤害，疾病和畸形

Surgery is the use of instruments during an operation to treat injuries, diseases, and deformities. *It is usually invasive and may require an opening being made into the body for accessing the area to be treated.*① Surgery is performed in clinics, physician's offices,

今天，由于药物和麻醉剂可以更快地恢复，手术是一种安全、有效的治疗选择

ambulatory surgical centers, and hospitals. Today, surgery is a safe, effective treatment option due to medication and anesthetics that allow a quicker recovery. *Laser and scope technology have led to the development of new procedures that offer less risk, less invasive, faster recovery, and reduced hospitalization.*②

Surgery is scheduled based on the urgency required for a successful outcome for the client. *Emergent or immediate surgery is needed when life or limb is suddenly threatened and any*

阑尾破裂或外伤性截肢

*delay in surgery would jeopardize the client's life or limb.*③ Examples of this are ruptured appendix, or traumatic limb amputation. Urgent surgery is the need for an operation within 24 to 30 hours. Example for this is fracture repair. *An elective surgery is one that can be planned and scheduled without any immediate time constraints.*④ Examples for this are joint replacement and hernia repair. *Optional surgery*⑤, such as cosmetic surgery, is done at the request of the client. Anyway, surgery is done for several reasons. *The purposes of surgery are to cure pathologies by removing abnormal tissues or secretion, and repairing damaged structures or inborn defects, to diagnose, to slow the progress of pathologic process, and to relieve the undesirable symptoms.*⑥ The surgical process threatens the client's ability to meet basic needs. All body systems are affected and all needs, physical, physiologic, and

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① 外科手术通常是侵入性的，指为了接近需治疗的病变区域而将其切开。
② 激光和腔镜技术引导了新的手术方式的发展，降低了手术的危险性，减少了侵入性，加速了康复并减少了住院时间。
③ 当生命或肢体突然受到威胁，而任何耽误都可能危及病人的生命和肢体时就需要紧急手术。
④ 择期手术是指那些没有紧急时间限制，可以按照计划预期安排的手术。
⑤ 自主选择性手术。
⑥ 手术的目的是：通过清除病变组织或分泌物、修复被破坏的或先天缺陷的组织结构来治疗疾病，通过手术诊断疾病、阻止疾病的恶化进程以及减轻不舒适症状。

psychosocial, are compromised. A wide variety of factors influence the client's response to the surgical process. These influences include the client's state of health prior to the surgery, level of growth and development, cognitive abilities, level of knowledge, level of anxiety, cultural and societal expectations, the meaning of the lost body part, and previous experiences with hospitalization and surgery. *The nurse, through understanding the process and appropriate intervention, can do much to promote, restore, and maintain the client's high-level wellness in the face of these threats.*①

There are three phases in the surgical process: preoperative, intraoperative, and postoperative. *These three phases are referred to collectively by the word perioperative, which is defined as the time surrounding and during surgery.*② The preoperative phase starts when the client is apprised of the need for surgery and ends when anesthesia is induced. The intraoperative phase begins with the induction of anesthesia and finishes when the surgical procedure is completed. The postoperative phase covers the time from when the surgery ends and continues until the healing process is completed.

In the preoperative phase, the client undergoes a series of diagnostic tests and physical examinations to ascertain his or her present state of health. The nurse obtains a nursing history to determine the client's nursing care needs and to identify actual or potential negative health problems. Once the nursing history has been obtained and negative health behaviors have been identified, the nurse can plan to intervene. All clients must sign an informed consent before any surgical process. If the client is under 18 years of age or incapable of signing the form, the closest relative does so. *This consent not only gives the surgeon permission to perform surgery but also states that the client has been informed of exactly what procedure is being done, its outcomes, and the possible complications.*③

Preoperative teaching serves to help relieve the fears and anxieties of the client as well as provides the client with the basic skills needed during the postoperative period. *General areas of preoperative teaching include methods and rationale for coughing and deep breathing, incision splinting, turning, ambulating, and exercising.*④ Some preparations are carried out the day before surgery, such as laxatives or enemas, and skin shaves and cleanness. The night before surgery is often especially stressful for the client. Sleeping medications are usually ordered the night before surgery to help promote sleep.

Final safety and preparatory checks are carried out on the day of surgery. *Special preoperative checklists are usually available to have a written record of the preparations that*

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① 当病人面临这些威胁时，护士可以通过对病变过程的掌握和提供恰当的护理措施来促进、恢复和维护病人的最佳健康状况。
② “围手术期”一词是这三个阶段的总称，指围绕手术前后和手术期中的整个阶段。
③ 手术同意书不仅表明对外科医生执行手术的许可，而且声明该病人已经被告知手术方法、预期结果以及可能出现的并发症。
④ 一般的术前健康教育内容包括咳嗽与深呼吸、切口的保护、翻身、下床活动以及功能锻炼的方法与原理。

are needed and the time they are done.① Preoperative medications are given approximately 30 minutes before the scheduled time of the surgery. These drugs reduce the chance of aspiration of secretions during and immediately after surgery and enhance the induction of the desired level of anesthesia.

The intraoperative phase begins when the client enters the surgical suite. *The surgical team usually consists of the surgeon, the surgeon's assistants, the anesthetist, the scrub nurse, and the circulation nurse*.② The operating room nurses are in the unique position of assessing the client's immediate physical, physiological, and psychosocial responses to operation process and communicating this information to those who care for the client postoperatively. Thus the operating room nurse acts as the bridge for the continuity of care. In addition, operating room nurses perform supervisory, managerial tasks and coordination of ancillary staff, as well as carry out the technical aspects of nursing care.

When the surgery is completed, the incision is appropriately dressed and the client is transferred to the postoperative recovery room. Records of the client's vital signs during surgery, medications received, blood transfused, and any other immediately pertinent information are transferred with the client. They provide data for comparison in the recovery room. *The goal of the recovery room nurse is to maintain adequacy of all vital functions, to prevent early complications of surgery, and to relieve discomfort*.③ The client remains in the recovery room until he or she has completely reacted from the anesthesia and vital signs are stable. And then, the client is transferred to his or her ward.

In the early postoperative period, the nurse assesses the passage and function of all drainage tubes and adjusts them as necessary. The client's levels of orientation, consciousness and feelings are assessed. *Because the client may still be very groggy, special measures must be taken to ensure safety*.④ Maybe the side rails are raised, the bed is in the lower position, and the call bell is available to the person. The nurse puts the client in the comfortable position and helps him or her change the position intermittently. It is common for clients to experience pain after surgery, so pain control should be concerned all the time. In order to decrease the client's anxiety or fear, any questions the client asks should be answered and the family members should be allowed to see the client.

As the initial postoperative period ends, the client begins to meet more of his or her own health care needs. Nursing actions are those that will help to maximize the client's recovery and assist the client to attain the highest level of well being within the usual life setting. *Physiologic care is still a focus in this phase, and it requires a great deal of energy as the*

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① 通常要列出专门的术前准备清单，上面记录需要完成的术前准备以及完成时间。
② 外科手术组成员通常包括外科医生及其助手、麻醉师、洗手护士和巡回护士。
③ 复苏室护士的护理目标是维持（病人）重要器官的功能、预防术后早期并发症、减轻不舒适感。
④ 因为病人可能仍然处于昏昏欲睡的状态，这时需要采用特殊措施来保证病人的安全。

healing process continues.① Protein, vitamin, and calorie intakes are increased to meet the energy needs of the healing process. The nurse can assist in this area by suggesting foods that will promote elimination, and by teaching the client the relationship between exercise and elimination. *Ambulation is steadily progressed to increase muscle strength and endurance as well as to promote healing*.② The nurse needs to continue careful observation of the wound, since many wound disruptions can occur at this time. Sutures are usually removed around 7 to 10 days after surgery, depending on the amount and type of tissues involved and the progression of healing.

Many new health behaviors may have to be learned during the postoperative period. Although the nurse has been teaching all along, this is when the major emphasis will be placed on the teaching-learning process. *Self-care skills, changes in habits, and modification of environment and lifestyle are some of the behaviors with which the client may require help*.③ The family should be included in this teaching-learning activity so that they too are aware of the client's capabilities and limitations and can assist the client in meeting health care needs. Many resources exist within the community to assist the client in the latter part of the postoperative period. *The nurse helps the client to make the plan for resuming activities of the daily living and contacts the community nurse in order to keep continuity of nursing care*.④

Daily Language

1. Would you sign this informed consent, please?
   请你签署这份手术同意书。
2. I will give you skin shave. Please take it easy.
   我要给你备皮，请放松。
3. Can you remove my sutures in five days?
   你可以在这 5 天内给我拆线吗？
4. The client's arm is in rupture, and she needs an emergency surgery.
   这个病人的手臂皮肤破损，需要做紧急手术。
5. Please take a deep breath.
   请做深呼吸。
6. You should take ambulation as early as possible.
   你应该尽早下床活动。
7. I can not sleep. May I have some sleeping pills?

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① 生理方面的护理依然是这个阶段的重点，在病人愈合过程中需要大量的能量。
② 逐渐进行下床活动，以增强肌肉的力量和耐力并促进康复。
③ 病人的某些行为，如自护技能、习惯的改变、环境和生活方式的改善需要别人的帮助来完成。
④ 护士要帮助病人制定出恢复日常生活行为的计划，并与社区护士联系以保持护理照顾的连续性。

Unit 33 Main Changes and Care of Woman during Pregnancy

1. State the main physical and psychological changes of woman during pregnancy;
2. Describe the abnormal signs that need immediate medical care.

Pregnancy brings both psychological and physical changes to the woman. The physiologic changes of pregnancy occur gradually but eventually affect all organs and systems of the woman's body. The changes occur in order for the woman to provide oxygen and nutrients for the growing fetus as well as extra nutrients for her own increased metabolism during the pregnancy. The changes of pregnant woman prepare her body for labor and birth and for lactation if she chooses to breast-feed once the baby is born. Although physical changes that occur with pregnancy are extensive, they are also temporary. When pregnancy ends, the woman's body returns nearly to its pre-pregnant state. A woman's psychological response to pregnancy changes with time. The families must *adapt to a new role*① in the family structure and deal with feelings that may result from the birth of infant. *Many pregnant women rely on nurses to provide accurate information and guidance throughout their pregnancy*.② To respond effectively, nurses must understand not only the physiologic changes but also the ways these changes affect the daily lives of *expectant mothers*③.

The major physical changes during pregnancy take place in *reproductive system*④. Uterine changes are the most obvious. The size of the uterus increases to accommodate the growing fetus. Over the nine-month pregnancy, the uterus increases in length from approximately 6.5 cm to 32 cm; in depth, from 2.5 to 22 cm; and in width, from 4 to 24 cm. Its weight increases from 50 g to 1,000 g. At the beginning of pregnancy, the uterine wall is about 1cm thick, and its cavity is barely large enough to hold 2 ml water. As pregnancy begins, the wall first hypertrophies and thickens to about 2 cm. Then it begins to turn thinner, so that by the end of pregnancy it is quite supple and only about 0.5 cm thick.

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① 适应新的角色。
② 许多孕妇依赖护理人员提供准确的信息，并进行整个孕期的指导。
③ 准妈妈。
④ 生殖系统。

*By term*①, the increase in the size of the uterus is so extensive that it can hold a 3, 000 g fetus plus 1, 000 ml of *amniotic fluid*②, or a total of about 4, 000 g.

The psychological changes of pregnant woman are different during three trimesters. During the first trimester, some women are surprised to find themselves pregnant and experience less pleasure and closer to disappointment or anxiety at the news. Fortunately, most women are able to change their attitude toward pregnancy by the time they feel the fetus move inside them. During the second trimester, the woman feels the first *fetal movement*③. *The fetal movement proves the existence of the child and the mother may think of the life inside her as an integral part of herself rather than as a separate entity*.④ She knows it is there; she eats to meet its needs and takes special vitamins to help it grow. She begins to imagine herself as a mother, perhaps teaching her baby the alphabet or how to ride a bicycle. At this time, the mother has accepted the baby. During the third trimester, the woman usually begins "*nest-building*" *activities*⑤, such as planning the infant蒂 sleeping arrangements, buying clothes, choosing names for the infant, and "*ensuring safe passage*"⑥ by learning about birth.

*Prenatal care*⑦ is essential for assuring the overall health of babies and their mothers. It is a major strategy to help reduce the number of *low-birth-weight babies*⑧ born yearly. During the first visit, an extensive health history and a complete physical examination, including a *pelvic examination*⑨, and blood and urine specimens for laboratory work are obtained. The nurse should tell the pregnant woman the dangerous signs during pregnancy. The dangerous signs to be noticed are vaginal bleeding, persistent vomiting, chills, fever, sudden escape of fluid from vagina, abdominal or chest pain, swelling of face or fingers, vision changes, diplopia, dimness or blurring of vision, severe and continuous headache, and increase or decrease in intensity or frequency of fetal movements. The nurse should also teach the mother how to count the number of the fetal movements. One popular way to approach this assessment is to ask the mother to *lie in a left recumbent position*瘢 after a meal and record how many fetal movements she feels over the next hour. Fetal movements can be felt by the mother at 18 to 20 weeks of pregnancy and reach a peak at 29 to 38 weeks. Normally, a fetus moves 3 to 5 times an hour. Presence of a fetal heart can be demonstrated by hearing its sound or seeing its beating by an ultrasound examination. The fetal heart is audible by auscultation of the abdomen with an ordinary stethoscope only at about 18 to 20 weeks of

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① 足月。
② 羊水。
③ 胎动。
④ 胎动证明了胎儿的存在，这时母亲可能将体内的胎儿当作自己身体的一部分，而不是一个独立的个体。
⑤ 筑巢活动。
⑥ 确保安全分娩。
⑦ 产前检查。
⑧ 低出生体重儿。
⑨ 骨盆检查。
瘢 左侧斜靠位。

pregnancy and it can be heard and counted as early as the 11th week of pregnancy by the use of an *ultrasonic Doppler technique*① . They are best heard when the position of the fetus is determined by palpation and the stethoscope placed over the area of the fetus' back . A fetal heart rate usually ranges between 120 and 160 beats per minute .

Nutrition, rest, and hygiene are very important to pregnant women . A pregnant woman who eats well and takes care of her own health will provide a healthy environment for the fetus to develop and grow . The nutritional state pre-pregnancy and during pregnancy has a direct effect on her health as well as on fetal growth and development . But the expectant mother may need instructions . As a result, nurses play a major role in education for promoting maternal and fetal health .

Daily Language

1 . Please lie on your back . I am going to listen to the fetal heart rate .
请仰卧。我要听一下胎心率。

2 . The normal range of fetal heart rate is from 120 to 160 beats per minute .
胎心率的正常范围是每分钟 120 至 160 次。

3 . You'd better take no medicine without physician's approval .
没有医生同意，您不能随便服药。

4 . If you have any abnormal signs such as increasing vaginal discharge, vaginal bleeding, abdominal or chest pain, severe headache, swelling of face or fingers, vision changes, increase or decrease in intensity or frequency of fetal movements, be sure to see a doctor as soon as possible .
如果有阴道分泌物增加、阴道出血、腹痛、胸痛、严重头痛、脸或手指肿胀、视觉改变，以及胎动强度或次数增加或减少等异常情况，一定要赶快来医院检查。

Vocabulary

1 . metabolism [mə'tæbəlizəm] *n* . 新陈代谢
2 . labor ['leibə] *n* . 分娩，阵痛
3 . lactation [læk'teiʃən] *n* . 哺乳
4 . uterus ['juːtərəs] *n* . 子宫
5 . accommodate [ə'kɔmədeit] *v* . 适应，容纳
6 . hypertrophy [hai'pəːtrəfi] *n* . 肥大，过度生长，过度增大
7 . supple ['sʌpl] *adj* . 柔软的，柔和的

① 超声波多普勒技术。

Unit 34 Main Changes and Care of Woman during Postpartum

1. Define the terms: puerperium and uterine involution;
2. State the changes of uterus during postpartum;
3. Describe the three phases of psychological adaptations during postpartum;
4. State the goals of nursing care during postpartum.

*Postpartum period, or puerperium, refers to the time between placenta delivering until all organs have returned to their pre-pregnant state.*① It refers to a 6-week period following childbirth. During this time, the mother experiences numerous physiologic and psychosocial changes. The physical postpartum care a woman receives can influence her health for the rest of her life. The emotional support she receives can influence the emotional health of her child and family and can extend into next generation.

The *uterine involution*② is a major change in postpartum. Uterine involution is the process whereby the reproductive organs return to their non-pregnant state. Healing at the placental site occurs more slowly and requires approximately 6 to 7 weeks. Immediately after birth, the uterus weighs about 1,000 g. At the end of the first week, it weighs 500 g. By the time involution is complete, it will weigh approximately 50 g, its pre-pregnant weight.

The fundus assessment is very important for uterine involution. *The bladder should be empty for accurate fundal assessment because a full bladder displaces the uterus and raises the height of the fundus.*③ *Press in and downward with one hand on the umbilicus until you are against a firm globular mass in the abdomen: the uterine fundus.*④ For the first hour after birth, the height of the fundus is at the umbilicus or even slightly above it. Then it decreases about 1 cm or 1 *finger breadth*⑤ per day. By 10 days postpartum, the fundus can be palpated abdominally. *The consistency of the fundus should be firm, with a round,*

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① 产褥期指胎盘娩出到全身各器官恢复到未怀孕前状态的这段时间。
② 子宫复旧。
③ 为了准确评估子宫底的高度，应排空膀胱，因为充满尿液的膀胱将推移膀胱，使子宫底升高。
④ 一手放在脐部轻轻按压子宫直到在腹部扪及一圆、硬的包块，这便是子宫底。
⑤ 一指宽。

smooth shape.① A soft or boggy fundus indicates atony. Lochia is a very important sign in uterine involution. Assessment of the lochia includes the amount, color, and odor. Lochia is dark red (*lochia rubra*②) in the first days after delivery and is usually moderate in amount. About the fourth day, it becomes more serous and pink (*lochia serosa*③) with a decrease in flow. A week to 10 days later, lochia becomes yellowish-white (*lochia alba*④) with scant flow. Lochia alba persists until 3 weeks postpartum and indicates normal progression of healing. The recurrence of fresh, red bleeding after lochia has progressed to serosa or alba may indicate infection or delayed hemorrhage. A foul odor suggests endometrial infection.

There are three phases of psychosocial adaptations in postpartum. In the early 1960s, Rubin identified *restorative phases*⑤ that the mother must go through to replenish the energy lost during labor and attain comfort in the role of mother. *The puerperal phases are called taking-in, taking-hold and letting-go*.⑥ During taking-in phase, the mother focuses primarily on her own basic needs for fluid, food, deep and restorative sleep. She also takes in every detail of the neonate, but seems content to allow others to make decisions. This phase lasts for approximately 2 ~ 3 days. In taking-hold phase, the mother becomes more independent. She exhibits concern about managing her own body function and assumes responsibility for her own self-care. When she feels more comfortable and in control of her body, she shifts her attention from her own needs to the performance of the infant. She compares her infant with other infants. The mother becomes strongly interested in caring for her baby. The taking-hold phase extends over several days. During this phase, the mother may express a great deal of anxiety about her competence as a mother. She may compare her care-taking skills unfavorably with those of the nurse. The third phase is called letting-go. The mother finally redefines her new role. She gives up her previous childless role and acknowledges the loss of a carefree lifestyle. Also, she gives up the fantasized image of her child and accepts the real one. In addition, some mothers may be disappointed at the size, gender, and characteristics of the infant who does not "match up" with the fantasy baby of pregnancy. For them, the feelings of loss and grief may be acute.

Care planning in postpartum should include enough time for health education. All the interventions in the postpartum should be family-centered so that the family will be drawn as closely together as possible during this important period of *parent-child bonding*⑦. Interventions should also be intended to increase the woman's self-esteem and help her to view herself as a new mother and the infant as part of her family.

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① 子宫底应该是硬、圆而光滑的。
② 血性恶露。
③ 浆性恶露。
④ 白色恶露。
⑤ 恢复阶段。
⑥ 产褥期心理变化将经历依赖期、依赖 - 独立期和独立期。
⑦ 父母与孩子的情感连结。

Shortly after the delivery, the woman may express a desire for something to eat or drink. *The postpartum diet should provide balanced nutrition with enough calories to supply the additional requirements for lactation, if the woman is to breast-feed her newborn.*① The mother usually has a good appetite and becomes hungry between meals, especially when she is breast-feeding. Between-meal snack, including milk or milk products, will help to supply the mother with the additional milk requirements. An increased intake of fluid is also essential for her.

During the puerperium, the mother needs adequate rest and should be encouraged to relax and sleep whenever possible. *Group assessments and care*②, and try to correlate them with time when the mother would be awake, such as the time before or after meals, infant feeding time, and visiting hours. If the room is shared, providing care for both women at the same time can reduce activities that interrupt sleep. A quiet, softly lit environment can also promote sleep.

It is important that the nurse explain the purpose and value of early ambulation to the mother or other decision makers. Activities should be gradually increased according to the mother蒂 strength. Early ambulation promotes circulation, reduces the risk of thrombophlebitis and prevents constipation. *Bladder and bowl functions can be improved by ambulation, reducing the need for catheterization and decreasing abdomen distention and constipation.*③

Taking a shower is refreshing and promotes hygiene. A mother with no complications is allowed to shower within a few hours of delivery. At the first time the mother takes a shower, the nurse should remain nearby for safety. At the same time, the nurse can provide self-care instructions for bathing and breast care.

Daily Language

1. You should try to get up from your bed and move about, because early ambulation can promote the blood circulation, improve bladder and bowl functions, and prevent constipation.
   你应该尽早从床上起来活动一下。因为产后活动可促进血液循环，有助于大小便功能的恢复，同时预防便秘。
2. You may have a rest or sleep whenever you feel tired and sleepy. Rest and sleep is important to your recovery.
   如果你觉得疲倦或困了，你就休息或睡觉。休息和睡眠对于你身体的康复很重要。

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① 对准备母乳喂养的产妇，产后必须提供足够热能的平衡饮食，以满足哺乳的需要。
② 集中评估和护理。
③ 产后活动能促进大小便功能的恢复，降低插管的需求，减少尿潴留和便秘的发生。

Unit 35 Breast-Feeding

1. State the nutritional traits of human milk;
2. Interpret the advantages of breast-feeding;
3. Instruct mother and family how to breast-feed.

Human milk is the most appropriate natural nutrient for the infant. *Breast-feeding*① is one of the methods of feeding infants. The first breast-feeding should be undertaken as soon as possible after delivery. The lactation of a healthy mother usually meets the infant's needs in the first 6 months of age.

Components of human milk

There are hundreds of components in human milk. *However, human milk does not have a constant component, and significant changes occur during the course of lactation, diurnally and during each feed.*② According to the course of lactation, World Health Organization divides human milk into 4 types, which are *colostrum*③ (4 days of postnatal), *transitional milk*④ (5～10 days of postnatal), *mature milk*⑤ (11 day～9 months), and *late milk*⑥ (after 10 months).

Protein: The figure of the protein content in human milk is 12 g/L compared to 35 g/L in cow's milk. The two main classes of protein are whey and casein. Human milk is whey predominant (about 60% of total protein). *In human whey, α-lactalbumin is the dominant protein followed by lactoferrin.*⑦ The ratio of casein to lactalbumin in human milk is 4:6, different from that in cow's milk (4:1). *Human milk casein yields a softer, more flocculent curd than cow's milk at low pH, which is more helpful for infant to digest.*⑧ The essential amino acid content is high in human milk. During early lactation milk protein is much higher than that in mature milk.

Fat: The fat content of milk varies from different mothers and during the course of

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① 母乳喂养。
② 母乳的成分并不是不变的，它随着乳汁分泌过程、昼夜交替和每一次喂哺而变化。
③ 初乳。
④ 过渡乳。
⑤ 成熟乳。
⑥ 晚乳。
⑦ 在人乳清中，以α-乳清蛋白含量高，其次为乳铁蛋白。
⑧ 在酸性环境下，人乳中酪蛋白凝块较牛乳中的软，多为絮状，更利于婴儿消化。

lactation. The fat concentration rises markedly from around 21 g/L to 41 g/L during a feed. Human milk has a higher proportion of *unsaturated fatty acids*① and a greater concentration of the *essential fatty acids*② than cow's milk. *Long-chain fatty acids*③ in human milk are esterified to glycerol, which improves their absorption. The property of human milk fat is markedly influenced by diet. Vegetarians, for instance, consume more long-chain fatty acids than those on a mixed diet.

Carbohydrate: Higher concentration of lactose is present in human milk (70 g/L) than in cow's milk (47 g/L). *Lactose enhances calcium absorption from the gut, promotes the growth of lactobacilli, and helps to create a favorable gut flora that protects against gastroenteritis.*④

Minerals: All major minerals are present in lower concentrations in human milk than in cow's milk, which reduces infant's *renal solute load*⑤. The appropriate phosphate/calcium ratio in human milk is beneficial to absorption. The absorption rate of iron in human milk is about 50%, although the concentration is very low. There is good bioavailability of zinc from human milk.

Vitamins: There are plenty of vitamin A, C, D and E in human milk, especially in colostrum. However, low vitamin K concentration (15 mg/L) in human milk may contribute to vitamin K deficiency of the infant *exclusively breast-fed*⑥.

Immune components: There are many immune components in human milk. *The concentration of secretory IgA is very high in human milk, especially in colostrums.*⑦ Secretory IgA is comparatively resistant to low pH and proteolytic enzymes, and protects infant from infection. There are other immune components present in human milk such as lactoferrin, lysozyme, and other antimicrobial factors.

Enzymes: Several kinds of enzymes are present in human milk such as lipase, amylase, and several proteases.

Advantages of breast-feeding

Advantages of breast-feeding are as follows:

Feeding infant easily: Breast milk is always readily available at the proper temperature and needs no time for preparation. It is also economical and healthy.

Reducing infant morbidity and mortality: *Human milk is fresh and free of contaminating bacteria, which reduces the chances of gastrointestinal disorders.*⑧ Breast-fed infants have less vulnerability for diarrhea because of the appropriate nutritive contents and

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① 不饱和脂肪酸。
② 必需脂肪酸。
③ 长链脂肪酸。
④ 乳糖可增加肠道钙的吸收，促进乳酸杆菌的生长，并有助于建立肠道正常菌群，防止胃肠炎。
⑤ 肾脏的溶质负荷。
⑥ 纯母乳喂养。
⑦ 母乳尤其是初乳中分泌型 IgA 浓度非常高。
⑧ 母乳新鲜，未受细菌污染，可减少胃肠道疾病。

proportions and much more immune factors in breast milk . Allergy and intolerance to cow蒜 milk may create significant disturbances such as diarrhea, intestinal bleeding, atopic eczema . *Low incidence of pneumonia as well as certain allergic and chronic diseases in the later life has been reported in breast-fed infants .*①

Meeting infant蒜 nutritional needs: Plenty of nutrients and their appropriate ratios in breast milk are helpful for infant to digest and absorb .

Promoting mother-infant attachment: Successful breast-feeding is a satisfying experience for both mother and infant . By feeding her baby, the mother may gain both a feeling of being essential and a sense of accomplishment . Moreover, the infant has a close and comfortable relationship with the mother . Breast-feeding offers increased opportunity for close contact between the mother and the infant .

Physical benefits for the mother: Breast-feeding may promote the resumption of uterus, inhibit ovulation, and reduce morbidity of breast cancer and ovary cancer .

Dialogue

1 . How often should I feed my baby ?
我应该间隔多长时间给孩子喂一次奶？

2 . There is no exact time . We encourage feeding the baby once he wakes up or looks hungry .
没有确定的时间规定。我们鼓励一旦孩子醒了或看起来饿了的时候就喂奶。

3 . Place a low stool under your foot to raise your knee . Then, hold the baby comfortably with your arm and let the baby face the breast . Use the other hand to support the breast .
在你的脚下放一张矮凳子使膝盖抬高。然后用你的胳膊把孩子抱舒服，让孩子的脸正对乳房，用另一只手托着乳房。

4 . After feeding the baby, put him upright, have him lean against your shoulder and burp him .
哺乳完毕，将婴儿竖抱，让他伏在你的肩上，轻轻拍拍他的背部直到他打嗝。

Vocabulary

1 . lactation [l ˈkeɪʃn] *n* . 泌乳，泌乳期，哺乳（期），授乳（期）

2 . diurnal [daɪˈɜːnl] *adj* . 每日的，一日间的，白天的，昼间的
diurnally *adv* .

① 据报道，母乳喂养的婴儿日后肺炎、过敏性疾病和慢性病的发生率低。

Unit 36 Contraception

1. List the common methods of contraception;
2. List the advantages and disadvantages of each of the methods of contraception.

There are a number of different methods of contraception. Each has its own advantages and disadvantages. *All forms of contraception work by preventing the fertilization of a woman's egg by a man's sperm*①, which can be achieved in various ways.

The male condom is really just a rubber tube. It covers penis and stops sperm from going into vagina. An advantage of using male condom is that it helps to protect against pregnancy as well as STDs such as AIDS.

Spermicides are chemical agents that stop sperm from traveling up into the cervix. *Spermicide comes in different forms including the sponge, vaginal pessaries, and foam.*② Spermicides are not very effective against pregnancy when used alone, but they can be used effectively with the male condom. The male condom and spermicides, when used together, is a good combination for providing effective protection against both pregnancy and STDs.

Contraceptive pills contain chemicals called hormones. One type of pill contains two hormones: oestrogen and progestogen. *The pill stops the release of an egg every month but doesn't stop periods.*③ The other type of pill contains only progestogen. *It works by altering the mucous lining of the vagina to make it thicker.*④ The sperm cannot then get through, and as a result, the girl can't get pregnant. Usually the girl has to take one pill every day for about three weeks every month. It is very important for the girl not to forget to take the pills. The disadvantage of the pill is that it does not provide any protection against STDs. For very good protection against both pregnancy and STDs, the contraceptive pills should be used together with the male condom.

*Depo-Provera*⑤ is the most popular form of *injectable hormonal contraception*⑥, which involves the girl having an injection once every twelve weeks. The injection contains the

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① 所有的避孕方法都是通过阻止卵细胞受孕而发挥作用的。
② 杀精剂有海绵状、阴道栓剂和泡沫状等不同的形状。
③ 口服避孕药能阻止卵细胞的释放，但却不会干扰正常的月经。
④ 它通过使阴道的黏膜内壁变厚而起作用。
⑤ 甲孕酮。
⑥ 激素类的注射型避孕药。

hormone progestogen . It works in the same way as the progestogen only pill but has the advantage that you do not have to remember to take a pill every day . However, it has the same disadvantage as the hormonal pill, in that it provides no protection against STDs .

Norplant ① (an implant) is *a prescription birth control method*② that provides prolonged protection against pregnancy . *Six match-sized capsules are inserted under the skin of the upper arm in a fan-like pattern, using a local anesthetic .*③ The capsules release a constant dose of a hormone, progestin, thereby protecting against pregnancy . It is a very effective contraceptive method . Once in place the contraceptive benefits last for up to five years . A disadvantage of the implant is that it provides no protection against STDs .

*Diaphragm/ Cervical Cap covers opening to uterus and contains contraceptive cream or jelly which destroys sperm .*④ To make it work well, exam is needed to measure correct size . Cervical cap or diaphragm with cream or jelly should be inserted into vagina before sexual intercourse and left in place at least 6 hours after intercourse, but not longer than 24 hours . Diaphragm or cervical cap can be reused many times if properly cared for .

*Natural Family Planning (periodic abstinence) is an fertility awareness method .*⑤ Women learn to recognize their fertile time (when they can get pregnant) and infertile time (when they can get pregnant) by checking temperature and cervical mucus daily and keeping a record of their periods . Natural Family Planning users do not have intercourse during fertile time if pregnancy is not desired . Special classes are needed to learn the method . It is a method allowed by most religious groups . However, it is difficult for women with irregular cycles .

Sterilization is a permanent method . Men have vasectomy, an operation which separates the tubes between the testes and penis so they can still ejaculate though there is no sperm in their semen . Women have tubal ligation, an operation which separates and blocks the tubes between the ovaries and uterus so that sperm cannot reach eggs . Sterilization requires surgery and is permanent . A disadvantage is that it is rarely reversible .

*Emergency contraception*⑥ can be used when people have had *unprotected sex*⑦ or the birth control method they are using fails . There are basically two types of emergency contraception available: the *Emergency Contraceptive Pill (ECP)*⑧ method and the *Intrauterine Device (IUD)*⑨ method . The ECP method uses a high dose of combined oral contraceptives (OCs) to prevent conception . It is effective if used within 72 hours after unprotected sex . The IUD method involves the insertion of an IUD into the uterus by the

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① 皮下埋植。
② 一种需要医生开具处方的避孕方法。
③ 通过局麻方法把 6 片火柴大小的胶囊像扇形一样埋植在上臂的皮肤下。
④ 子宫帽盖住子宫的入口，并含有能杀灭精子的乳脂或胶状物。
⑤ 安全期避孕法是一种回避受孕时间的避孕方法。
⑥ 紧急避孕法。
⑦ 没有保护的性行为。
⑧ 紧急口服避孕药，缩写为 ECP。
⑨ 子宫内避孕器，缩写为 IUD。

doctor . The IUD creates an unfriendly environment for egg and sperm . The IUD must be inserted within 7 days of unprotected sex .

Vocabulary

1 . contraception [kɔntrəsepʃən] *n* . 避孕，避孕法
contraceptive [kɔntrəseptiv] *adj* . 避孕的
contraceptive pill 口服避孕药
2 . fertilization [fəːtilaizeiʃən] *n* . 授精
3 . egg [eg] *n* . 卵细胞
4 . sperm [spəːm] *n* . 精液，精子
5 . condom [kɔndəm] *n* . 避孕套
6 . penis [piːnis] *n* . 阴茎
7 . vagina [vəˈdʒainə] *n* . [动] 鞘，[解] 阴道
8 . spermicide [spəːmisaid] *n* . 杀精子剂
9 . cervix [səːviks] *n* . [解] 颈部，子宫颈
10 . pessary [pesəri] (*pl* . pessaries) *n* . 子宫套，阴道栓剂
11 . hormone [hɔːmoun] *n* . 荷尔蒙，激素
12 . oestrogen [iːstrədʒən] *n* . 雌激素
13 . progestogen [prəˈdʒestəʃən] *n* . 孕激素
14 . capsule [kæpsjuːl] *n* . 胶囊
15 . anesthetic [ænisˈθetik] *n* . 麻醉剂，麻药 *adj* . 麻醉的
16 . diaphragm [daiəfræm] *n* . 横膈膜，子宫帽避孕套
17 . cream [kriːm] *n* . 乳脂状
18 . jelly [ˈdʒeli] *n* . 胶状物
19 . intercourse [ˈintəkɔːs] *n* . 性交
20 . fertility [fəˈtiliti] *n* . 受精能力，生育力
fertile [ˈfəːtail; ˈfəːtil] *adj* . 有繁殖力的，能生育的
infertile *adj* . 不育的
21 . mucus [ˈmjuːkəs] *n* . 黏液，胶
22 . sterilization [sterilaizeiʃən] *n* . 杀菌，绝育
23 . vasectomy [vəˈsektəmi] *n* . 输精管切除术
24 . testis [ˈtestis] (*pl* . testes) *n* . 睾丸
25 . ejaculate [ˈkjuleit] *vi* . 射出，射精 *n* . 一次射出的精液
26 . semen [ˈsiːmen] *n* . 精液，精子
27 . ligation [laiˈgeiʃən] *n* . 结扎
tubal ligation 输卵管结扎术

Unit 37 Sexually Transmitted Diseases (STDs)

1. State the names of common STDs;
2. Tell the transmission route of STDs;
3. List the ways to prevent the spread of STDs;
4. Identify the goals of care for STDs patients.

*Sexually transmitted diseases (STDs) are a diverse group of infections acquired through sexual activity with an infected person.*① Examples of STDs are syphilis, gonorrhea, chancroid, granuloma inguinale, human papillomavirus, herpes simplex virus, chlamydial infections, and trichomonal vaginitis. *Acquired immunodeficiency syndrome (AIDS)*② is not a true STD, but one of its transmission routes is by *sexual intercourse*③.

Certain elements and behaviors contribute to the transmission of STDs. Examples of these include: unknown carriers, casual sex practices, absence of laws that require the reporting of all STDs, unfaithful sexual partner, length of time between exposure to an STD and the appearance of symptoms, failure to recognize the signs and symptoms of an STD followed by failure to seek early treatment as well as to *refrain from sexual activity*④ until treatment is complete, lack of proper knowledge regarding STDs and their prevention, and failure of the sexually active person to heed the warnings about STDs or adhere to methods of prevention.

Many clients with STDs might experience pain, hyperthermia, diarrhea, anxiety, and fear. *Some of them might face altered sexuality patterns, high risk for infection transmission, ineffective individual coping, noncompliance, and high risk for ineffective management of therapeutic regimen related to insufficient knowledge of modes of transmission, treatment, and prevention.*⑤ The major goals of nursing may include relief of pain or discomfort, relief of symptoms, control of infection, prevention of transmission, and avoidance of another infection. To make these goals measurable, more specific criteria

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① 性传播疾病是一组通过与感染者发生性关系而获得的疾病。
② 获得性免疫缺陷综合症，缩写为 AIDS。
③ 性交。
④ 避免性活动。
⑤ 某些病人可能面临性交方式的改变、高危感染传播、个人应对无效、不依从，以及由于缺乏疾病的传播、治疗及预防知识而带来的治疗管理无效等问题。

must be added as follows .

Pain is a common symptom for clients with STDs . *Reassure the client that the pain or discomfort is relieved when the infection is eradicated .*① Report severe discomfort or pain to the physician because an analgesic may be necessary . Some STDs, for example gonorrhea and nongonorrheal urethritis, invade the genitourinary tract and cause dysuria and difficulty in urinating . These symptoms usually disappear a few days after treatment begins .

Fever may be associated with some STDs and may persist until the infection is brought under control . Report any temperature elevation at the time of the physical assessment as well as a history of a fever to the physician . An antipyretic is usually effective in controlling a fever .

Diarrhea may occur as an adverse effect of some antimicrobial agents, especially when the drug is taken orally over an extended period . *Until the infection is controlled, instruct these clients to increase their fluid intake as well as attain a normal food intake .*② If diarrhea is severe, instruct the client to notify the physician .

People might express various emotions when they discover that they have an STD . They may be unable to accept the diagnosis, experience varying degrees of anxiety, fear the consequences that the infection may have on their marriage or their sexual partners, and fear that the infection may not be cured . Moreover, a client may have difficulty explaining or discussing this problem . A nurse should establish an atmosphere of mutual trust with the client . Allow clients to express the fears, anxieties, and emotions that accompany them, try to identify any discomfort the client may have, and attempt to dispel those discomfort by means of explanation, education, and, possibly, referral to a social agency, such as a marriage counseling service or a group that deals with teens with an STD .

All clients with STDs are capable of transmitting the infection, so they need to refrain from sexual activity until the infection is cured . When the STD cannot be cured and periodically recurs, as for example, an HSV infection, encourage the client to use a condom . *This alteration in sexual activity may not be accepted by all persons;*③ therefore, emphasize the importance of changing the pattern of sexual activity . *Advise these clients to return for examination and laboratory tests to be sure they are no longer able to transmit the infection to others .*④

Many people have wrong or no information about STDs . Some people disregard the information they do receive . A nurse should first determine what these clients know and then add to their knowledge or correct any misinformation . Give information or explanations in a positive manner to help the client understand the importance of seeking treatment, completing a course of therapy, preventing the transmission of the infection to others, and using methods to avoid infection in the future . Some cities have hot lines that adolescents and young adults

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① 告诉病人疼痛或不适将随着感染的控制而减轻，以打消其疑虑。
② 感染控制以前，指导病人增加液体摄入量并同时保持正常的食物摄入。
③ 可能不是所有的人都能接受性活动方式的改变。
④ 建议病人复查以确定他们不会再传染别人。

with STDs can call for answers to their questions and that can help people with an STD get in touch with a support group . The phone numbers of these hot lines can be given to the client .

Daily Language

1 . How does someone catch an STD ?
一个人是怎么感染上性传播疾病的？

2 . STDs are transmitted through having unprotected sex with an infected partner . One won't contract an STD from holding hands, hugging, or sharing a glass with others .
性传播疾病是通过与感染者发生无保护措施的性行为传播的。与他人握手、拥抱或共用杯子不会传染上性传播疾病。

3 . As STDs are infectious, it is better for you to refrain from sexual activity until the infection is cured .
因为性传播疾病具有传播性，所以你们最好节制性行为直到感染痊愈。

Vocabulary

1 . syphilis [ˈsifilis] *n* . 梅毒
2 . gonorrhea [ˌgɔnəˈriə] *n* . 淋病
3 . chancroid [ˈʃæŋkroid] *n* . 软性下疳
4 . granuloma inguinale [ˌgrænjuˈləumə ˈiŋgwineil] *n* . 肉芽肿，肉芽瘤
5 . human papillomavirus [ˈhjuːmən ˌpæpiləuˈmə ˈvaiərəs] *n* . 人类乳头瘤病毒
6 . herpes [ˈhəːpiːz] *n* . 疱疹
7 . chlamydia [kləˈmidiə] *n* . 衣原体
8 . trichomonal [ˌtrikəuˈmɔnəl] *adj* . 毛滴虫的，毛滴虫引起的
9 . vaginitis [ˌvædʒiˈnaitis] *n* . 阴道炎
10 . hyperthermia [ˌhaipəˈθəːmiə] *n* . 高热，体温过高
11 . genitourinary [ˈdʒenitəuˈjuərinəri] *adj* . 泌尿生殖器的
12 . dysuria [disˈjuəriə] *n* . 排尿困难
13 . urinate [ˈjuərineit] *vi* . 小便，撒尿

Questions on Reading

1 . What can contribute to the transmission of sexually transmitted diseases (STDs) ?
2 . What are the ways of preventing and controlling the spread of STDs ?
3 . What are the major goals of nursing the clients with STDs ?
4 . Can you list the nursing interventions for those problems such as pain, hyperthermia, diarrhea, anxiety, fear, and knowledge deficit ?